Bio-303

Biochemistry - II



Lecture No. 1 to 70
Midterm complete Handouts

Composed By

M. Habib

BS. Zoology 5th semester

Bioenergetics

Bioenergetics is the quantitative study of the **energy transductions that is changes of one form of e into another— that** occur in living systems, and of the nature and function of the chemical proc underlying these transductions.

Process of Bioenergetics

- **Glycolysis** is a biological process in which glucose is broken down into **pyruvate** thus generatin molecules of ATP (Adenosine triphosphate) per molecule of glucose
- **Gluconeogenesis** is another process in which the cell synthesizes glucose from biomolecules su proteins, amino acids and fats
- **Citric acid cycle** generates molecules of ATP by oxidation of energy stored in food material

Laws of thermodynamics

There are two fundamental laws of thermodynamics.

1. First law

The first law is the principle of the conservation of energy: it states that

"for any physical or chemical change, the total amount of energy in the universe remains constant; e may change form or it may be transported from one region to another, but it cannot be created or destro

Entropy

- The term "entropy," was first used in 1851 by Rudolf Clausius, which literally means "a cl within,"

Entropy = disorder

Entropy is a quantitative expression for the randomness or disorder in a system. When the product reaction are less complex and more disordered than the reactants, the reaction is said to proceed with a g entropy.

Denoted by **S**

ΔS has a +ve sign when entropy increase and –ve when entropy decreases

units of entropy are joules/mole.Kelvin (J/mol.K) (joules/mole **multiply by** Kelvin)

2. Second law

The second law of thermodynamics states that "*the universe always tends towards increasing disord all natural processes, the entropy of the universe increases.*

Enthalpy (*H*)

- Enthalpy is the heat content of the reacting system.
- It reflects the number and kinds of chemical bonds in the reactants and products
- When a chemical reaction releases heat, it is said to be **exothermic** and **value of H is negative**
- When a chemical reaction takes up heat from surroundings, it is called as endothermic and **positive values of *H***

Gibbs free energy *(G)*

- **Gibbs free energy, *G*** expresses the amount of energy capable of doing work during a reacti constant temperature and pressure
- When a reaction proceeds with the release of free energy the free-energy change, ***G*, has a neg value** and the reaction is said to be **exergonic**
- In endergonic reactions, **the system gains free energy and *G* is positive**
- The units of ***G* and *H*** are joules/mole or calories/mole
- (1 cal is equal to 4.184 Joules)
- Under the conditions existing in biological systems (including constant temperature and pres changes in free energy, enthalpy, and entropy are related to each other quantitatively by the equa

$$\Delta G = \Delta H - T \Delta S$$

ΔG: is the change in Gibbs free energy

ΔH: the change in enthalpy

T: the absolute temperature

Δ S: change in entropy

Cells require sources of Free Energy (G)

amount of energy capable of doing work during a chemical reaction (G)
Cells work at constant temperature and pressure (isothermal)
Heat flow is NOT energy source
Change in Gibbs Free energy (*ΔG)*

Significance of Free-Energy Changes (*ΔG)*

- Actual free-Energy changes depend on Rea and Product Concentrations
- Allows prediction of the direction of chemical reactions, their exact equilibrium position an amount of work done
- The free-energy change for a reaction is independent of the pathway by which the reaction occu
- **Free-energy changes are additive;** the net chemical reaction that results from successive rea sharing a common intermediate has an overall free-energy change that is the sum of the *ΔG* valu the individual reactions

Standard free-energy *G°*

- The free energy of reaction at under standard conditions
- When Temperature: *298K or 25℃*, when reactants are initially present at 1 M concentrations c gases, at partial pressures of 101.3 kilopascals (kPa), or 1 atm, the force driving the system to equilibrium is defined as the standard free-energy change, *Δ G°*

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

Key conventions in biochemical reactions

- Most biochemical reactions occur near pH 7 (buffered)
- Standard biochemical state
  - (H^+) is 10^{-7} M
  - concentration of water is 55.5 M
- Both the pH and the concentration of water are essentially constant
- Under these conditions, **Δ*G*** *is written as* **Δ*G'°*** ***(with prime)***
- **Concentration of water, H+ and Mg+ are not included in equations for simplicity**

Molar concentration of water

- Density of pure water = 1000 g/L
- Molar mass of water = 18 g/mol (2H + 1 Oxygen=2+16=18)
- Molar concentration of water = d/m
  = 1000 / 18.02
  = 55.5 mol/L

Difference between Δ*G* and Δ*G*°?

- the actual free-energy change, *Δ G*
- the standard free-energy change, *Δ G'°*
- Each chemical reaction has a characteristic standard free-energy change, which may be pos negative, or zero, depending on the **equilibrium constant** of the reaction
- **Thus *Δ G'°* is a constant: it has a characteristic, unchanging value for a given reaction**
- But the *actual* free-energy change, *Δ G*, is a function of reactant and product ***concentrations a the temperature prevailing during the reaction,*** none of which will necessarily match the sta conditions as defined above
- **The criteria for spontaneity of a reaction is the value of ΔG not $\Delta G'^0$**
- **if DG is negative (if the free energy of products is less than that of reactants).. A reacti spontaneous..** the reaction tends to go in the forward direction
- **if DG is positive (if the free energy of products is more than that of reactants).. A reaction is spontaneous…** the reaction tends to go in the reverse direction
- **if DG is ZERO (if the free energy of products is equal to that of reactants).. A reaction equilibrium**

Note :

The **standard conditions**
T: 298K (25^OC) **pH:** 7
Concentration: 1M
Partial pressures: 101.3 kilopascals (kPa), or 1 atm

The Standard Free-Energy is directly related to the Equilibrium constant

What is Equilibrium constant (K)?

- The composition of a reacting system tends to continue changing until **equilibrium** is reached
- At the equilibrium **concentration of reactants and products**, the **rates of the forward and re reactions are exactly equal** and no further net change occurs in the system.
- The number that expresses the relationship between the amounts of products and reactants pr at **equilibrium** in a reversible chemical reaction is known as the equilibrium constant, *K*eq
- In the general reaction

$$\mathbf{aA + bB \quad cC + dD}$$

How to calculate Equilibrium constant?

- The equilibrium constant is given by following equation where [A], [B], [C], and [D] are the concentrations of the reaction components at the point of equilibrium, equilibrium constant c written as:

$$\mathbf{Keq} = \frac{\mathbf{[C]^c\,[D]^d}}{\mathbf{[A]^a\,[B]^b}}$$

Standard free-energy change

The standard free-energy change of a chemical reaction is an alternative mathematical way of expre its equilibrium constant.

- ΔG is characteristic for a given reaction and can be calculated from the equilibrium constant f reaction:

$$\Delta G'^{\circ} = -RT \ln K'\text{eq}$$

where

R is the gas constant (8.31 cal/Mol.K)
T is standard temperature (25 C or 298K)
K'eq is the equilibrium constant

$$\boldsymbol{\Delta G'^{\circ} = -RT \ln K'\text{eq}}$$

- **If the *K*eq is equal to 1.0**

the standard free-energy change (ΔG^{O}) of that reaction is 0.0 (the natural logarithm **(ln)** of 1 zero)…**EQUILIBRIUM**

- **If *K*eq of a reaction is greater than 1.0**

ΔG^{O} is negative, **forward direction is favored**

- **If *K*eq is less than 1.0**

ΔG^{O} is positive, **reverse direction is favored**

WORKED EXAMPLE 13–1 Calculation of $\Delta G'^{\circ}$

Calculate the standard free-energy change of the reaction catalyzed by the enzyme phosphoglucomutase

Glucose 1-phosphate $\rightleftharpoons$ glucose 6-phosphate

given that, starting with 20 mM glucose 1-phosphate and no glucose 6-phosphate, the final equilibrium mixture at 25°C and pH 7.0 contains 1.0 mM glucose 1-phosphate and 19 mM glucose 6-phosphate. Does the reaction in the direction of glucose 6-phosphate formation proceed with a loss or a gain of free energy?

Solution: First we calculate the equilibrium constant:

$$K'_{eq} = \frac{[\text{glucose 6-phosphate}]}{[\text{glucose 1-phosphate}]} = \frac{19\text{ mM}}{1.0\text{ mM}} = 19$$

We can now calculate the standard free-energy change:

$$\begin{aligned}\Delta G'^{\circ} &= -RT \ln K'_{eq}\\ &= -(8.315\text{ J/mol}\cdot\text{K})(298\text{ K})(\ln 19)\\ &= -7.3\text{ kJ/mol}\end{aligned}$$

Because the standard free-energy change is negative, the conversion of glucose 1-phosphate to glucose 6-phosphate proceeds with a loss (release) of free energy. (For the reverse reaction, $\Delta G'^{\circ}$ has the same magnitude but the *opposite* sign.)

The Standard Free-Energy is directly related to the Equilibrium constant …cont

ΔG and $\Delta G'^{\circ}$ are related

- The actual free-energy change, ΔG is a variable that depends on $\Delta G'^{\circ}$ and on the concentratic reactants and products:

aA + bB → cC + dD

$$\Delta G = \Delta G'^{\circ} - RT \ln ([\text{products}]/[\text{reactants}])$$

$$\Delta G = \Delta G'^{\circ} + RT \ln \frac{[C]^c [D]^d}{[A]^a [B]}$$

(equation 13-4… according to Lehninger 6th Edition—chapter No 13)

- The concentration terms in this equation express the effects commonly called mass action, and the $[C]^c[D]^d/[A]^a[B]^b$ is called the **mass-action ratio**, ***Q***.
- Thus the equation can be written a follows:

$$\Delta G = \Delta G'^{\circ} + RT \ln Q$$

- Lets suppose that the reaction is under standard conditions..
- But the conc. Of reactants and products is NEITHER equal to eachother… Nor the conc. Is 1M
- **non-standard concentrations**
- **Actual conc. Of reactants and products but standard values of $\Delta G'^{\circ}$, *R and T***
- **Value of ΔG *at equilibrium* (for spontaneous reaction)??**

Standard Free-Energy Changes Are Additive

- Consider two sequential chemical reactions
- A → B → $\Delta G'^{0}_1$
- B → C → $\Delta G'^{0}_2$
- Since the two reactions are sequential, we can write the overall reaction as
- A → C → $\Delta G'^{0}_{total}$
- The $\Delta G'^{0}$ values of sequential reactions are additive.
- $\mathbf{\Delta G'^{0}_{total} = \Delta G'^{0}_1 + \Delta G'^{0}_2}$

Example

- For example, the synthesis of glucose 6-phosphate is the first step in the utilization of glucose by organisms:
- Glucose + Pi → glucose6-phosphate + H2O
- G'° = 13.8 kJ/mol
- The positive value of $\Delta G'^{\circ}$ predicts that under standard conditions the reaction will tend not to pr spontaneously in the direction written.
- Hydrolysis of ATP to ADP and Pi, is very exergonic:

  ATP + H2O = ADP + Pi

  $\Delta G'^{\circ}$ = -30.5 kJ/mol
- These two reactions share the common intermediates Pi and H2O and may be expressed as sequ reactions:

(1) Glucose + Pi → glucose6-phosphate + H2O ____reaction (1)

(2) ATP + H2O → ADP + Pi ____reaction (2)

- *Sum:* ATP + Glucose → ADP + glucose6-phosphate
- The overall standard free-energy change is obtained by adding the $\Delta G'^{\circ}$ values for indiv reactions:
- $\Delta G'^{\circ}$ =13.8 kJ/mol + (-30.5 kJ/mol) = -16.7 kJ/mol

As $\Delta G'^{\circ}$ is a way of expressing the equilibrium constant for a reaction. For reaction (1) above,

$$K'_{eq_1} = \frac{[\text{glucose 6-phosphate}]}{[\text{glucose}]\,[P_i]} = 3.9 \times 10^{-3}\ \text{M}^{-1}$$

- **The equilibrium constant for the hydrolysis of ATP is**

$$K'_{eq_2} = \frac{[\text{ADP}]\,[P_i]}{[\text{ATP}]} = 2.0 \times 10^{5}\ \text{M}$$

- **The equilibrium constant for the two coupled reactions is**

$$K'_{eq_3} = \frac{[\text{glucose 6-phosphate}]\,[\text{ADP}]\,[P_i]}{[\text{glucose}]\,[P_i]\,[\text{ATP}]}$$
$$= (K'_{eq_1})(K'_{eq_2}) = (3.9 \times 10^{-3}\ \text{M}^{-1})(2.0 \times 10^{5}\ \text{M})$$
$$= 7.8 \times 10^{2}$$

- This calculation illustrates that equilibrium constants: although the $\Delta G'^{\circ}$ values for two reaction sum to a third, overall reaction are *additive*, the *K*.eq for the overall reaction is the *product* individual *K*.eq values for the two reactions.
- **Equilibrium constants are *multiplicative.***
- Energy stored in ATP is used to drive to synthesis of glucose 6-phosphate, eventhough its form from glucose and P_i is endergonic.
- **This strategy works only if compounds such as ATP are continuously available.**
- **Homogeneous chemical reaction chemical reactions** in which the reactants and products are same phase.
- **Heterogeneous chemical reaction** have reactants in two or more phases.
- **Le Châtelier Principle** ("The Equilibrium Law") states that when a system experiences a distur (such as concentration, temperature, or pressure changes), it will respond to restore a equilibrium state.

Types of chemical reactions in cells

5 types of chemical reactions in cells

The reactions going on in biological systems can be categorized into 5 types:

1. Reactions that make or break carbon-carbon bonds
2. Internal rearrangements, isomerizations, and eliminations
3. Free-radical reactions
4. Group transfer
5. Oxidation-reduction reactions

2 basic chemical principles

(1) A covalent bond can be broken in two general ways

- Homolytic cleavage
- Heterolytic cleavage
- A **Covalent bond**, also called a molecular *bond*, involves the sharing of electron pairs
- between atoms.
- These electron pairs are known as shared pairs or *bonding* pairs, and the stable balance of attra and repulsive forces between atoms, when they share electrons, is known as *covalent bonding*.

2 mechanisms for breaking a
C-C or C-H bond

- **Homolytic cleavage: each atom leaves the bond as a radical, carrying one unpaired electro**
- **Heterolytic cleavage (more common): one atom retains both bonding electrons**

Second principle

2. Many biochemical reactions involve interactions between nucleophiles and electrophiles

- Nucleophiles: functional groups rich in and capable of donating electrons
- (Nucleophiles are Lewis bases)
- Electrophiles: electron-deficient functional groups that seek electrons

Common nucleophiles and electrophiles in biochemical reactions

Electron pushing: The movement of a pair of electrons (a lone pair of electrons or a bond) from an ele rich site to an electron poor site.

Significance of Carbonyl group

- Carbonyl groups are particularly important in the chemical transformations of metabolic pathwa

(1) The carbon of a **carbonyl group has a partial positive** charge due to the electron-withdra property of the **carbonyl oxygen→ electrophilic carbon**

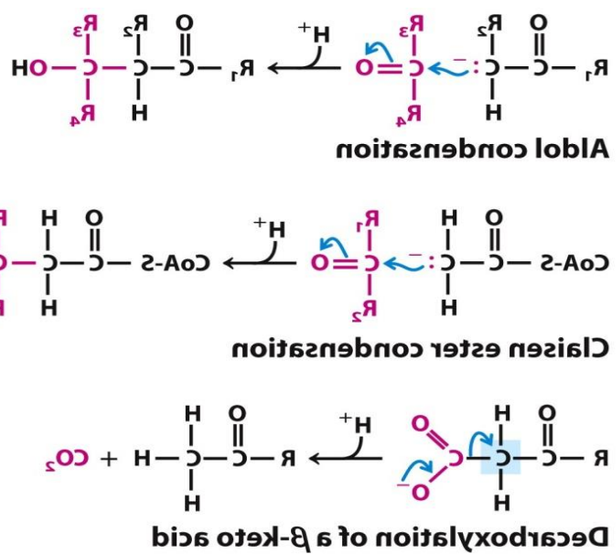
(2) Carbonyl group facilitate the formation of a carbanion's on an adjoining carbon by delocalizin carbanion's negative charge

(3) An imine group can serve a similar function

1. Types of reactions that make or break C-C bonds

- Three major class of reaction in which **carbon-carbon** bonds are formed or broken

(1) Aldol condensation
(2) Claisen condensation
(3) Decarboxylation



Common nucleophiles and electrophiles in biochemical reactions

Electron pushing: The movement of a pair of electrons (a lone pair of electrons or a bond) fro electron rich site to an electron poor site.

Significance of Carbonyl group

- Carbonyl groups are particularly important in the chemical transformations of metabolic pathwa

(1) The carbon of a **carbonyl group has a partial positive** charge due to the electron-withdra property of the **carbonyl oxygen→ electrophilic carbon**

(2) Carbonyl group facilitate the formation of a carbanion's on an adjoining carbon by delocalizir carbanion's negative charge

(3) An imine group can serve a similar function

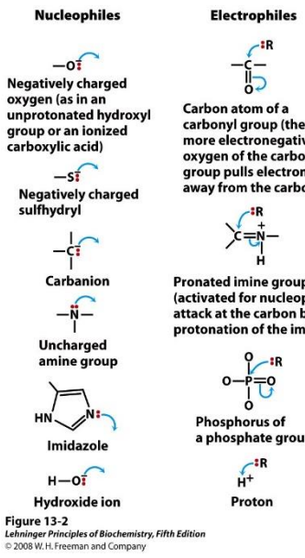
1. Types of reactions that make or break C-C bonds

- Three major class of reaction in which **carbon-carbon** bonds are formed or broken

(1) Aldol condensation

(2) Claisen condensation

(3) Decarboxylation



1. **Aldol condensation:**
   When the enolate of an aldehyde or a ketone reacts at the α-c with the carbonyl of another molecule to obtain β- hy aldehyde or β-hydroxy ketone
2. **Claisen condensation:**

A carbon–carbon bond forming **reaction** that occurs between two este one ester and another carbonyl compound in the presence of a strong resulting in a β-keto ester.

- In **Aldol condensation and Claisen condensation** a carb serves as nucleophile and the carbon of a carbonyl group serv electrophile, but **,** in **Claisen condensation,** the carbanion stabilized by the carbonyl of an adjacent thioester

3. **Decarboxylation:** involves the formation of a carb stabilized by a carbonyl group. eg, during formation of k bodies during fatty acid metabolism

   Decarboxylation: involves the formation of a carbanion stabilized by a carbonyl group. eg, d formation of ketone bodies during fatty acid metabolism

- **ENOL:** also known as, alkenol, is a type of reactive structure or intermediate in organic chemistr is represented as an alkene with a **hydroxyl group attached to one end of the alkene double b**
- **ESTER:** is a chemical compound derived from an acid (organic or inorganic) in which at least OH (hydroxyl) group is replaced by an –O–alkyl (alkoxy) group. Usually, **esters** are derived b reaction between a carboxylic acid and an alcohol.
- **ALDOL:** An **aldol** (from "Aldehyde alcohol") is a hydroxy ketone or aldehyde, and is the pr of **aldol** addition (as opposed to **aldol** condensation.

Types of chemical reactions in cells...continued

5 types of chemical reactions in cells

The reactions going on in biological systems can be categorized into 5 types:

1. **Reactions that make or break carbon-carbon bonds**
2. **Internal rearrangements, isomerizations, and eliminations**
3. **Free-radical reactions**
4. **Group transfer**
5. **Oxidation-reduction reactions**

2. Internal rearrangements, isomerizations, and eliminations

- **Intramolecular rearrangement**: redistribution of electrons results in alterations of many dif types without a change in the overall oxidation state of the molecule
  - different group in the molecule may undergo oxidation-reduction, with no net change in oxidatio of the molecule
  - groups at a double bond may undergo a cis-trans rearrangement
  - the positions of double bonds may be transposed

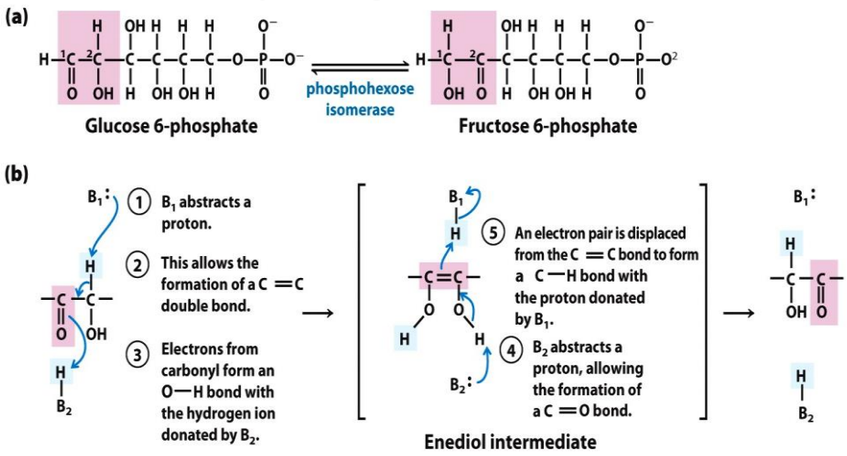


Figure 13-6
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

(b) Cis-trans rearrangement

- In general, Peptide bonds formed between amino acid residues with alpha carbons in configuration… (more than 99.95% have **trans configuration**.)
-
- For peptide bonds involving the **imino nitrogen of proline**, **peptide bonds** are able to exist both **and trans configuration.. (6% are in the cis configuration)** through peptidyl-prolyl (pr isomerase enzyme

(c) Elimination

- **The loss of water from an alcohol, resulting in the introduction of a C=C** bond b/w 2 ad carbon atoms
- **An elimination reaction with no changes in oxidation state**

3) Free-radical reactions

- Homolytic cleavage of covalent bond → produce free radical

1) **Isomerizations** that make use of adenosylcobalamin (vitamin B12) for methylmalonyl-CoA m (MCM) reactions
2) Radical-initiated **decarboxylation** reaction
3) Some **reduction** reaction, such as that catalyzed by **ribonucleotide reductase (co ribonucleotides to deoxyribonucleotides)**
4) Some **rearrangement reactions**, such as that catalyzed by **DNA photolyase**

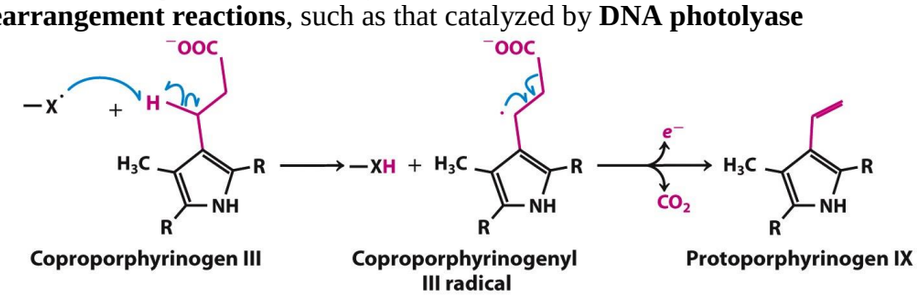


Figure 13-7
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Oxygen-independent **coproporphyrinogen III oxidase** (HemN) catalyzes decarboxylation

Types of chemical reactions in cells...continued

- **Group Transfer Reactions**
  The transfer of **acyl, glycosyl, and phosphoryl** is common in living cells
- **Acyl transfer**: acyl group transfer generally involves the addition of a nucleophile to the car carbon of an **acyl group** to form a tetrahedral intermediate, eg. Chymotrypsin reaction

2) Phosphoryl group activates the metabolic intermediate for subsequent reactio

- **Nucleophilic substitutions reactions and attachment of a phosphoryl group to functional gr like (-OH) occur in hundreds of metabolic reactions**
- **A nucleophile (OH) on c-6 of glucose attacks ATP, it displaces ADP..and a pentacov intermediate can be formed..**
- A phosphoryl group ($-PO_3^{2-}$) from ATP is transferred to an alcohol (→ forming a phosphate est
- When nucleophile attacks the electrophilic phosphorus atom in ATP, a relatively stable pentacov structure forms as a reaction intermediate.
- **Phosphorous can form five covalent bonds**
- Because oxygen is more electronegative than phosphorus, the sharing of electrons is unequa central phosphorus bears a partial positive charge and can therefore act as an electrophile
- **Kinases: enzymes that catalyze phosphoryl group transfer) using ATP as donor)**
- Thioalcohols (thiols): the oxygen atom of an alcohol is replaced with a sulfur atom

5) Oxidation-reduction reactions

Oxidation
Addition of oxygen
loss of electrons
Loss of hydrogen
Reduction
Removal of oxygen
Gain of electrons
Gain of hydrogen

- Carbon **atoms can exist in five oxidation states**
- The transition of these status are of crucial importance in metabolism
- Biological reaction (commonly), a compound loses two electrons and two hydrogen **dehydrogenations (catalyzed by dehydrogenases)**
- In some biological oxidation, a carbon atom becomes covalently bonded to an oxygen atom (a en catalyze these reaction calls **oxidases**)
- Biological reaction (commonly), a compound loses two electrons and two hydrogen dehydrogenations (catalyzed by dehydrogenases)
- In some biological oxidation (but not all), a carbon atom becomes covalently bonded to an ox atom (a enzyme catalyze these reaction called **oxidases**)

Oxygenases: if oxygen is derived directly from molecular oxygen (O_2).

Vinyl group
In chemistry, vinyl or ethenyl is the functional group with the formula −CH=CH2. It is the eth (IUPAC ethene) molecule ($H_2C=CH_2$) less one hydrogen atom.

Acyl group
It contains a double bonded oxygen atom and an **alkyl group**. In organic chemistry, the **acyl g** (IUPAC name: alkanoyl) is usually derived from a carboxylic acid. **Acyl group is ele withdrawing due to the electronegativity of the oxygen atom.**

Phosphoryl group
A *phosphoryl group* is the chemical entity PO_3^{2-}. The term is usually used for the compounds in v the *phosphoryl group* is attached to other atoms, e.g. *phosphoryl* chloride, or in the descripti catalytic mechanism.

Enediol
An **alkene enol** with a **hydroxyl group attached to both carbon atoms of the carbon double b** A reducing sugar can form an enediol.

Enone:
An enone, also called an α, β-unsaturated carbonyl, is a type of organic compound consisting alkene conjugated to a ketone. The simplest enone is methyl vinyl ketone or $CH_2=CHCOCH_3$.

Acryloyl group

In organic chemistry, the **acryloyl group is form of enone** with structure $H_2C=CH–C(=O)–$**.** It **acyl group derived from acrylic acid**. The preferred IUPAC name for the group is prop-2-enoy it is also known as **acrylyl or simply acryl.**

Phosphoryl Group Transfers and ATP

- **Energy currency** of living cells
- ATP is used for **Phosphoryl transfer reactions**
- Heterotrophic organisms obtain energy from ATP by breakage of covalent bond, and can underg of the two following reactions

A) ATP → ADP+ P_i OR
B) ATP → AMP+ 2 P_i

ATP donates some of its chemical energy to

1. Endergonic processes such as the synthesis of **metabolic intermediates** and macromolecules smaller precursors
2. The **transport of substances across membranes** against concentration gradients
3. Mechanical motion (**muscle contraction**), etc.
4. **Energy donation involves group transfer and hydrolysis of ATP/energy rich phos compounds**

The Free-Energy Change for ATP Hydrolysis Is Large and Negative

1. The hydrolytic cleavage of the terminal **phosphoanhydride bond** in ATP separates one of the negatively charged phosphates and thus relieves some of the electrostatic repulsion in ATP
2. The terminal two **phosphoanhydride bonds**, are known as high energy **bonds** because their cle releases tremendous amount of free energy
3. Released P_i and ADP are highly soluble

- ADP^{2-} is the other product of hydrolysis and it immediately ionizes, releasing H^+ into a medi very low $[H^+]$ (~10^{-7} M)
- the hydrolysis of ATP is highly **exergonic ($\Delta G'^0$ = -30.5 kj/mol)**, the ATP is stable at pH 7
- Rapid hydrolysis of ATP occurs only when catalyzed by an enzyme
- Actual free energy of hydrolysis (ΔG) of ATP in living cells is very different because

(1) the cellular concentration of ATP, ADP, and Pi are not identical
(2) ATP, ADP, and Pi concentration are much lower than the standard condition of 1M
(3) Mg^{2+} in the cytosol binds to ATP and ADP

For ATP as phosphoryl group donor, the true substrate is MgATP-2.

Phosphorylation potential: The actual free energy of hydrolysis of ATP under intracellular conditi often called its **phosphorylation potential**, ΔG_p.

WORKED EXAMPLE 13–2 Calculation of ΔG_p

Calculate the actual free energy of hydrolysis of ATP, ΔG_p, in human erythrocytes. The standard free energy of hydrolysis of ATP is −30.5 kJ/mol, and the concentrations of ATP, ADP, and P_i in erythrocytes are as shown in Table 13–5. Assume that the pH is 7.0 and the temperature is 37 °C (body temperature). What does this reveal about the amount of energy required to *synthesize* ATP under the same cellular conditions?

Solution: The concentrations of ATP, ADP, and P_i in human erythrocytes are 2.25, 0.25, and 1.65 mM, respectively. The actual free energy of hydrolysis of ATP under these conditions is given by the relationship (see Eqn 13–4)

$$\Delta G_p = \Delta G'^{\circ} + RT \ln \frac{[ADP][P_i]}{[ATP]}$$

Substituting the appropriate values we get

$$\begin{aligned}\Delta G_p &= -30.5\ \text{kJ/mol} + \left[(8.315\ \text{J/mol}\cdot\text{K})(310\ \text{K}) \ln \frac{(0.25\times10^{-3})(1.65\times10^{-3})}{(2.25\times10^{-3})}\right] \\ &= -30.5\ \text{kJ/mol} + (2.58\ \text{kJ/mol}) \ln 1.8\times10^{-4} \\ &= -30.5\ \text{kJ/mol} + (2.58\ \text{kJ/mol})(-8.6) \\ &= -30.5\ \text{kJ/mol} - 22\ \text{kJ/mol} \\ &= -52\ \text{kJ/mol}\end{aligned}$$

(Note that the final answer has been rounded to the correct number of significant figures (52.5 rounded to 52), following rules for rounding a number that ends in a 5 to the nearest even number.) Thus ΔG_p, the actual free-energy change for ATP hydrolysis in the intact erythrocyte (−52 kJ/mol), is much larger than the standard free-energy change (−30.5 kJ/mol). By the same token, the free energy required to *synthesize* ATP from ADP and P_i under the conditions prevailing in the erythrocyte would be 52 kJ/mol.

Phosphorylated compounds

- Phosphoenolpyruvate
- 1,3-bisphosphoglycerate
- Phosphocreatine
- ADP
- ATP
- AMP
- PPi
- Glucose 1-phosphate
- Fructose 6-phosphate
- Glucose 6-phosphate

Large Free Energies of Hydrolysis Phosphorylated Compounds and Thioesters

- **Other compounds that have large free energy of Hydrolysis**

1. Phosphorylated compounds
2. Thioesters (Acetyl-CoA)

Phosphoryl Group Transfers and ATP ...continued

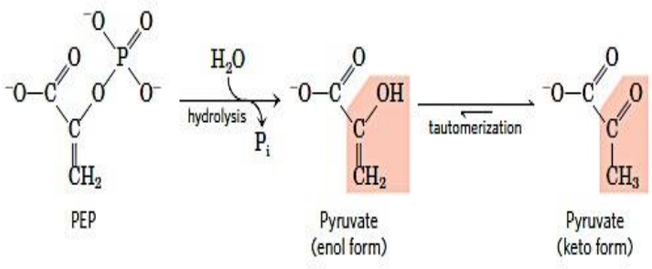
Other Phosphorylated Compounds Have Large Free Energies of Hydrolysis

- **Other compounds that have large free energy change**
- Phosphorylated compounds
- Thioesters (Acetyl-CoA)

Phosphoenolpyruvate (PEP)

- Phosphoenolpyruvate contains a **phosphate ester bond** that undergoes hydrolysis to yield to **enol of pyruvate**
- The **enol** form of pyruvate can immediately tautomerize to the more stable **keto** form of pyruvat
- Because phosphoenolpyruvate has only one form (enol) and the product, pyruvate, has two po forms, the product is more stabilized relative to the reactant.
- This is the greatest contributing factor to the high standard free energy change of hydroly phosphoenolpyruvate ($\Delta G'^0$ = -61.9 kj/mol)

FIGURE 13-13 Hydrolysis of phosphoenolpyruvate (PEP). Catalyzed by pyruvate kinase, this reaction is followed by spontaneous tautomerization of the product, pyruvate. Tautomerization is not possible in PEP, and thus the products of hydrolysis are stabilized relative to the reactants. Resonance stabilization of P_i also occurs, as shown in Figure 13-11.



$$PEP^{3-} + H_2O \longrightarrow pyruvate^- + HPO_4^{2-}$$
$$\Delta G'^0 = -61.9\ kJ/mol$$

1,3-bisphosphoglycerate

- 1,3-bisphosphoglycerate contains an **anhydride bond** between the carboxyl group at C-1 phosphoric acid.
- Hydrolysis of this **acyl phosphate** is accompanied by a large, negative, standard free energy cl ($\Delta G'^0$ = -49,3 kj/mol)
- This large, negative $\Delta G'^0$ can, again, be explained in terms of the structure of reactants and proc

Addition of water to anhyhride bond of 1,3-**bisphosphoglycerate, one of the direct product phosphoglyceric acid,** immediately leads to the lose a proton to give the **carboxylate io** phosphoglycerate, which has two equally probable resonance forms

Removal of a direct product, **3-phospho-glyceric acid**, and formation of resonance-stabilized ion fav forward reaction.

Phosphocreatine (Pcr)

- In the **phosphocreatine**, the P-N bond can be hydrolyzed to generate free creatine and P_i.
- The release of P_i and the resonance stabilization of creatine favor the forward reaction.
- Pi is also resonance stabilized.
- The standard free energy change of phosphocreatine is large and negative ($\Delta G'^0$ = -43 kj/mol).

Thioesters

- Thioesters **contain a sulfur** atom in the position occupied

by an oxygen

- Thioesters have large, negative standard free energy change

of hydrolysis.

- **Hydrolysis of acetyl-coenzyme A..** important in metabolism and has a large, negative, standar energy of hydrolysis.
- **The acyl group in these compounds is activated for trans-acylation**, condensation or oxid reduction reactions.
- Hydrolysis of the **ester bond generates a carboxylic acid**
- $\Delta G'^0$ = -31,4 kj/mol for acetyl-CoA hydrolysis

Hydrolysis of acetyl-coenzyme A. Acetyl-CoA is a thioester with a large, negative, standard free ene hydrolysis.

Summary for hydrolysis reactions

For hydrolysis reactions with large, negative standard free energy changes, the products are more stabl the reactants for one or more of the following reasons:

1. The bond strain in reactants due to electrostatic repulsion is relieved by charge separation, as for ATI
2. The products are stabilized by **ionization**, as for ATP, acyl phosphates, thioesters
3. The products are stabilized by isomerization (tautomerization) as for phosphoenolpyruvate
4. The products are stabilized by **resonance** as for creatine released from phosphocreatine, **carboxyla released from acyl phosphates** and thioesters and phosphate released from **anhydride or ester linkag**

- The phosphate compounds found in living organisms can be arbitrarily divided into two groups on their standard free energy changes of hydrolysis.

1. 'High-energy' compounds have a $\Delta G'^0$ of hydrolysis more negative than -25 kj/mol
2. 'low-energy' compounds have a less negative $\Delta G'^0$

- ATP with a $\Delta G'^0$ of hydrolysis of -30 kj/mol is a high-energy compound
- glucose 6-phosphate is a low-energy compound ($\Delta G'^0$ = -13,8 kj/mol)

TABLE 13–6 Standard Free Energies of Hydrolysis of Some Phosphorylated Compounds and Acetyl-CoA (a Thioester)

	$\Delta G'^\circ$ (kJ/mol)	$\Delta G'^\circ$ (kcal/mol)
Phosphoenolpyruvate	−61.9	−14.8
1,3-bisphosphoglycerate (→ 3-phosphoglycerate + P_i)	−49.3	−11.8
Phosphocreatine	−43.0	−10.3
ADP (→ AMP + P_i)	−32.8	−7.8
ATP (→ ADP + P_i)	−30.5	−7.3
ATP (→ AMP + PP_i)	−45.6	−10.9
AMP (→ adenosine + P_i)	−14.2	−3.4
PP_i (→ 2P_i)	−19.2	−4.0
Glucose 3-phosphate	−20.9	−5.0
Fructose 6-phosphate	−15.9	−3.8
Glucose 6-phosphate	−13.8	−3.3
Glycerol 3-phosphate	−9.2	−2.2
Acetyl-CoA	−31.4	−7.5

Source: Data mostly from Jencks, W.P. (1976) in *Handbook of Biochemistry and Molecular Biology,* 3rd edn (Fasman, G.D., ed.), *Physical and Chemical Data*, Vol. 1, pp. 296–304, CRC Press, Boca Raton, FL. The value for the free energy of hydrolysis of PP_i is from Frey, P.A. & Arabshahi, A. (1995) Standard free-energy change for the hydrolysis of the α-β-phosphoanhydride bridge in ATP. *Biochemistry* **34**, 11,307–11,310.

Table 13-6
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

ATP Provides Energy by Group Transfers, Not by Simple Hydrolysis

ATP hydrolysis in two steps.

- The contribution of ATP to a reaction is often shown as a single step, but is almost always a tw process.
- Shown here is the reaction catalyzed by ATP-dependent **glutamine synthetase.**
- A phosphoryl group is transferred from ATP to glutamate
- The phosphoryl group is displaced by NH_3 and released as Pi.

Hydrolysis accompanied biological processes

- No covalent binding of ATP (or GTP), followed by its hydrolysis ADP (or GDP) and Pi, (to cycle some proteins between two conformations), in muscle contracti

- In the movement of enzymes along DNA
- In the movement of ribosomes along messenger RNA
- The energy-dependent reactions catalyzed by helicases and some topoisome GTP-binding proteins that act in signaling pathways to drive conformational changes

The phosphate compounds found in living organisms can be divided into two groups based on their sta free energy changes of hydrolysis.

- 'High-energy' compounds have a $\Delta G'^0$ of hydrolysis more negative than -25 kj/mol
- 'low-energy' compounds have a less negative $\Delta G'^0$

ATP with a $\Delta G'^0$ of hydrolysis of -30 kj/mol is a high-energy compound
glucose 6-phosphate is a low-energy compound ($\Delta G'^0$ = -13,8 kj/mol)

- The term "high-energy phosphate bond," used to describe the P-O bond broken in hydrolysis reac is incorrect and misleading as wrongly suggests that the bond itself contains the energy.
- the breaking of all chemical bonds requires an *input* of energy
- The additivity of free energy changes of sequential reactions
- any phosphorylated compound can be synthesized by coupling the synthesis to the breakdo another phosphorylated compound with a more negative standard free energy change of hydroly
- **Example:** $\Delta G'^0$
- **PEP + H_2O → Pyruvate + P_i -61,9**
- **ADP+ P_i → ATP+ H_2O +30,5**
- **PEP + ADP → Pyruvate + ATP -31,4**
- **Clevage of P_i from PEP releases more energy than is needed to drive to condensation of P ADP, the direct donation of a phosphoryl group from PEP to ADP is thermodynamically fe**
- Notice that while the energy of overall reaction is represented as the algebraic sum of firs reactions, the overall reaction (third) does not involve P_i ; PEP donates a phosphoryl group direc ADP.
- On the basis of their standard free energy changes of hydrolysis, phosphorylated compounds as h a high or low **phosphoryl group transfer potential**.
- The transfer of a phosphoryl group to a compound effectively puts free energy into that compoun can be used in subsequent metabolic transformations.

ATP Provides Energy by Group Transfers, Not by Simple Hydrolysis.. cont

Ranking of biological phosphate compounds by standard free energies of hydrolysis

- Flow of phosphoryl groups, **represented by P**, from **high-energy phosphoryl group donors via to acceptor molecules** (such as glucose and glycerol) to form their low-energy phosphate deriva
- This flow of phosphoryl groups is catalyzed by **kinases**, and proceeds with an overall loss o energy under intracellular conditions.
- **in aqueous solution ATP is thermodynamically unstable and is therefore a good phosp group donor, it is kinetically stable.**
- Because of high activation energies required for uncatalyzed reaction ATP does not spontane donate phosphoryl groups to water or to the other potential acceptors in the cell.
- ATP hydrolysis occurs only when specific enzymes which lower the energy of activation are pre
- **thermodynamically unstable**: when energy level of products is lower than that of reactants
- Kinetically stable is when reaction has high activation energy due to some strong bonds.

ATP Donates Phosphoryl, Pyrophosphoryl, and Adenylyl Groups

Nucleophilic displacement reactions of ATP.

- Any of the three P atoms (α,β orγ) may serve as the electrophilic target for nucleophilic attack.
- By the labeled nucleophile R-^{18}O: The nucleophile may be an alcohol (ROH), a carboxyl (RCOO$^-$), or a phosphoanhydride (a nucleoside mono- or diphosphate, for example).
- When the oxygen of the nucleophile attacks the position, the bridge oxygen of the product is lal indicating that the group transferred from ATP is a phosphoryl ($2PO_3^{-2}$), not a phosphate (OPO_3
- Attack on the *position* displaces AMP and leads to the transfer of a pyrophosphoryl (not pyrophos group to the nucleophile.
- Attack on the position displaces PPi and transfers the adenylyl group to the nucleophile.
- 5-**phosphoribosyl-1-pyrophosphate** is a key intermediate in nucleotide synthesis
- Hydrolysis of a-b phosphoanhydride bond: ~46 kJ/mol
- Hydrolysis of b-g phosphoanhydride bond: ~31 kJ/mol
- PPi → 2 Pi: -19.2 kJ/mol: total = -64.8 kJ/mol (**Adenylation: produces Ppi hydrolysed to 2 inorganic pyrophosphatase enzyme)**

Activation of fatty acids

- First, adenylate (AMP) is transferred from ATP to the carboxyl group of the fatty acid, forming a r anhydride (fatty acyl adenylate) and liberating PPi.
- The thiol group of coenzyme A then displaces the adenylyl group and forms a thioester with the acid.
- sum of these two reactions is energetically equivalent to the exergonic hydrolysis of ATP to AM Ppi
- The formation of fatty acyl–CoA is endergonic but made energetically favorable (exergoni hydrolysis of the PPi by inorganic pyrophosphates.
- Thus, in the activation of a fatty acid, both phosphoanhydride bonds of ATP are broken.

Luciferin

- William McElroy and his colleagues at the Johns Hopkins University isolated the principal bioche components: luciferin, a complex carboxylic acid, and luciferase, an enzyme.
- The generation of a light flash requires activation of luciferin by an enzymatic reaction invc pyrophosphate cleavage of ATP to form luciferyl adenylate.
- In **the presence of molecular oxygen and luciferase**, the luciferin undergoes a multistep **oxic decarboxylation to oxyluciferin**.
- This process is accompanied by emission of light.
- **Luciferin is regenerated from oxyluciferin** in a subsequent series of reactions.
- **In the laboratory, pure firefly luciferin and luciferase are used to measure minute quantit ATP by the intensity of the light flash produced. As little as a few picomoles (10^{-12} pico m ATP can be measured in this way.**
- **Next-gen pyrosequencing of DNA relies on flashes of light from the luciferin-luciferase rea to detect the presence of ATP after addition of nucleotides to a growing strand of DNA**

Assembly of Informational Macromolecules Requires Energy

- **Pi and Ppi groups from ATP**
- The precursors for DNA and RNA synthesis are nucleoside triphosphates
- Polymerization of DNA and RNA is accompanied by cleavage of the **phosphoanhydride lir** between **α and β phosphates, with the release of PPi**
- The moieties transferred to the growing polymer in these reactions are **adenylate (AMP), guar (GMP), cytidylate (CMP), or uridylate (UMP)** for RNA synthesis, and their deoxy analogs TMP in place of UMP) for DNA synthesis.

Adenylyl groups from ATP

- Required for the **activation of amino acids for protein synthesis,** several steps on the riboson also accompanied by **GTP hydrolysis**.

ATP energizes active transport and muscle contraction

- Transport processes are major consumers of energy
- In brain and kidney, two third of total is consumed to pump sodium and potassium ions a plasma membrane (Na + K+ ATPase)
- Na+ dependent phosphorylation of Na+ K+ ATPase changes protein conformation dependent dephosphorylation brings back original conformation.
- Phosphoryl group transfer: to enzyme, not to substrate

ATP energizes active transport and muscle contraction

- Direct hydrolysis of ATP (ATP → ADP+Pi) is the source of energy in the conformational ch that produce muscle contraction
- ATP bind with head of myosin (not covalent) → dissociate from actin → ATP hydrolysis in conformational change (still ADP and Pi bound tightly) → Pi release → conformational change head of myosin → actin move

Transphosphorylations between nucleotides occur in all cell types

- ATP is considered as the cell's energy currency and donor of phosphoryl groups, all other nucle triphosphates (GTP, UTP, CTP) and all the deoxynucleoside triphosphates (dATP, dGTP, dTT dCTP) are energetically equivalent to ATP
- The free energy changes associated with hydrolysis of their **phosphoanhydride linkages** are nearly identical with those for ATP.
- In preparation for their various biological roles, other nucleotides are generated as the nucle triphosphate (NTP) forms by phosphoryl group transfer to the corresponding nucleoside diphosp (NDPs) and monophosphates (NMPs)
- ATP is the primary high-energy phosphate compound produced by catabolism in the process glycolysis, oxidative phosphorylation. Several enzymes carry phosphoryl groups from ATP to the nucleotides
- Nucleoside diphosphate kinases, found in all cells, catalyzes the reaction

Mg^{2+}

- ATP+NDP(or dNDP) ADP+NTP(or dNDP)
- Although this reaction is fully reversible the relatively high ATP/ADP ratio in cells normally d the reaction to the right, with the net formation of NTPs and dNTPs
- Phosphoryl group transfers from ATP results in an accumulation of ADP. eg, when mus concracting vigorously ADP accumulates and interferes with ATP-dependent contraction.
- During intense demand for ATP, the cell lowers the ADP concentration, and at the same time acc ATP, by the action of adenylate kinase:

Mg^{2+}

2ADP ATP + AMP

Ping-pong mechanism of nucleoside diphosphate kinase

Ping-pong mechanism of nucleoside diphosphate kinase: The enzyme binds its first substrate (ATP i example), and a phosphoryl group is transferred to the side chain of a **His** residue. ADP departs, and ar nucleoside (or deoxynucleoside) diphosphate replaces it, and this is **converted to the correspo triphosphate by transfer of the phosphoryl group** from the phosphohistidine residue.

- This reaction is fully reversible, so after the intense demand for ATP ends, the enzyme can re AMP by converting it to ADP which can then be phosphorylated to ATP in mitochondria
- A similar enzyme **guanylate kinase, converts GMP to GDP** at the expense of ATP.
- **By these pathways energy conserved in the catabolic production of ATP is used to supply th with all required NTPs and dNTPs**
- **Phosphocreatine (PCr)** serves as a ready source of phosphoryl groups for the quick synthesis of from ADP. The phosphocreatine concentration in skeletal muscle is considerably higher than th the other tissues. The enzyme **creatine kinase** catalyzes the reversible reaction.

Mg^{2+}

- ADP+ PCr ATP+Cr

$\Delta G'^{0}$ = -12,5 kj/mol

Inorganic polyphosphate is a donates phosphoryl group

- Inorganic polyphosphate ((polyP)n): linear polymer of Pi … Present in simple organisms of other like yeast
- In yeast, polyP accumulated in vacuoles (= 200 mM)
- Serve as a **phosphagen (Reservoir of phosphoryl group).. Like PCr**
- The shortest polyP is PPi (n=2): can serve as the energy source for active transport of H^+ in plan (across vacuolar membrane)
- **In bacteria, the enzyme polyphosphate kinase-1 (PPK-1) catalyzes the reversible reaction**
- **polyphosphate kinase-2** (PPK-2), catalyzes the reversible synthesis of GTP (or ATP) polyphosphate and GDP (or ADP):

Biological oxidation-reduction reactions

- Central feature of metabolism

1) Phosphoryl group transfer
2) Electron transfer in oxidation-reduction

- The flow of electrons in oxidation-reduction reactions is responsible for all work done by organisms

Oxidation	**Reduction**
Addition of oxygen	Removal of oxygen
loss of electrons	**Gain** of electrons
Loss of hydrogen	**Gain** of hydrogen

Biological oxidation-reduction reactions

- In nonphotosynthetic organism, source of electron: food
- In photosythetic organism, source of electron: chemical compound excited by the absorption of
- Electrons move from various metabolic intermediates to specialized electron carriers in enz catalyzed reaction → energy release
- **The flow of electrons can do biological work**

- **Electromotive force (emf):** (measured in volts), is the voltage developed by any source of elec energy such as a battery… In which two chemical species differ in their affinity for electrons, s electrons flow spontaneously through the circuit, due to the difference in electron affinity
- **Living cells have an analogous biological "circuit",** with relatively reduced compound su glucose (the source of electrons)
- Glucose → enzymatically oxidized → release electron → spontaneously flow through electron-c intermediates to another chemical species, such as O_2 this electron flow is **exergonic reaction be**
  O_2 has a higher affinity for electrons than do electron-carrier intermediates

- The resulting emf provides energy to a variety of **molecular energy transducers** (**enzymes and proteins**) that do biological work

The flow of electrons can do biological work

- In the mitochondrion, membrane-bound enzymes couple electron flow → a productic **transmembrane pH difference** and a **transmembrane electrical potential** → which **hel accomplish osmotic and electrical work**
- The **proton gradient** thus formed has potential energy, sometimes called the **proton-motive for** analogy with electromotive force.
- Another enzyme, ATP synthase in the inner mitochondrial membrane, uses the proton-motive fo do chemical work: synthesis of ATP from ADP and Pi as protons flow spontaneously acros membrane.

- Bacterial flagellar motion: use proton-motive force
- Thus, the proton potential has potential energy: electron-motive force

Oxidation-Reductions Can Be Described as Half-Reactions

Half-Reactions:

- A **half reaction** is either the oxidation or reduction reaction component of a redox reaction.
- A half reaction is obtained by considering the change in oxidation states of individual subst involved in the redox reaction

Although oxidation and reduction occur together, we consider the two halves of an oxidation-redu reaction separately.

- For example, the oxidation of ferrous ion by cupric ion is written as follows,

$$\mathbf{Fe^{+2} + Cu^{+2} \leftrightarrow Fe^{+3} + Cu^{+}}$$

- Same can be described as 2 half reactions in the following form:

1. $\mathbf{Fe^{+2} \leftrightarrow Fe^{+3} + e^{-}}$
2. $\mathbf{Cu^{+2} + e^{-} \leftrightarrow Cu^{+}}$

Conjugate redox pair

- The electron-donating molecule in an oxidation-reduction reaction is called the reducing age reductant
- the electron-accepting molecule is the oxidizing agent or oxidant.
- A given agent, such as an iron cation existing in the ferrous (Fe^{+2}) or ferric (Fe^{+3}) state function conjugate reductant-oxidant pair (redox pair), just as an acid and corresponding base functior **conjugate acid-base pair.**
- Acid-base reactions can be written as a general equation:
- proton donor ↔ H^{+} + proton acceptor
- In redox reactions we can write a similar general equation:
- electron donor ↔ e^{-} + electron acceptor
- In the reversible half-reaction (1) above, Fe^{+2} is the electron donor and Fe^{+3} is the electron acc together, Fe^{+2} and Fe^{+3} constitute a **conjugate redox pair**.

- The oxidation of reducing sugar by cupric acid
- Aldehydes and ketones with anomeric carbon can act **reducing sugars.. Having capability to r copper ions in medium..** The reducing sugar reduces the copper(II) ions in these test solutic copper(I), which forms a brick red copper(I) oxide precipitate
- Basis of qualitative carbohydrate test (Benedict's test)
- **consider the oxidation of a reducing sugar by cupric ion:**
- **This overall reaction can be expressed as two half reactions:**
- **Because two electrons are removed from the aldehyde carbon, the second half-reaction (the electron reduction of cupric to cuprous ion) must be doubled to balance the overall equatic**

Biological Oxidations Often Involve Dehydrogenation

- In biological Systems oxidation (loss of electrons) is often synonymous with **dehydrogenatio** many enzymes that catalyze oxidation reactions are **dehydrogenases**.
- The carbon in living cells exists in a range of oxidation status
- Electronegativity**: H<C<S<N<O**

OXIDATION STATE

It is calculated by counting all the electrons bonded to carbon and add them in following way:

- for attached C atoms, *i.e.* C-C bonds electrons shared, count 0
- for attached X atoms, *i.e.* C-X bonds (X more electronegative), count -1 (per bond)
- for attached H atoms, *i.e.* C-H bonds (H is less electronegative than C), count +1

- Add the total for atoms attached to the C in question, then switch the sign.

Oxidation Levels of carbon compounds

For the figure given on right:

Different levels of oxidation of carbon compounds in the biosphere.

- In order to estimate the level of oxidation of these compounds, focus on the red carbon atom a bonding electrons.
- **When this carbon is bonded to the less electronegative H atom, both bonding electrons (rec assigned to the carbon.**
- When carbon is bonded to another carbon, bonding electrons are shared equally, one of the electrons is assigned to the red carbon. When the red carbon is bonded to the **more electronegat atom**, the bonding electrons are assigned to the oxygen.
- The number to the right of each compound is the number of electrons "owned" by the red carb rough expression of the degree of oxidation of that compound. As the red carbon undergoes oxic (loses electrons), the number gets smaller.

Electrons are transferred from one molecule to another in 4 ways

1. **Directly as electrons: Fe^{2+}/Fe^{3+} redox pair can transfer an electron to the Cu/Cu^{2+} redox**

$$Fe^{2+} + Cu^{2+} \leftrightarrow Fe^{3+} + Cu^{+}$$

2. As hydrogen atoms: hydrogen atom consists of a proton (H^+) and a single electron (e^-)

$$AH_2 \leftrightarrow A + 2e^- + 2H^-$$

AH_2/A: conjugate redox pair → can reduce another compound B

$$AH_2 + B \leftrightarrow A^+ + BH_2$$

3. As a hydride ion ($:H^-$) .. has two electrons

This reaction occurs in NAD-linked dehydrogenase

4. Through direct combination with oxygen, where Oxygen covalently incorporated into product

$$R\text{-}CH_3 + 1/2O_2 \leftrightarrow R\text{-}CH_2\text{-}OH$$

Reduction potential measures affinity for electrons

- **Reduction potential:**

Reduction potential (redox potential, oxidation/reduction potential), is a measure of the tendency chemical species to acquire electrons and thereby get **reduced**. Units: Volts (V), or millivolts (mV).. de by ***E*.**

- **Standard reduction potential:**

The tendency for a chemical species to be reduced, **at standard conditions. The more po the potential is the more likely it will be reduced…** denoted by E^O**.**

A Half-cell

- **Reduction potential of a half-cell depends on the activity of reduced and oxidized species v is approximated by their concentrations**
- **On the anode, oxidation takes Zn Zn^{2+} + 2 Zn is the reducing agent, and Zn^{2+} the oxidizing agent.**
- **On the cathode, reduction takes place.**

Cu^{2+} + 2 e

Cu^{2+} is the oxidizing agent and Cu the reducing agent.

A standard Half-cell

- Standard Hydrogen electrode..(SHE) is the primary reference electrode with platinum electrode d in solution with H ions under standard conditions
- The electromotive force (emf) of this electrode is designated 0.00 V. (At pH 7, 25 ^{O}C)
- Using observed emf (of test solution) and the known emf of the reference cell, the emf of the tes with the redox pair can be calculated.
- **The cell that gains electrons has, by convention, the more positive reduction potential.**
- **The electrode at which this half-reaction occurs (called a half-cell) is arbitrarily assigned a of 0.00 V.**

Standard reduction potentials can be used to calculate free-energy change

- Electrons tend to flow to the half-cell with the more positive *E*, and the strength of that tende proportional to **Δ** *E*, the difference in reduction potential
- We can then predict the direction in which electrons will tend to flow when the two half-cel connected through an external circuit
- Electrons tend to flow to the half-cell with the more positive *E* → the strength of flow depends c (difference in reduction potential)

- The energy made available by this spontaneous electron flow (the free-energy change, **Δ G**, f oxidation-reduction reaction) is proportional to **Δ** *E*:

$$\Delta G = -n \Delta E \quad \text{or} \quad \Delta G'^{o} = -n \Delta E'^{o}$$

- where *n* is the number of electrons transferred in the reaction
- **F is Faraday** (**unit** of electricity), used in the study of electrochemical reactions and equal amount of electric charge that liberates one gram equivalent of any ion from an electrolytic solu
- With this equation we can calculate the actual free-energy change for any oxidation-reduction re from the values of E'^{o} (reduction potentials) and the concentrations of reacting species
- When acetaldehyde is reduced by the biological electron carrier NADH, ΔG = -35.3 kJ/mol

TABLE 13–7 Standard Reduction Potentials of Some Biologically Important Half-Reactions

Half-reaction	E'° (V)
$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$	0.816
$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	0.771
$NO_3^- + 2H^+ + 2e^- \longrightarrow NO_2^- + H_2O$	0.421
Cytochrome *f* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *f* (Fe^{2+})	0.365
$Fe(CN)_6^{3-}$ (ferricyanide) + $e^- \longrightarrow Fe(CN)_6^{4-}$	0.36
Cytochrome a_3 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome a_3 (Fe^{2+})	0.35
$O_2 + 2H^+ + 2e^- \longrightarrow H_2O_2$	0.295
Cytochrome *a* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *a* (Fe^{2+})	0.29
Cytochrome c (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *c* (Fe^{2+})	0.254
Cytochrome c_1 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome c_1 (Fe^{2+})	0.22
Cytochrome *b* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *b* (Fe^{2+})	0.077
Ubiquinone + $2H^+ + 2e^- \longrightarrow$ ubiquinol + H_2	0.045
$Fumarate^{2-} + 2H^+ + 2e^- \longrightarrow succinate^{2-}$	0.031
$2H^+ + 2e^- \longrightarrow H_2$ (at standard conditions, pH 0)	0.000
Crotonyl-CoA + $2H^+ + 2e^- \longrightarrow$ butyryl-CoA	−0.015
$Oxaloacetate^{2-} + 2H^+ + 2e^- \longrightarrow malate^{2-}$	−0.166
$Pyruvate^- + 2H^+ + 2e^- \longrightarrow lactate^-$	−0.185
Acetaldehyde + $2H^+ + 2e^- \longrightarrow$ ethanol	−0.197
FAD + $2H^+ + 2e^- \longrightarrow FADH_2$	−0.219*
Glutathione + $2H^+ + 2e^- \longrightarrow$ 2 reduced glutathione	−0.23
$S + 2H^+ + 2e^- \longrightarrow H_2S$	−0.243
Lipoic acid + $2H^+ + 2e^- \longrightarrow$ dihydrolipoic acid	−0.29
$NAD^+ + H^+ + 2e^- \longrightarrow NADH$	−0.320
$NADP^+ + H^+ + 2e^- \longrightarrow NADPH$	−0.324
Acetoacetate + $2H^+ + 2e^- \longrightarrow$ β-hydroxybutyrate	−0.346
α-Ketoglutarate + $CO_2 + 2H^+ + 2e^- \longrightarrow$ isocitrate	−0.38
$2H^+ + 2e^- \longrightarrow H_2$ (at pH 7)	−0.414
Ferredoxin (Fe^{3+}) + $e^- \longrightarrow$ ferredoxin (Fe^{2+})	−0.432

Source: Data mostly from Loach, R.A. (1976) In *Handbook of Biochemistry and Molecular Biology*, 3rd edn (Fasman, G.D., ed.), *Physical and Chemical Data*, Vol. 1, pp. 122–130, CRC Press, Boca Raton, FL.
* This is the value for free FAD; FAD bound to a specific flavoprotein (e.g., succinate dehydrogenase) has a different E'° that depends on its protein environment.

Table 13-7
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

- **Conjugate redox pair**: An electron donor and its corresponding electron acceptor form; for exa Cu+ (donor) and Cu2+ (acceptor), or NADH (donor) and NAD+ (acceptor) form **Conjugate pair.**
- **Half cell:** A **half cell** is one of the two electrodes in a galvanic **cell** or simple battery. For exam the Zn-Cu battery, the two **half cells** make an oxidizing-reducing couple.

Universal Electron Carriers

Cellular oxidation of glucose to carbon dioxide requires specialized electron carriers

- The principles of oxidation-reduction energetics → can apply to the many metabolic reaction involve electron transfers
- Cells does not convert to CO_2 in a single, high energy releasing reaction but rather in a ser controlled reactions, some of which are oxidation for the ATP synthesis from ADP through NAD FAD

A Few Types of Coenzymes and Proteins Serve as Universal Electron Carriers

- The nucleotides NAD and NADP move readily from one enzyme to another
- The **flavin nucleotides FMN and FAD** are usually very tightly bound to the enzymes, **flavoproteins**, for which they serve as **prosthetic groups**
- **Lipid-soluble quinones: ubiquinone and plastoquinone** act as **electron carriers and proton d in the non-aqueous environment** of membranes
- **Iron-sulfur proteins and cytochromes**, which have tightly bound prosthetic groups that un reversible oxidation and reduction, serve as electron carriers in many oxidation-reduction reactic
- **NAD, NADP, FMN, and FAD** are water-soluble coenzymes and undergo reversible oxidatio reduction during many of the electron-transfer reactions of metabolism.

NAD and NADP are soluble electron carriers that act with dehydrogenase enzymes

- Activated electron carriers, involved in oxidation-reduction and coupling reactions

- NAD+ (nicotinamide adenine dinucleotide)
- NADP+ (nicotinamide adenine dinucleotide phosphate)
- **For NAD: → NAD^+: oxidized form**
- **For NADP:→ $NADP^+$: oxidized form**

- Joined by phosphoanhydride bond
- Pyridine nucleotides
- Nicotinamide moiety comes from **NIACIN Vitamin**
- Both coenzymes undergo reversible reduction of the nicotinamide ring
- As a substrate molecule undergoes oxidation (dehydrogenation), giving up two hydrogen atom oxidized form of the nucleotide (NAD^+ or $NADP^+$) accepts a hydride ion (:H2, the equivalen proton and two electrons) and is reduced (to NADH or NADPH)
- The second proton removed from the substrate is released to the aqueous solvent. The half-rea for these nucleotide cofactors are:
- **$NAD^+ + 2e^- + 2H^+ +$ NADH** **NADH + H^+**
- **$NADP^+ + 2e^- + 2H^+ +$ NADH** **NADPH + H^+**

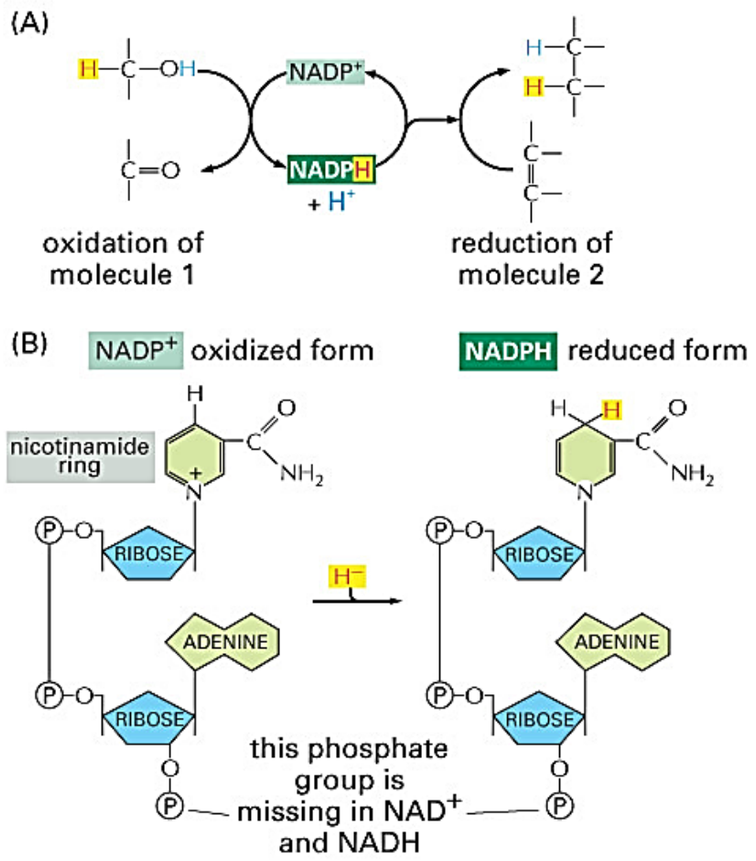


Figure 3-35 Essential Cell Biology, 2/e. (© 2004 Garland Science)

- Reduction of NAD or NADP converts the benzenoid ring of the nicotinamide moiety (with a positive charge on the ring nitrogen) to the **quinonoid** form (with no charge on the nitrogen).
- The reduced nucleotides absorb light at **340 nm**; the oxidized forms do not (used for detection)
- Plus sign with NAD and NADP does *not* indicate the net charge (both are negatively charged) r it indicates that the nicotinamide ring is in its oxidized form, with a positive charge on the **Nitr atom**
- **"H" in NADH and NADPH, the denotes the added hydride ion**
- Total concentration of NAD^+ + NADH in most tissues: 10^{-5} M
- Total concentration of $NADP^+$ + NADPH in most tissues: 10^{-6} M
- Concentration ratio: NAD^+ > NADH → favoring hydride transfer from a substrate to NAD^+ to NADH → function in oxidation, occur in mitochondrial matrix
- Concentration ratio: $NADP^+$ < NADPH → favoring hydride transfer from NADPH to a substra function in reduction → occur in cytosol
- More than 200 enzymes are known to catalyze reactions in which NAD+ (or NADP+) accepts a hy ion from a reduced substrate or NADPH (or NADH) donates a hydride ion to an oxidized substr
- **General reaction**
- **$AH_2 + NAD^+$ $A + NADH + H^+$ (NADH is Oxidized substrate) (AH_2 is reduced)**
- **$A + NADPH + H^+$ $AH_2 + NADP^+$ ($NADP^+$ is reduced substrate, A is oxidized)**

$CH_3CH_2OH + NAD^+$ $CH_3CHO + NADH + H^+$
Ethanol acetaldehyde

NAD and NADP:

- NAD and its phosphorylated analog NADP undergo reduction to NADH and NADPH, accep hydride ion from an oxidizable substrate.
- The hydride ion is added to either

the front (the A side) or the back (the B side) of the planar nicotinamide ring

TABLE 13–8 Stereospecificity of Dehydrogenases That Employ NAD^+ or $NADP^+$ as Coenzymes

Enzyme	Coenzyme	Stereochemical specificity for nicotinamide ring (A or B)	Text page
Isocitrate dehydrogenase	NAD^+	A	643
α-Ketoglutarate dehydrogenase	NAD^+	B	644
Glucose 6-phosphate dehydrogenase	$NADP^+$	B	577
Malate dehydrogenase	NAD^+	A	647
Glutamate dehydrogenase	NAD^+ or $NADP^+$	B	702
Glyceraldehyde 3-phosphate dehydrogenase	NAD^+	B	553
Lactate dehydrogenase	NAD^+	A	563
Alcohol dehydrogenase	NAD^+	A	565

TABLE 11–7 Some Diseases Resulting from Ion Channel Defects

Ion channel	Affected gene	Disease
Na^+ (voltage-gated, skeletal muscle)	*SCN4A*	Hyperkalemic periodic paralysis (or paramyotonia congenita)
Na^+ (voltage-gated, neuronal)	*SCN1A*	Generalized epilepsy with febrile seizures
Na^+ (voltage-gated, cardiac muscle)	*SCN5A*	Long QT syndrome 3
Ca^{2+} (neuronal)	*CACNA1A*	Familial hemiplegic migraine
Ca^{2+} (voltage-gated, retina)	*CACNA1F*	Congenital stationary night blindness
Ca^{2+} (polycystin-1)	*PKD1*	Polycystic kidney disease
K^+ (neuronal)	*KCNQ4*	Dominant deafness
K^+ (voltage-gated, neuronal)	*KCNQ2*	Benign familial neonatal convulsions
Nonspecific cation (cGMP-gated, retinal)	*CNCG1*	Retinitis pigmentosa
Acetylcholine receptor (skeletal muscle)	*CHRNA1*	Congenital myasthenic syndrome
Cl^-	*CFTR*	Cystic fibrosis

Rossmann fold

- Most dehydrogenases that use NAD or NADH contain a conserved protein binding domain call Rossmann fold
- Rossmann fold: consists of a six-stranded parallel beta sheet and four associated alpha helices
- Nucleotide binding domain of lactate dehydrogenase with NAD is shown
- NAD is bound to paired beta-alpha motifs of Rossmann fold
- The association between dehydrogenases and cofactor is loose → the coenzyme readily diffuse one enzyme to another → act as a water-soluble carrier of electron form one metabolite to anoth
- In overall reaction, there is not net production or consumption of NAD^+ or NADH → no concent change of NAD^+ or NADH .. and NAD^+ or NADH get recycled repeatedly

  (1) Glyceraldehyde 3-phosphate + NAD^+ 3-phosphoglycerate + NADH + H^+

  (2) Acetaldehyde + NADH + H^+ ethanol + NAD^+

  Glyceraldehyde 3-phosphate + Acetaldehyde 3-phosphoglycerate + ethanol

Dietary deficiency of Niacin causes Pellagra

- **Most coenzyme: derived from vitamins**
- **NAD and NADP: derived from vitamin niacin (=nicotinic acid)**

- **Human cannot synthesize sufficient quantities of niacin**
- **Niacin derived from tryptophan**
- **In the laboratory, nicotinic acid was first produced by oxidation of the natural pro nicotine—thus the name**
- **Niacin deficiency: affects all the NAD(P)-dependent dehydrogenase → causes the serious h disease pellagra (=rough skin): characterized by the Three Ds (dermatitis, diarrhea dementia)**

→ death

- **similar disease black tongue in dogs**
- **Niacin is curative agent (scientists)**
- **Too much alcohol drinking → reduce niacin absorption from the intestine**
- **Both nicotinic acid and nicotinamide cure pellagra, but nicotine (from cigarettes or elsew has no curative activity**

Flavoproteins

- **Flavoproteins: enzymes that catalyze oxidation-reduction reactions using coenzyme**

1. **Flavin mononucleotide (FMN)**
2. **Flavin adenine dinucleotide (FAD)**
3. **Flavin nucleotides → derived from the vitamin riboflavin**
4. **Flavoproteins act in a quite different role, as light receptors**

- **The fused ring structure of the flavin nucleotide (isoalloxazine ring)**
- **Isoalloxazine ring undergoes reversible reduction, accepting either one or two electrons i form of one or two hydrogen atoms (each atom an electron plus a proton) from a rec substrate**
- **When a fully oxidized flavin nucleotide accepts only one electron (one hydrogen atom) semiquinone form of the isoalloxazine ring is produced**
- **The fully reduced forms are abbreviated FADH2 and FMNH2**

Oxidized and reduced forms of FAD and FMN

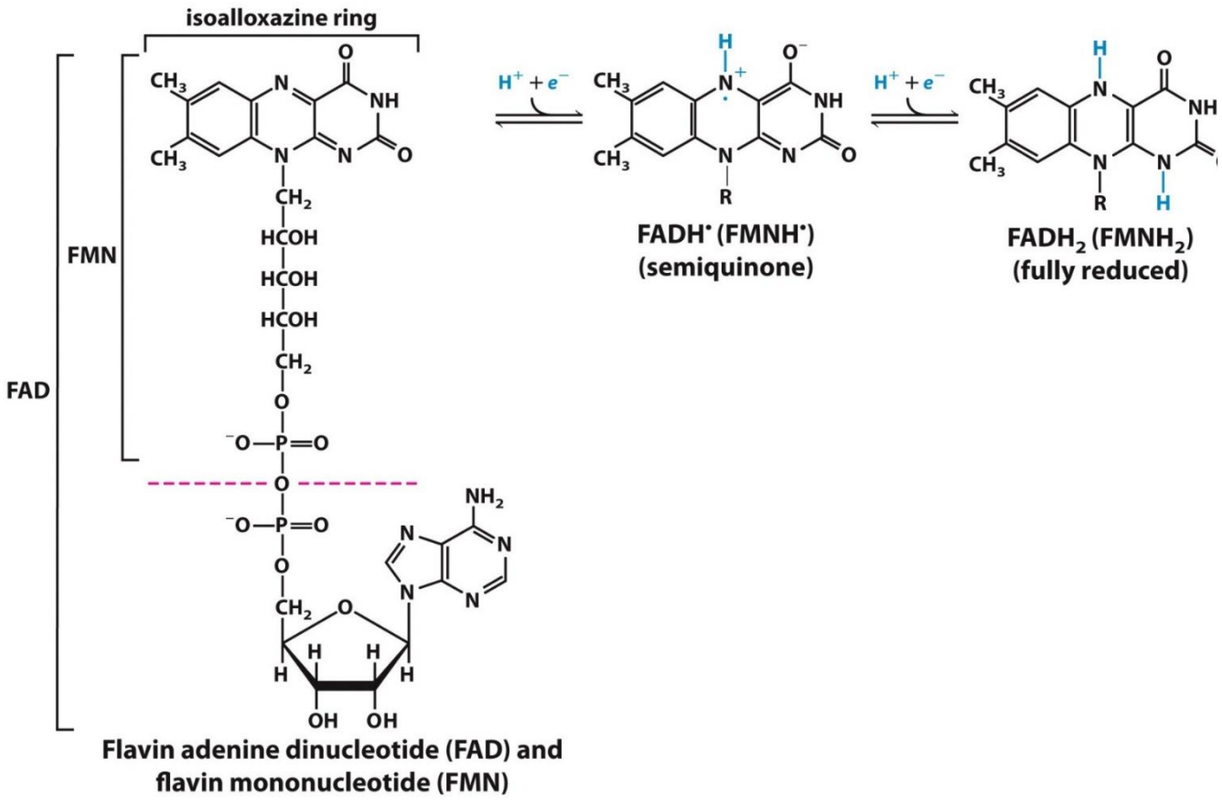


Figure 13-27
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

FMN consists of the structure above the dashed line on the FAD (oxidized form). The flavin nucle accept two hydrogen atoms (two electrons and two protons), both of which appear in the flavi system. When FAD or FMN accepts only one hydrogen atom, the semiquinone, a stable free ra forms.

Some enzymes use flavin nucleotides for oxidation-reduction

- **Flavin nucleotides are tightly bound in flavoproteins**

TABLE 13–9 Some Enzymes (Flavoproteins) That Employ Flavin Nucleotide Coenzymes

Enzyme	Flavin nucleotide	Text page(s)
Acyl-CoA dehydrogenase	FAD	673
Dihydrolipoyl dehydrogenase	FAD	637
Succinate dehydrogenase	FAD	646
Glycerol 3-phosphate dehydrogenase	FAD	759
Thioredoxin reductase	FAD	917
NADH dehydrogenase (Complex I)	FMN	738–739
Glycolate oxidase	FMN	813

Table 13-9
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

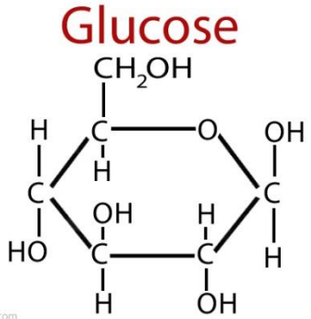
Cryptochromes

- **Family of flavoproteins,**
- **Found in both bacteria and eukaryotes**
- **mediate the effects of blue light on plant development and the effects of light on mamm circadian rhythms (oscillations in physiology and biochemistry, with a 24-hour period)**
- **Homologs of Photolyases**
- **Photolyases: Use energy of light to repair defects in DNA**

GLUCOSE METABOLISM

In animals and vascular plants, glucose has three major fates:

1. It may be stored (as a polysaccharide , i-e., glycogen or starch)
2. Oxidized to a three-carbon compound (pyruvate) via glycolysis to provide ATP and metabolic intermed
3. Oxidized via the pentose phosphate (phosphogluconate) pathway to yield ribose 5-phosphate for n acid synthesis and NADPH for reductive biosynthetic processes.



Glucose is a simple sugar with the molecular formula $C_6H_{12}O_6$, which means that it is a molecule that is of six carbon atoms, twelve hydrogen atoms, and six oxygen atoms. Glucose circulates in the blood of an as blood sugar.

- **Organisms that do not have access to glucose from other sources must make it**
- **Photosynthetic organisms make glucose by first reducing atmospheric CO2 to trioses, converting the trioses to glucose.**
- **Non-photosynthetic cells** make glucose from **simpler three and four-carbon precursors b process of gluconeogenesis**, effectively reversing glycolysis in a pathway that uses many glycolytic enzymes.

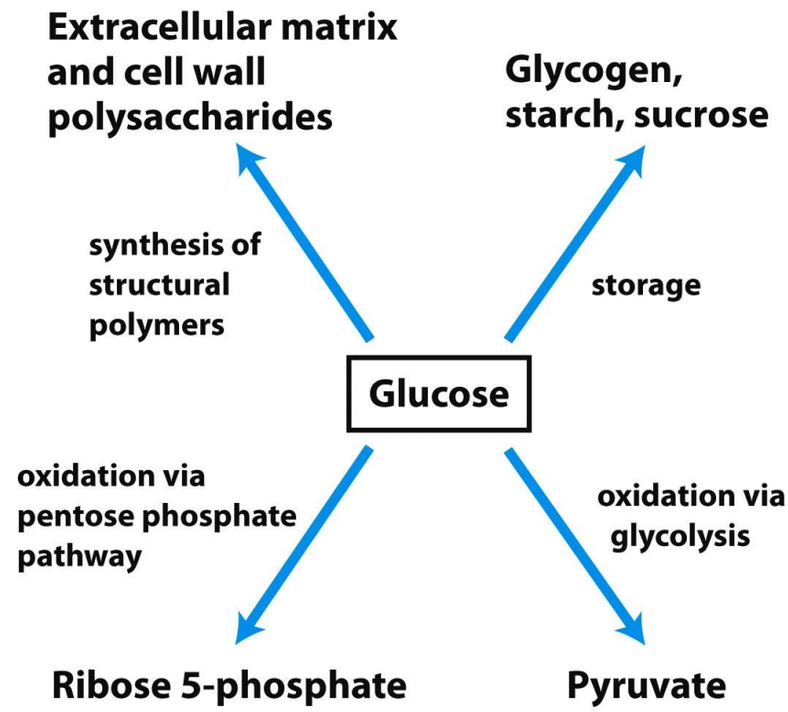


Figure 14-1
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

- In glycolysis (from the Greek *glykys*, "sweet" or "sugar," and *lysis*, "splitting"), a molecule of gl $C_6H_{12}O_6$is degraded in a series of enzyme-catalyzed reactions to yield two molecules of the carbon compound pyruvate $CH_3COCOO^- + H^+$.
- During glycolysis, the free energy released is used to form the high-energy compounds (adenosine triphosphate) and NADH (reduced nicotinamide adenine dinucleotide).
- The pathway of glycolysis was explained by Otto Warburg and Hans von Euler-Chelpin **in yeas** in **muscle** by Gustav Embden and Otto Meyerhof in1930s.
- The most common type of glycolysis is the *Embden–Meyerhof–Parnas (EMP pathway)*, whic discovered by Embden and Meyerhof and J.K. Parnas.
- Glycolysis occurs in most organisms in the **cytosol** of the cell.
- **It is an oxygen independent metabolic pathway**, meaning that it does not use molecular oxyge atmospheric oxygen) for any of its reactions.
- **The products of glycolysis, that is, pyruvate and NADH + H^+)** are sometimes disposed off atmospheric oxygen.
- Aerobic Glycolysis: When molecular oxygen is used in the disposal of the products of glycolys process is usually referred to as Aerobic Glycolysis.
- Anaerobic Glycolysis: When no oxygen is used, the process is said to be Anaerobic.

2 phases of GLYCOLYSIS

1. **The Preparatory Phase – in which ATP is consumed, hence also known as the investment p**
2. **The Pay Off Phase – in which ATP is produced**

- **Occurs in 10 steps, the first 5 of which constitute the *preparatory phase.***

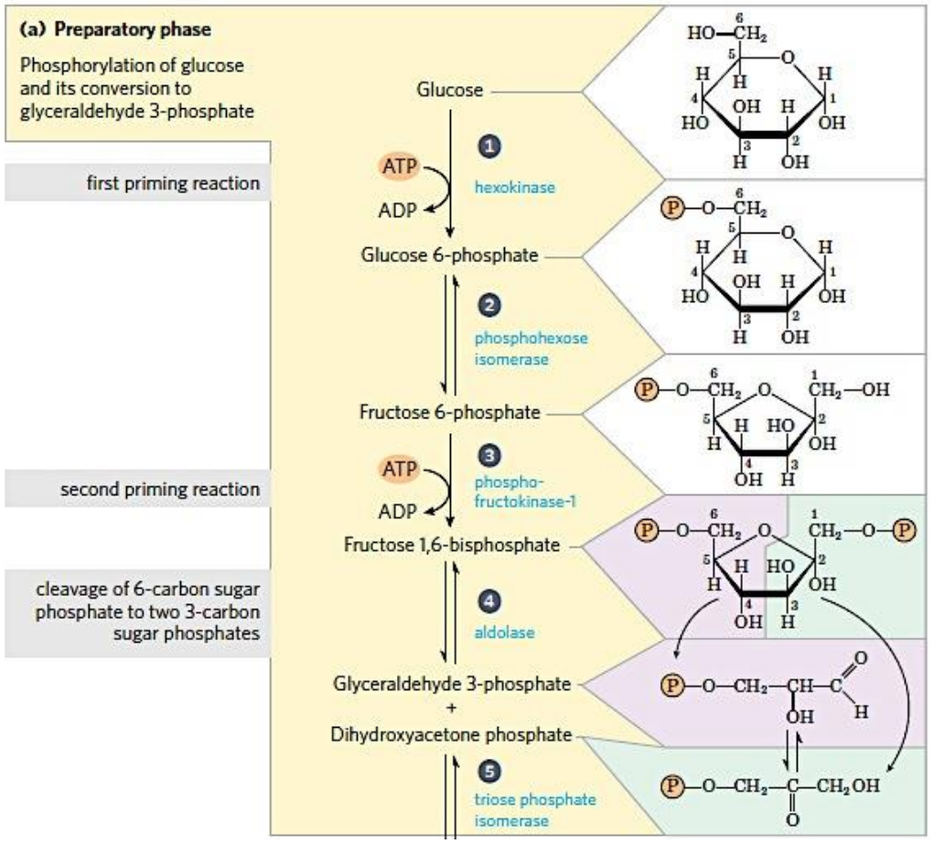
The Preparatory Phase

- **step 1:** The first step in glycolysis is phosphorylation of glucose at the hydroxyl group on C-6 family of enzymes called hexokinases to form glucose-6-phosphate (G6P).
- **step 2:** The D-glucose 6-phosphate thus formed is converted to D-fructose 6-phosphate,
- step 3: which is again phosphorylated, this time at C-1, to yield D-fructose 1,6-bisphosphate
- For both phosphorylations, ATP is the phosphoryl group donor.
- **step 4:** Fructose 1,6-bisphosphate is split to yield two three-carbon molecules, dihydroxyac phosphate and glyceraldehyde 3-phosphate (step 4); this is the "lysis" step that gives the pathw name.
- **step 5:** The dihydroxyacetone phosphate is isomerized to a second molecule of glyceraldehy phosphate (step 5), ending the first phase of glycolysis.

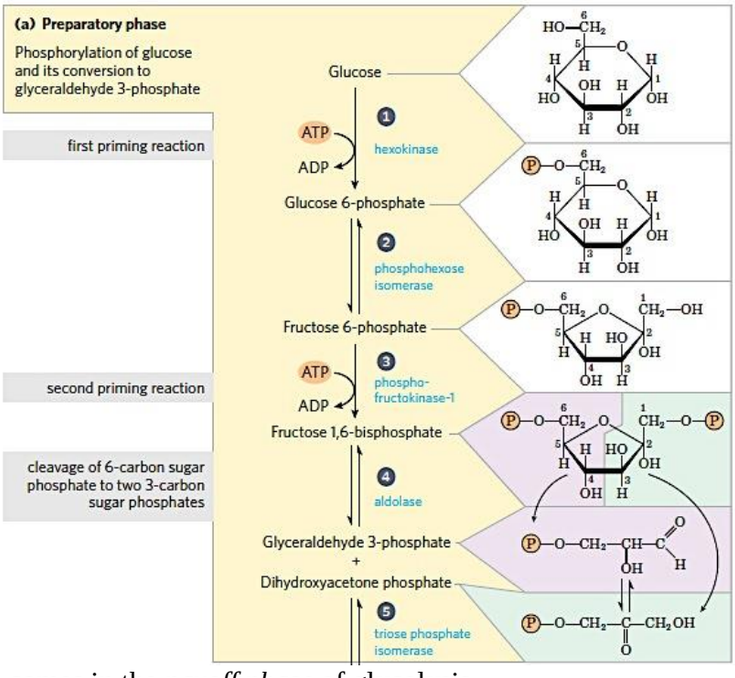
To summarize:

In the preparatory phase of glycolysis the energy of ATP is invested, raising the free-energy content intermediates, and the carbon chains of all the metabolized hexoses are converted to a common pr glyceraldehyde 3-phosphate.

Note that during preparatory phase, 2 molecules of ATP are invested before the cleavage of glucose int
three-carbon pieces.



GLYCOLYSIS
The Pay-off Phase



- The energy gain comes in the *payoff phase* of glycolysis
- step 6: Each molecule of glyceraldehyde 3-phosphate is oxidized and phosphorylated by inor phosphate (*not* by ATP) to form 1,3-bisphosphoglycerate
- Energy is also conserved in the formation of **two molecules of the electron carrier NADH** molecule of glucose
- steps 7 through 10: Much of this energy is conserved by the coupled phosphorylation of four mole of ADP to ATP.
- The net yield is two molecules of ATP per molecule of glucose, because two molecules of ATP invested in the preparatory phase.
- Pyruvate is the end product of the second phase of glycolysis.
- For each glucose molecule, two ATP are consumed in the preparatory phase and four ATP are pro in the payoff phase,

- This gives a net yield of two ATP per molecule of glucose converted to pyruvate.
- each phosphoryl group, represented here as P, has two negative charges

Energy yield during Glycolysis

- Two ATP are consumed in the preparatory phase and four ATP are produced in the payoff p giving a net yield of two ATP per molecule of glucose converted to pyruvate. Note that each phosp group, represented as P, has two negative charges ($2PO_3^{-2}$)
- Much of this energy is conserved by the coupled phosphorylation of four molecules of ADP to A
- Energy is also conserved in the payoff phase in the formation of two molecules of the electron c NADH per molecule of glucose.
- In the sequential reactions of glycolysis, three types of chemical transformations are noteworthy

(1) degradation of the carbon skeleton of glucose to yield pyruvate
(2) phosphorylation of ADP to ATP by compounds with high phosphoryl group transfer potential, fc during glycolysis
(3) Transfer of a hydride ion to NAD+ forming NADH.

Preparatory phase of Glycolysis Requires ATP

Experiment by A. Harden and W. Young

- Preparatory phase of glycolysis, two molecules of ATP are invested and the hexose chain is cl into **two triose phosphates**
- In 1906, Arthur Harden and William Young tested their hypothesis that **inhibitors of prote enzymes would stabilize the glucose fermenting enzymes in yeast extract.**
- They added blood serum **(known to contain inhibitors of proteolytic enzymes)** to yeast extrac observed the predicted stimulation of glucose metabolism.
- However, in a control experiment with the boiling serum, they discovered that boiled serum wa as effective at stimulating glycolysis
- **Harden and Young** soon discovered that glucose added to their yeast extract was converte hexose bisphosphate (the "Harden- Young ester," eventually identified as **fructose 1,6-bisphosp**
- Careful examination revealed that inorganic phosphate was responsible for the stimulation.
- This was the beginning of investigations on the role of organic esters and anhydrides of phosph biochemistry, which has led to our current understanding of the central role of **phosphoryl g transfer in biology.**

Preparatory phase of Glycolysis Requires

1. ***Phosphorylation of Glucose***

- In the first step of glycolysis, glucose is activated for subsequent reactions by its phosphorylation 6 to yield **glucose 6-phosphate**, with ATP as the phosphoryl donor:
- This reaction is irreversible under intracellular conditions, and catalyzed by **hexokinase**.
- The acceptor in the case of **hexokinase is a hexose, normally D-glucose**, hexokinase also cata the phosphorylation of other common hexoses, such as D-fructose and D-mannose, in some tissu
- Hexokinase requires Mg^{+2} for its activity, because the true substrate of the enzyme is not ATP^{-4} b $MgATP^{-2}$ complex
- Hexokinase and other nine enzymes of are soluble cytosolic proteins

Note:

- **Isozymes**

Two or more enzymes that catalyze the same reaction but are encoded by different genes are called **isoz**

- **Kinase:**
- Kinases are a subclass of transferases...catalyse transfer of phosphate group from phosphate dor molecule to specific substrates

3. Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate

The enzyme phosphohexose isomerase (phosphoglucose isomerase) catalyzes the reversible isomerizati glucose 6-phosphate, an aldose, to fructose 6- phosphate, a ketose.

The mechanism for this reaction involves an enediol intermediate
Enediol: An enediol is an alkene enol with a hydroxyl group attached to both carbon atoms of the c double bond.

- 3 steps for ***Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate*** **(phosphohexose isom reaction)**

1. opening of ring
2. Isomerization of glucose to fructose through formation of enediol intermediate
3. Closing of fructose ring

Fig 14-5: The phosphohexose isomerase reaction.

The ring opening and closing reactions (steps 1 and 4) are catalyzed by an active-site **His residu** mechanisms omitted here for simplicity.

The proton (light red) initially at C-2 is made more easily abstractable by electron withdrawal by the adj carbonyl and nearby hydroxyl groups. After its transfer from C-2 to the active-site **Glu residue (a weak** the proton is freely exchanged with the surrounding solution; that is, the proton abstracted from C-2 in s is not necessarily the same one that is added to C-1 in step 3.

Preparatory phase of Glycolysis Requires ATP - 2

- ***Phosphorylation of Fructose 6-Phosphate to Fructose 1,6-Bisphosphate***

Phosphofructokinase-1 (PFK-1) catalyzes the transfer of a phosphoryl group from ATP to fructo phosphate to yield **fructose 1,6-bisphosphate**.

KEY CONVENTION:

Compounds that contain two phosphate or phosphoryl groups attached at different positions in the mol are named ***bisphosphates* (or *bisphospho* compounds);** for example, fructose 1,6-bisphosphate and bisphosphoglycerate. **Compounds with two phosphates linked together as a pyrophosphoryl grou named *diphosphates*;** for example, adenosine diphosphate (ADP). **Similar rules apply for the nami *trisphosphates*** (such as inositol 1,4,5-trisphosphate;) and *triphosphates* (such as adenosine triphosphate,

- The **PFK-1 reaction** is essentially irreversible under cellular conditions, and it is the first "comm step in the glycolytic pathway
- Glucose 6-phosphate and fructose 6-phosphate have other possible fates, but **fructose bisphosphate is targeted for glycolysis**
- Some **bacteria and protists and perhaps all plants have a phosphofructokinase that pyrophosphate (PPi), not ATP**, as the phosphoryl group donor in the synthesis of fructose bisphosphate:
- In metabolic pathways, enzymes catalyzing essentially irreversible reactions are potential si control
- **Phosphofructokinase-1** is subject to complex **allosteric regulation**
- The PFK1 has got binding sites for both, ATP-Mg2+ and F6P binding sites
- **The activity of PFK-1 enzyme and glycolysis is inhibited whenever the cell has high levels of**
- In this way, the cell can increase or decrease the rate of glycolysis in response to the cell's e requirements
- In Eukaryotes, **fructose 2,6-bisphosphate** (not to be confused with the **PFK-1 reaction pro fructose 1,6-bisphosphate**) is a potent allosteric activator of PFK-1
- Ribulose 5-phosphate, an intermediate in the pentose phosphate pathway also act phosphofructokinase indirectly.
- **Imine:** An **imine** is a functional group or chemical compound containing a carbon–nitrogen d bond. The nitrogen atom can be attached to a hydrogen or an organic group.
- **Schiff base intermediate:**

They can be considered a sub-class of imines, being either secondary ketimines or secondary aldi depending on their structure. (R-CH=NR' where R' ≠ H)

Preparatory phase of Glycolysis Requires ATP - 2

- In metabolic pathways, enzymes catalyzing essentially irreversible reactions are potential si control
- **Phosphofructokinase-1** is subject to complex **allosteric regulation**
- The PFK1 has got binding sites for both, ATP-Mg2+ and F6P binding sites
- **The activity of PFK-1 enzyme and glycolysis is inhibited whenever the cell has high levels of**
- In this way, the cell can increase or decrease the rate of glycolysis in response to the cell's e requirements
- In Eukaryotes, **fructose 2,6-bisphosphate** (not to be confused with the **PFK-1 reaction pro fructose 1,6-bisphosphate**) is a potent allosteric activator of PFK-1
- Ribulose 5-phosphate, an intermediate in the pentose phosphate pathway also act phosphofructokinase indirectly.
- ***Cleavage of Fructose 1,6-Bisphosphate***

The enzyme fructose 1,6-bisphosphate aldolase, often called simply aldolase, catalyzes a reversible condensation (see Fig. 13–4..Lehninger Edition 6).

Fructose 1,6-bisphosphate is cleaved to yield two different triose phosphates, glyceraldehyde 3-phospha aldose, and dihydroxyacetone phosphate, a ketose.

Two classes of aldolases

- There are two classes of aldolases.
- **Class I,** found in animals and plants, form the Schiff base intermediate
- **Class II**, in fungi and bacteria, do not form the Schiff base intermediate

- **Imine:** An **imine** is a functional group or chemical compound containing a carbon–nitrogen d bond. The nitrogen atom can be attached to a hydrogen or an organic group.
- **Schiff base intermediate:**

They can be considered a sub-class of imines, being either secondary ketimines or secondary aldi depending on their structure. (R-CH=NR' where R' ≠ H)

- **MECHANISM**
- FBPA, class I aldolase reaction. The reaction shown here is the reverse of **an aldol condensatio**
- Cleavage between C-3 and C-4 depends on the presence of the **carbonyl group at C-2**, wh converted to an imine on the enzyme.
- A and B represent amino acid residues that serve as general acid (A) or base (B).
- **Imine:** An **imine** is a functional group or chemical compound containing a carbon–nitrogen d bond. The nitrogen atom can be attached to a hydrogen or an organic group.
- **Schiff base intermediate:**

They can be considered a sub-class of imines, being either secondary ketimines or secondary aldi depending on their structure. (R-CH=NR' where R' ≠ H)

- **Secondary amine:**

Secondary amines have two organic substituents (alkyl, aryl or both)
bound to the nitrogen together with one hydrogen.

- **Enamine:**

An enamine is an unsaturated compound derived
by the condensation of an aldehyde or ketone with
a secondary amine.

4. Interconversion of the Triose Phosphates

Only one of the two triose phosphates formed by aldolase, glyceraldehyde 3-phosphate, can be di degraded in
the subsequent steps of glycolysis.
The other product, dihydroxyacetone phosphate, is rapidly and reversibly converted to glyceraldehy phosphate by the fifth enzyme of the glycolytic sequence, triose phosphate isomerase:

- The reaction mechanism is similar to the reaction promoted by phosphohexose isomerase in ste glycolysis (Fig. 14–5, Lehninger; edition 6).
- After the triose phosphate isomerase reaction, the carbon atoms derived from C-1, C-2, and C-3 starting glucose are chemically indistinguishable
- from C-6, C-5, and C-4, respectively, the two "halves" of glucose have both yielded glyceraldehy phosphate.
- This reaction completes the preparatory phase of glycolysis.
- The hexose molecule has been phosphorylated at C-1 and C-6 and then cleaved to form two mole of glyceraldehyde 3-phosphate.

Fate of the glucose carbons in the formation of glyceraldehyde 3-phosphate.

(a) The origin of the carbons in the two 3-carbon products of the aldolase and triose phosphate isom reactions. The end product of the two reactions is glyceraldehyde 3-phosphate (two molecules).
(b) Each carbon of glyceraldehyde 3-phosphate is derived from either of two specific carbons of glu Note that the numbering of the carbon atoms of glyceraldehyde 3-phosphate differs from that glucose from which it is derived. In glyceraldehyde 3-phosphate, the most complex functional (the carbonyl) is specified as C-1. This numbering change is important for interpreting experi with glucose in which a single carbon is labeled with a radioisotope.

Radioisotopes:
Atomic number (Z): no. of protons
Mass number (A): the total **number** of protons and neutrons in an atomic nucleus (together kno nucleons).

- Isotopes are the atoms in an element that have the same atomic number but a different atomic that is, the same number of protons and thus identical chemical properties, but different numb neutrons and consequently different physical properties.
- Isotopes can be stable or unstable or radioisotopes.
- In the latter, their nuclei have a special property: they emit energy in the form of ionizing rad while searching for a more stable configuration.

The Payoff Phase of Glycolysis Yields ATP and NADH

Reference Book: Lehninger Principles of Biochemistry. Edition 6

The Payoff Phase of Glycolysis Yields ATP and NADH

- The payoff phase of glycolysis includes the energy-conserving phosphorylation steps

- The chemical energy of the glucose molecule is conserved in the form of **ATP and NADH**
- The conversion of two molecules of glyceraldehyde 3-phosphate to two molecules of pyruv accompanied **by the formation of four molecules of ATP from ADP.**
- the net yield of ATP per molecule of glucose degraded is only two, because two ATP were inves the preparatory phase of glycolysis to phosphorylate the two ends of the hexose molecule.

5. Oxidation of Glyceraldehyde 3-Phosphate to 1,3-Bisphosphoglycerate

The first step in the payoff phase is the oxidation of glyceraldehyde 3-phosphate to 1,3-bisphosphoglyc catalyzed by glyceraldehyde 3-phosphate dehydrogenase

This is the first of the two energy-conserving reactions of glycolysis that eventually lead to the formati ATP.

The aldehyde group of glyceraldehyde 3-phosphate (G3P) is oxidized, not to a free carboxyl group bu carboxylic acid anhydride with phosphoric acid.

This type of anhydride, called an acyl phosphate, has a very high standard free energy of hydrolysis (kJ/mol)

- Much of the free energy of oxidation of the aldehyde group of glyceraldehyde 3-phosphate is cons by formation of the acyl phosphate group at **C-1 of 1,3-bisphosphoglycerate.**
- NAD+ is the e- acceptor (bound to Rossmann fold of dehydrogenase)
- G3P is covalently bound to the dehydrogenase
- The aldehyde group of glyceraldehyde 3-phosphate reacts with the —SH group of an essentia **residue** in the active site, in a reaction analogous to the formation of a **hemiacetal** produc *thio*hemiacetal.
- Reaction of the essential Cys residue with a heavy metal such as Hg+2 irreversibly inhibits the enz
- The amount of NAD+ in a cell is far smaller than the amount of glucose metabolized in a few mi
- Glycolysis would soon come to a halt if the NADH formed in this step of glycolysis wer continuously reoxidized and recycled.

7. Phosphoryl Transfer from 1,3-Bisphosphoglycerate to ADP

- The enzyme **phosphoglycerate kinase** transfers the high-energy phosphoryl group from the car group of 1,3-bisphosphoglycerate to ADP, forming ATP and **3-phosphoglycerate**.
- phosphoglycerate kinase is named for the reverse reaction, in which it transfers a phosphoryl enz it catalyzes the reaction in both directions.
- In glycolysis, the reaction it catalyzes proceeds in the direction of ATP synthesis.
- Steps 6 and 7 of glycolysis together constitute an energy-coupling process in which bisphosphoglycerate is the common intermediate; it is formed in the first reaction (which wou endergonic in isolation), and its acyl phosphate group is transferred to ADP in the second rea (which is strongly exergonic). Overall reaction is exergonic.. The sum of these two reactions is

$$\text{Glyceraldehyde 3-phosphate} + \text{ADP} + \text{P}_\text{i} + \text{NAD}^+ \rightleftharpoons \text{3-phosphoglycerate} + \text{ATP} + \text{NADH} + \text{H}^+ \qquad \Delta G'^\circ = -12.2\ \text{kJ/mol}$$

- The outcome of these reactions, both reversible under cellular conditions, is that the energy rel on **oxidation of an aldehyde to a carboxylate** group is conserved by the coupled formation of from ADP and Pi.
- The formation of ATP by phosphoryl group transfer from a substrate such as 1,3-bisphosphogly is referred to as a **substrate-level phosphorylation**, to distinguish this mechanism from **respira linked phosphorylation**.
- **Substrate-level phosphorylation**
- **Substrate-level phosphorylation** is a metabolic reaction that results in the formation of ATP or by the direct transfer of a phosphoryl (PO_3) group to ADP or GDP another **phosphorylated** compound.
- Substrate-level phosphorylations involve soluble enzymes and chemical intermediates bisphosphoglycerate in this case).
- Respiration-linked phosphorylations, involve membrane-bound enzymes and transmembrane gra of protons.

Acyl group

It contains a double bonded oxygen atom and an **alkyl group**. (R-C=O **group** is called **Acyl Grou** organic chemistry, the **acyl group** (IUPAC name: alkanoyl) is usually derived from a carboxylic acid.

Acid anhydride

An acid anhydride is a compound that has two acyl groups bonded to the same oxygen atom.

Thus, $(CH_3CO)_2O$ is called **acetic anhydride**. Mixed (or unsymmetrical) **acid anhydrides**, as **acetic** formic **anhydride.**

Acyl phosphate

A phosphate group is connected to the acyl group by a single bond and that the bond occurs between c the oxygen atoms of the phosphate and the carbonyl carbon of the acyl group.

GLYCOLYSIS

ATP Formation Coupled to Glycolysis

- ***ATP Formation Coupled to Glycolysis*** **During glycolysis some of the energy of the glucose mol is conserved in the form of ATP, while much remains in the product, pyruvate.**
- **The overall equation for glycolysis is**

Glucose + 2NAD$^+$ + 2ADP + 2Pi → 2 pyruvate + 2NADH + 2H$^+$ + 2ATP + 2H$_2$O (Eqn I

- **For each molecule of glucose degraded to pyruvate, two molecules of ATP are generated ADP and Pi.**
- **We can now resolve the equation of glycolysis into two processes:**

(1) **the conversion of glucose into pyruvate is exergonic:**

Glucose + 2NAD$^+$ → 2 pyruvate + 2NADH + 2H$^+$,$\Delta G°'_1$ = -146 kJ/mol (Eqn II)

(2) and the formation of ATP from ADP and Pi is endergonic:

2ADP + 2Pi → 2ATP + 2H$_2$O , $\Delta G°'_2$ = **2(30.5 kJ/mol) = 61 kJ/mol (Eqn III)**

- **If we now write the sum of Equations (II) and (III), we can also determine the overall stan free-energy change of glycolysis (Eqn I), including ATP formation, as the algebric sum, ΔG' ΔG°'1 and ΔG°'2**

$$\Delta G°'_s = \Delta G°'_1' + \Delta G°'_2 = -146 \text{ kJ/mol} + 61 \text{ kJ/mol} = -85 \text{ kJ/mol}$$

Under standard and intracellular conditions, glycolysis is an essentially irreversible process, driv completion by this large net decrease in free energy.

Energy Remaining in the Pyruvate Produced by Glycolysis

- The two molecules of pyruvate formed by glycolysis still contain most of the chemical potential e of glucose
- **This energy that can be extracted by oxidative reactions in the citric acid cycle and oxid phosphorylation**
- 4 electrons..in the form of 2 hydride ions are transferred from 2 G3P to 2NAD+
- 2 molecules of NADH are reoxidized to NAD+ by transfer of electrons to electron transport (ETC) within mitochondria
- ETC passes these electrons to oxygen, that provides energy for ATP synthesis during respiration

"Pasteur effect"

- The ATP yield from glycolysis under anaerobic conditions (2 ATP per molecule of glucose) is smaller than that from the complete oxidation of glucose to CO2 under aerobic conditions (30 ATP per glucose)
- About 15 times more glucose would be consumed in anaerobic conditions to produce same amo energy (ATP)
- This was discovered by Louis Pasteur while studying process of fermentation in yeast, and is k as the "Pasteur effect"

"Warburg effect"

- The German biochemist Otto Warburg first observed in 1928 that tumors of nearly all types can glycolysis at a much higher rate than normal tissue, *even when oxygen is available*.
- This is known as "Warburg effect" and is the basis for several methods of detecting and treating c
- Isolated 7 enzymes of glycolysis from tumor tissues
- Warburg manometer: experimental tool to measure amount oxygen consumed ..so oxidative acti
- Nobel prize of Physiology in 1931

Why glycolysis in tumor tissue?

- Most tumor cells grow under hypoxic conditions (i.e., with limited oxygen supply) because, at initially, they lack the capillary network to supply sufficient oxygen.
- Cancer cells located more than 100 to 200 *u*m from the nearest capillaries must depend on glyc alone (without further oxidation of pyruvate) for much of their ATP production. The energy yi ATP per glucose) is far lower than can be obtained by the complete oxidation of pyruvate to C mitochondria (about 30 ATP per glucose)
- To make the same amount of ATP, tumor cells must take up much more glucose than do normal converting it to pyruvate and then to lactate as they recycle NADH.

- In general, the more aggressive the tumor, the greater is its rate of glycc This increase in glycolysis is achieved at least in part by increased synthesis of the glycolytic enz and of the plasma membrane transporters GLUT1 and GLUT3 that carry glucose into cells.
- The **hypoxia-inducible transcription factor (HIF-1)** is a protein stimulates the production of a eight glycolytic enzymes and the glucose transporters when oxygen supply is limited
- Another protein induced by HIF-1 is the peptide hormone VEGF (vascular endothelial growth fa which stimulates the growth of blood vessels (angiogenesis) toward the tumor
- **This heavier reliance of tumors than of normal tissue on glycolysis suggests a possibilit anticancer therapy: inhibitors of glycolysis might target and kill tumors by depleting their su of ATP. Three inhibitors of hexokinase have shown promise as chemotherapeutic agen deoxyglucose, lonidamine, and 3-bromopyruvate.**
- **Glucose Uptake Is Deficient in Type 1 Diabetes Me** The metabolism of glucose in mammals is limited by the rate of glucose uptake into cells a phosphorylation by **hexokinase.**
- Glucose uptake from the blood is mediated by the **GLUT family** of glucose transporters
- The transporters of hepatocytes (GLUT1, GLUT2) and of brain neurons (GLUT3) are always pr in plasma membranes.
- In contrast, the main glucose transporter in the cells of skeletal muscle, cardiac muscle, and ad tissue (GLUT4) is sequestered in small intracellular vesicles and moves into the plasma mem only in response to an insulin signal
- Thus in skeletal muscle, heart, and adipose tissue, glucose uptake and metabolism
- depend on the normal release of insulin by **pancreatic beta cells** in response to elevated blood gl

Individuals with type 1 diabetes mellitus have too few cellsand cannot release sufficient insulin to t glucose uptake by the cells of skeletal muscle, heart, or adipose tissue.

Thus, after a meal containing carbohydrates, glucose accumulates to abnormally high levels in the blc condition known as hyperglycemia. Unable to take up glucose, muscle and fat tissue use the fatty ac stored triacylglycerols as their principal fuel.

In the liver, acetyl-CoA derived from this fatty acid breakdown is converted to "ketone bodies"—acetoa and -hydroxybutyrate—which are exported and carried to other tissues to be used as fuel

These compounds are especially critical to the brain, which uses ketone bodies as alternative fuel when gl is unavailable. (Fatty acids cannot pass through the blood-brain barrier and thus are not a fuel for neurons.)

In untreated type 1 diabetes, overproduction of acetoacetate and hydroxybutyrate leads to their accumu in the blood, and the consequent lowering of blood pH produces **ketoacidosis**, a life-threatening conditi

Insulin injection reverses this sequence of events:

GLUT4 moves into the plasma membranes of hepatocytes and adipocytes, glucose is taken up into the and phosphorylated, and the blood glucose level falls, greatly reducing the production of ketone bodies.

Importance of Phosphorylated Intermed

Each of the nine glycolytic intermediates between glucose and pyruvate is phosphorylated. The phos groups appear to have three functions.

1. The phosphate groups are ionized at pH 7, thus giving each of the intermediates of glycolysis negative charge.
2. Because the plasma membrane is impermeable to molecules that are charged, the phosphor intermediates cannot diffuse out of the cell.
3. After the initial phosphorylation, the cell does not have to spend further energy in reta phosphorylated intermediates despite the large difference between the intracellular and extrace concentrations of these compounds.

Phosphate groups are essential components in the enzymatic conservation of metabolic energy.

Energy released in the breakage of phosphoric acid anhydride bonds (such as those in ATP) is pa conserved in the formation of phosphate esters such as glucose-6 phosphate.

High-energy phosphate compounds formed in glycolysis (1,3-bisphosphoglycerate and phosphoenol pyru donate phosphate groups to ADP to form ATP.

- **Binding of phosphate groups to the active sites of enzymes provides binding energy contributes to lowering the activation energy and increasing the specificity of enzyme-cata reactions.**
- **The phosphate groups of ADP, ATP, and the glycolytic intermediates form complexes with and the substrate binding sites of many of the glycolytic enzymes are specific for these complexes.**
- **Nearly all the glycolytic enzymes require Mg^{2+} for activity.**

GLYCOLYSIS

Importance of Phosphorylated Intermediates

Importance of Phosphorylated Intermediates

Each of the nine glycolytic intermediates between glucose and pyruvate is phosphorylated. The phos groups appear to have three functions.

- The phosphate groups are ionized at pH 7, thus giving each of the intermediates of glycolysis negative charge.
- Because the plasma membrane is impermeable to the phosphorylated sugars and charged moi cannot diffuse out of the cell.
- After the initial phosphorylation, the cell does not have to spend further energy in reta phosphorylated intermediates despite the large difference between the intracellular and extrace concentrations of these compounds.

2. Phosphate groups are essential components in the enzymatic conservation of metabolic energy.

- **Energy released in the breakage of phosphoric acid anhydride bonds (such as those in AT partially conserved in the formation of phosphate esters such as glucose-6 phosphate.**
- **High-energy phosphate compounds formed in glycolysis (1,3-bisphosphoglycerate phosphoenol pyruvate) donate phosphate groups to ADP to form ATP.**

3. Binding of phosphate groups to the active sites of enzymes:

- **provides binding energy that contributes to lowering the activation energy and increasin specificity of enzyme-catalyzed reactions.**
- **The phosphate groups of ADP, ATP, and the glycolytic intermediates form complexes with and the substrate binding sites of many of the glycolytic enzymes are specific for these complexes.**
- **Nearly all the glycolytic enzymes require Mg^{2+} for activity.**

Glucose Uptake Is Deficient in Type 1 Diabetes Mellitus

- The metabolism of glucose in mammals is limited by the rate of glucose uptake into cells a phosphorylation by **hexokinase.**
- Glucose uptake from the blood is mediated by the **GLUT family** of glucose transporters
- The transporters of hepatocytes (GLUT1, GLUT2) and of brain neurons (GLUT3) are always pr in plasma membranes.
- In contrast, the main glucose transporter in the cells of skeletal muscle, cardiac muscle, and ad tissue (GLUT4) is sequestered in small intracellular vesicles and moves into the plasma mem only in response to an insulin signal
- Thus in skeletal muscle, heart, and adipose tissue glucose uptake and metabolism depend on the n release of insulin by **pancreatic beta cells** in response to elevated blood glucose
- Patients with type 1 diabetes mellitus have too few cells and cannot release sufficient insulin to t glucose uptake by the cells of skeletal muscle, heart, or adipose tissue
- Thus, after a meal containing carbohydrates, glucose accumulates to abnormally high levels i blood, a condition known as hyperglycemia
- Unable to take up glucose, muscle and fat tissue use the fatty acids of stored triacylglycerols as principal fuel
- In the liver, acetyl-CoA derived from this fatty acid breakdown is converted to "ketone bodi acetoacetate and –hydroxybutyrate
- These compounds are especially critical to the brain, which uses ketone bodies as alternative fuel glucose is unavailable. (Fatty acids cannot pass through the blood-brain barrier and thus are not for brain neurons)
- In untreated type 1 diabetes, overproduction of acetoacetate and hydroxybutyrate leads to accumulation in the blood, and the consequent lowering of blood pH produces **ketoacidosis**, a threatening condition.
- **Insulin injection reverses this sequence of events**:
- GLUT4 moves into the plasma membranes of hepatocytes and adipocytes, glucose is taken up in cells and phosphorylated, and the blood glucose level falls, greatly reducing the production of k bodies.

Fates of Pyruvate

- The pyruvate formed by glycolysis is further metabolized via one of three catabolic routes.
- In aerobic organisms or tissues, under aerobic conditions, glycolysis is only the first stage i complete degradation of glucose

1. Oxidation of Pyruvate:

Pyruvate is oxidized, with loss of its carboxyl group as CO_2, to yield the acetyl group of acetyl-coenzyı the acetyl group is then oxidized completely to CO_2 by the citric acid cycle. The electrons from these oxid are passed to O_2 through a chain of carriers in mitochondria, to form H_2O. The energy from the ele transfer reactions drives the synthesis of ATP in mitochondria

2. Reduction to lactate

The second route for pyruvate is its reduction to lactate via lactic acid fermentation.

When vigorously contracting skeletal muscle must function under low oxygen conditions (hypoxia), N cannot be reoxidized to NAD+, but NAD+ is required as an electron acceptor for the further oxidati pyruvate.

Under these conditions pyruvate is reduced to lactate, accepting electrons from NADH and th regenerating the NAD+ necessary for glycolysis to continue. Certain tissues and cell types (retin erythrocytes, for example) convert glucose to lactate even under aerobic conditions.

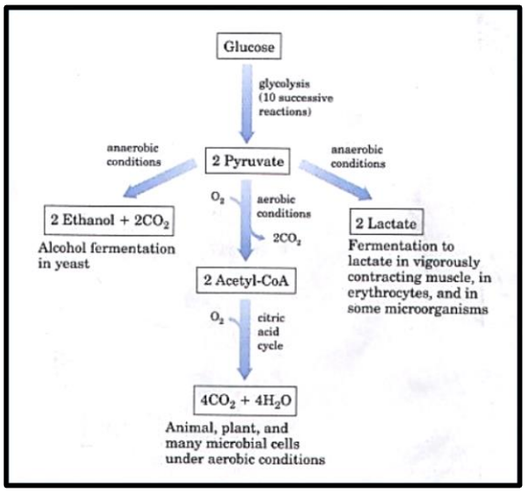
Lactate is also the product of glycolysis under anaerobic conditions in some microorganisms.

3. ethanol (alcohol) fermentation

The third major route of pyruvate catabolism leads to formation of ethanol.

In some plant tissues and in certain invertebrates, protists, and microorganisms such as yeast, pyruv converted under hypoxic or anaerobic conditions to ethanol and CO2, a process called ethanol (alc fermentation

The oxidation of pyruvate is an important catabolic process, but pyruvate has anabolic fates as well. I provide the carbon skeleton for the synthesis of the amino acid alanine or for the synthesis of fatty acids



1. **Fates of Pyruvate under Aerobic Conditions**
   - **Under Aerobic conditions**, the pyruvate formed in the final step of glycolysis is oxidized to **ac (acetyl- CoA), which enters the citric acid cycle** and is oxidized to CO2 and H2O.
   - The NADH formed by **dehydrogenation of glyceraldehyde 3-phosphate** is ultimately **reoxidiz NAD+** by passage of its electrons to O2 in mitochondrial respiration.
2. **Fates of Pyruvate under Anaerobic Conditions**
   - **Under hypoxic (low-oxygen) conditions**, as in very active skeletal muscle, in submerged plant ti in solid tumors, or in lactic acid bacteria**—NADH generated by glycolysis cannot be reoxidiz O2.**
   - Failure to regenerate NAD+ can leave the cell with **no electron acceptor** for the oxidati glyceraldehyde 3-phosphate, and the energy-yielding reactions of glycolysis would stop.
   - **There must be another way to regenerated NAD+**
   - Modern organisms continually regenerate NAD+ during anaerobic glycolysis **by transfe electrons from NADH to form a reduced end product such as lactate or ethano fermentation).**

Fermentation

- **Fermentation actually comes from Latin word "fervere" which means "to boil"**
- **So the word "Fermentation" is referred to physical state of boiling/bubbling**
- **The bubbling appearance in fermentation is due to the production of CO2 bubbles caus anaerobic catabolism of sugars**

Fermentation is a metabolic process that converts sugars to **acids, gases, or alcohol**

- The science of fermentation is known as **ZYMOLOGY**

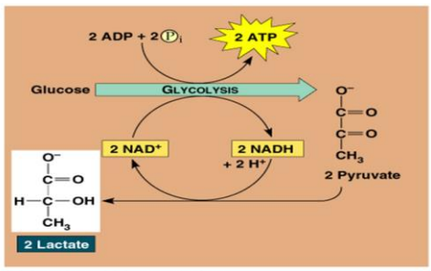
- French microbiologist Louis Pasteur is often remembered for his insights into fermentatio its microbial causes.
- It occurs in yeast and bacteria, and also in oxygen-starved muscle cells
- **In terms of BIOCHEMISTRY Fermentation relates to the generation of energy by catab of organic compounds**
- Industrial Fermentation is used to refer to the bulk growth of microorganisms on a growth me often with the goal of producing a specific chemical product.

Lactic Acid Fermentation

- Lactic acid fermentation is done by some fungi, some bacteria like in yogurt, and sometimes b muscles.

Glucose → carbon dioxide + lactic acid

- In the process of lactic acid fermentation, the 3-carbon pyruvic acid molecules are turned into acid
- Allows glycolysis to continue
- Normally our muscles do cellular respiration like the rest of our bodies, using O_2 supplied by our and blood. However, under greater exertion when the oxygen supplied by the lungs and blood s can't get there fast enough to keep up with the muscles' needs, our muscles can switch over a **lactic acid fermentation**.



(b) Lactic acid fermentation
©1999 Addison Wesley Longman, Inc.

Pyruvate Is the Terminal Electron Acceptor in Lactic Acid Fermentation

- NAD+ is regenerated from NADH by the reduction of pyruvate to **lactate**.
- Some tissues and cell types (such as erythrocytes, which have no mitochondria and cannot oxidi pyruvate to CO2) produce lactate from glucose even under aerobic conditions.
- The reduction of pyruvate in this pathway is catalyzed by **lactate dehydrogenase**, which forms lactate at pH 7.

- **Under aerobic conditions NADH transfers its electrons eventually to oxygen, regenerating NAD^+.**
- **Under anaerobic conditions pyruvate is reduced to lactate to regenerate NAD^+ for glycolys continue.**

- The overall equilibrium of the reaction strongly favors lactate formation, as shown by the large negative standard free-energy change.
- In glycolysis, dehydrogenation of the two molecules of glyceraldehyde 3-phosphate derived fror each molecule of **glucose converts two molecules of NAD+ to two of NADH.**
- Because the **reduction of two molecules of pyruvate** to two of lactate regenerates **two molecul NAD+,** there is no net change in NAD+ or NADH.
- The lactate formed by active skeletal muscles (or by erythrocytes) can be recycled; it is carried i blood to the liver, where it is converted to glucose during the recovery from strenuous muscular activity.
- When lactate is produced in large quantities during vigorous muscle contraction (during a sprint example), the acidification that results from ionization of lactic acid in muscle and blood limits t period of vigorous activity.
- **Although conversion of glucose to lactate includes two oxidation-reduction steps**, there is nc change in the oxidation state of carbon; in glucose (C6H12O6) and lactic acid (C3H6O3), the H ratio is the same.

Lactic Acid Fermentation

Homolactic Fermentation

- HOMOLACTIC In homolactic fermentation, one glucose molecule is converted into two molecu of lactic acid
- $C_6H_{12}O_6 \rightarrow 2\ CH_3CHOHCOOH$

Heterolactic Fermentation

- In Heterolactic Fermentation, one glucose molecule is converted into one molecule of lactic acic molecule of ethanol and one molecule of carbondioxide

- $C_6H_{12}O_6 \rightarrow CH_3CHOHCOOH + C_2H_5OH + CO_2$

Ethanol Fermentation

- Alcohol fermentation is done by yeast and some kinds of bacteria.
- Products: glucose undergoes fermentation to produce ethanol and CO2, instead of lactate.
- Substrate: glucose
- two-step process:
- Yeast contain the enzyme pyruvate decarboxylase that decarboxylates pyruvate to form acetalde
- .. Acetaldehyde is reduced to ethanol under anaerobic conditions.

Note: Ethanol Is the Reduced Product in Ethanol Fermentation

Final equation:

- As in lactic acid fermentation, there is no net change in the ratio of hydrogen to carbon atoms
- When glucose (H: C ratio = 12/6 = 2) is fermented to two ethanol and two CO2 (combined H: C = 12/6 = 2).
- In all fermentations, the H: C ratio of the reactants and products remains the same.
- Alcohol dehydrogenase is present in many organisms that metabolize ethanol, including humans
- In the liver it catalyzes the oxidation of ethanol, either ingested or produced by inte microorganisms, with the reduction of NAD+ to NADH.
- Pyruvate decarboxylase is present in brewer's and baker's yeast (*Saccharomyces cerevisiae*) and all other organisms that ferment glucose to ethanol,
  including some plants.
  The CO2 produced by pyruvate decarboxylation in brewer's yeast is responsible for the characte carbonation of champagne (ethanol fermentation).
- In baking, CO2 released by pyruvate decarboxylase when yeast is mixed with a fermentable sug causes dough to rise.
- The enzyme is absent in organisms that carry out lactic acid fermentation.

Use of Fermentation In Industry

Industrial Fermentation

- Oldest form of microbiology and biotechnology which was used to make wine, beer, bread with bacteria and yeasts without knowing scientific basis
- Fermentation has different meaning to biochemistry and industrial microbiologists.
- Biochemical meaning relates to the generation of energy by the catabolism of organic compounc
- Microbiologists have extended the term - fermentation - to describe: "A process that genera product by the mass culture of microorganism."
- In Industrial Biotechnology, fermentation means a process in which microorganisms that are cu on a large-scale under aerobic or anaerobic conditions, convert a substrate into a product wh useful to man.
- The industrial microorganisms are grown under controlled conditions with an aim of optimizir growth of the organism for production of a target microbial product.
- In 1910 Chaim Weizmann (later to become the first president of Israel) discovered that the bact *Clostridium acetobutyricum* ferments starch to butanol and acetone.
- This discovery opened the field of industrial fermentations, in which some readily available ma rich in carbohydrate (corn starch or molasses, for example) is supplied to a pure culture of a sp microorganism, which ferments it into a product of greater commercial value.

Some important fermentation products

- Fermentations Are Used to Produce Some Common Foods and Industrial Chemicals

Product	Organism	Use
Ethanol	*Saccharomyces cerevisiae*	Industrial solvents, beverages
Glycerol	*Saccharomyces cerevisiae*	Production of explosives
Lactic acid	*Lactobacillus bulgaricus*	Food and pharmaceutical
Acetone and butanol	*Clostridium acetobutylicum*	Solvents
α-amylase	*Bacillus subtilis*	Starch hydrolysis

- Yogurt, is produced when the bacterium *Lactobacillus bulgaricus* ferments the carbohydrate in producing lactic acid; the resulting drop in pH causes the milk proteins to precipitate, producing thick texture and sour taste of unsweetened yogurt.

- Cheese is produced by using *Propionibacterium freudenreichii*, which ferments milk to produce *propionic acid and CO2*; the propionic acid
- precipitates milk proteins, and bubbles of CO2 cause the holes characteristic of Swiss Cheese.

Commercial Yogurt

Contains 2 species of bacteria specialized to grow well in milk (but can't survive inside the human body These bacteria work in symbiosis. Each bacterium stimulates the growth of the other => acidifies the n more rapidly than either partner on its own.

First, Streptococcus *thermophilus* is more active, then slows down when acidity reaches 0.5%

Next, Lactobacillus *bulgaricus* is more acid tolerant and takes over until acidity ≥1%

Milk: Casein protein micelles (bundles)10^{-7} meters in diameter

Yogurt: Bacteria produce acid

Acid causes Casein bundles to fall apart into separate casein molecules.

These rebind to each other in a network that traps water. => makes a gel

Use of Fermentation In agriculture

- In agriculture, plant byproducts such as corn stalks are preserved for use as animal feed by packi them into a large container (a silo) with limited access to air; microbial fermentation produces ac that lower the pH.
- The silage that results from this fermentation process can be kept as animal feed for long time without spoilage

Fermentation Products and steps of industrial fermentation

There are five major groups of commercially important fermentations:

- Those that produce microbial cells (or biomass) as the product.
- Those that produce microbial metabolites.
- Those that produce microbial enzymes.
- Those that produce recombinant products.

Those that modify a compound which is added to the fermentation – the transformation process

Fermentation Products

- Biomass Production
- Baker's Yeast
- SCP (Single Cell Protein)
- Probiotics
- Bio fertilizers
- Vaccines
- Bioweapons
- **1. Upstream processing**

  Upstream processing includes *formulation of the fermentation medium*, *sterilisation* of air, fermentation medium and the fermenter, *inoculum preparation* and *inoculation* of the medium.

  The *fermentation medium* should contain an energy source, a carbon source, a nitrogen source ar micronutrients required for the growth of the microorganism along with water and oxygen, if necessary.

  A medium which is used for a large scale fermentation, in order to ensure the sustainability of th operation, should have the following characteristics;

1. It should be cheap and easily available
2. It should maximize the growth of the microorganism, productivity and the rate of formation of th desired product
3. It should minimize the formation of undesired products

- Usually, waste products from other industrial processes, such as molasses, cheese whey and corn steep liquor, after modifying with the incorporation of additional nutrients, are used as the substr for many industrial fermentation
- ***Sterilisation*** is essential for preventing the contamination with any undesired microorganisms. A **sterilised** by membrane filtration while the medium is usually heat sterilised. Any nutrient component which is heat labile is **filter-sterilised** and later added to the sterilised medium. The fermenter may be sterilised together with the medium or separately.
- ***Inoculum build up*** is the preparation of the seed culture in amounts sufficient to be used in the l fermenter vessel. This involves growing the microorganisms obtained from the pure stock cultur several **consecutive fermenters**. This process cuts down the time required for the growth of microorganisms in the fermenter, thereby increasing the rate of productivity. Then the seed cultu obtained through this process is used to inoculate the fermentation medium.

2. The fermentation process

- The fermentation process involves the *propagation of the microorganism* and *production of the desired product*. The fermentation process can be categorised depending on various parameters.

  It can be either aerobic fermentation, carried out in the presence of oxygen or anaerobic ferment carried out in the absence of oxygen. Many industrial fermentation are carried out under aerobic conditions where a few processes such as ethanol production by yeast require strictly anaerobic environments.

3. Downstream Processing

- Downstream Processing includes the recovery of the products in a pure state and the effluent treat Product recovery is carried out through a series of operations including *cell separation* by se centrifugation or filtration; *product recovery* by disruption of cells (if the product is pro intracellularly); *extraction* and *purification* of the product. Finally, the *effluents are treat* chemical, physical or biological methods.

General Aspects of Fermentation Processes-

What is fermentation technique?

Techniques for large-scale production of microbial products. It must both provide an optimum enviro for the microbial synthesis of the desired product and be economically feasible on a large scale.

Fermentation technique can be divided into following on the basis of moisture requirement

1. **surface (emersion) fermentation**
2. **submersion fermentation technique**

The latter may be run in batch, fed batch, continuous reactors

1. Surface techniques:

In the surface techniques, the microorganisms are cultivated on the surface of a liquid or solid substrate. These techniques are very complicated and used in industry for producing bread, cocoa

2. Submersion processes

- In the submersion processes, the microorganisms grow in a liquid medium provided with the sub
- Except in traditional beer and wine fermentation, the medium is held in fermenters and stirred to a homogeneous distribution of cells and medium.
- Most processes are aerobic, and for these the medium must be vigorously aerated.
- Used for pickling vegetables and producing soy sauce.
- All important industrial processes (production of biomass and protein, antibiotics, enzymes are c out by submersion processes.

Fermenter/Bioreactors

- The heart of the fermentation process is the fermenter.
- A biorector is a device in which the organisms are cultivated and motivated to form a desired pr
- Closed vessel or containment designed to give a right environment for optimal growth and met activity of the organism
- Fermenter: for microbes/ Bioreactor: for eukaryotic cells
- Size variable ranging from 20-250 million litres or Large scale production (..to1000-million L cap

Design of a Fermenter

The success of a fermentation process is highly dependent on environmental factors

Factors to consider when designing a fermenter

- Aseptic and long-term reliability
- Adequate for aeration and agitation
- Low power consumption
- Temperature and pH controls
- Sampling facilities
- drain or overflow
- control systems
- sensors
- cooling to achieve maximum microbial yield

(14 L fermenter shown is a copyright of New Brunswick Scientific)

Types of Bioreactors

- Simple fermenters (batch and continuous)
- Fed batch fermenter
- Air-lift or bubble fermenter
- Cyclone column fermenter
- Tower fermenter
- Packed bed bioreactor
- Photobioreactor

Basic modes for operating a fermenter

Types of fermentation on the basis of Process

- **Batch fermentation**
- Closed system
- Batch fermentation refers to "A technique used to grow microorganisms or cells.
- A limited supply of Sterile nutrients (substrate) for growth is inoculated **(starter culture)** when the used up, or some other factor becomes limiting, the **culture** declines.
- The time for which fermentation occurs is called 'fermentation time' or 'batch time'
- The only material added and removed during the course of a batch fermentation is oxygen (in the fo air, if the process is aerobic) and pH control solutions.
- The culture is allowed to grow until no more of the product is being made
- Cells, or products that the organisms have made, are "harvested" and fermenter is cleaned out for ar run.
- The **principal disadvantage** of batch processing is the high proportion of unproductive time (down- between batches, comprising the charge and discharge of the fermenter vessel, the cleaning, sterili: and re-start process
- We take the example of *Aspergillus niger* and *Lactobacillus* are the microbes used to commer produce citric acid and lactic acid, respectively.
- The production takes place in a batch fermenter.
- Rules of fermentation are applied on batch fermentation
    - *Microbial Growth Kinetics*
    - *Media for Industrial Fermentations*
    - *Sterilization*
    - *The Development of Inocula for Industrial Fermentations*
    - *Design of a Fermenter*
    - *Instrumentation and Control*
    - *Aeration and Agitation*

Phases of Batch fermentation

- **lag phase** (adapt to their surroundings)
- **exponential growth** (grow in numbers)
- **stationary phase** (stop growing)
- **death phase**

Microbial Growth Kinetics

- **Lag Phase**
    - This is the first phase in the fermentation process
    - The cells have just been injected into a new environment and they need time to adjust accorc
    - Cell growth is minimal in this phase.
- **Exponential Phase**
    - The second phase in the fermentation process
    - The cells have adjusted to their environment and rapid growth takes place
    - Cell growth rate is highest in this phase
    - At some point the cell growth rate will level off and become constant
    - The most likely cause of this leveling off is substrate limited inhibition
    - Substrate limited inhibition means that the microbes do not have enough nutrients in the me to continue multiplying.
- **Stationary phase**
    - This is the third phase in the fermentation process
    - The cell growth rate has leveled off and become constant
    - The number of cells multiplying equals the number of cells dying
- **Death phase**
    - The fourth phase in the fermentation process
    - The number of cells dying is greater than the number of cells multiplying
    - The cause of the death phase is usually that the cells have consumed most of the nutrients medium and there is not enough left for sustainability

2. Continuous Fermentation

- A **Continuous fermentation** is a process in which fresh medium (substrate) is **continuously** ad the bioreactor , and biomass or products containing left over nutrients and microorganism **continuously** removed at the same rate to keep the **culture** volume constant.
- Under these conditions the cells remain in the logarithmic phase of growth

3. Fed-batch Fermentation

- The fed-batch technique was originally devised by yeast producers in the early 1900s to regula growth in batch culture of *Saccharomyces* Yeast producers observed that in the presence of concentrations of malt, a by-product - ethanol - was produced, while in low concentrations of ma yeast growth was restricted. The problem was then solved by a controlled feeding regime, so that growth remained substrate limited.
- The concept was then extended to the production of other products, such as some enzymes, antibi growth hormones, microbial cells, vitamins, amino acids and other organic acids.
- ***Fed-batch culture*** is, in the broad sense, defined as an operational technique in biotechnol processes where one or more nutrients (substrates) are fed (supplied) to the bioreactor d cultivation and in which the **product(s)** remain in the bioreactor until the end of the run.
- An alternative description of the method is that of a culture in which "a base medium supports i cell culture and a feed medium is added to prevent nutrient depletion".
- It is also a type of **semi-batch culture**. In some cases, all the nutrients are fed into the bioreacto
- ***Advantage of the fed-batch culture*** is that one can control concentration of fed-substrate in the c liquid at arbitrarily desired levels (in many cases, at low levels).
- Generally speaking, fed-batch culture is superior to **conventional batch culture** when contr concentrations of a nutrient (or nutrients) affect the yield or productivity of the desired metaboli Basically, cells are grown under a batch regime for some time, usually until close to the end exponential growth phase.
  At this point, the reactor is fed with a solution of substrates, without the removal of culture fluid This feed should be balanced enough to keep the growth of the microorganisms at a desired sp growth rate and reducing simultaneously the production of by-products (that can be growth or pr production inhibitory and make the system not as effective).

GLUCONEOGENESIS

The process of synthesis of glucose from non-carbohydrate sources is known as Gluconeogenesis (abbrev GNG) ("new formation of sugar"), OR
Gluconeogenesis is the pathway, which converts pyruvate and related non-carbohydrate carbon compou glucose.

Why the body needs Gluconeogenesis?

- For the human brain and nervous system, as well as the erythrocytes, testes, renal medulla embryonic tissues, glucose from the blood is the sole or major fuel source.
- The brain alone requires about 120 g of glucose each day
- more than half of all the glucose stored as glycogen in muscle and liver
- between meals and longer fasts, after vigorous exercise, glycogen is depleted
- For these times, organisms need a method for synthesizing glucose from noncarbohydrate precu

Site of Gluconeogenesis?

Organisms doing Gluconeogenesis: all animals, plants, fungi, and microorganisms
In many microorganisms, gluconeogenesis starts from simple organic compounds of two or carbons, such as acetate, lactate, and propionate, in their growth medium.
Precursors of glucose in Gluconeogenesis: The important precursors of glucose in animals are carbon compounds such as lactate, pyruvate, and glycerol, as well as certain amino acids

Organs doing Gluconeogenesis:

- mainly in the liver
- To a lesser extent in renal cortex
- in the epithelial cells that line the inside of the small intestine
- In cytosol and mitochondria depending upon the substrate

Where does this glucose go?

The glucose produced passes into the blood to supply other tissues.
Nature: Anabolic: 4 ATP are required for synthesis of 1 glucose from 1 pyruvate or lactate

Pathways involved in Gluconeogenesis

- Reverse glycolysis (PEP to G6P)
- Tri carbocyclic acid cycle (TCA) or Kreb's cycle
- Cori cycle
- Glucose-alanine cycle

Precursors of glucose

- Gluconeogenesis occurs in all animals, plants, fungi, and microorganisms.
- The reactions are essentially the same in all tissues and all species.

Biomedical importance of Gluconeogenesis

- To maintain the blood glucose concentration during prolonged fasting or starvation when sufficient carbohydrate is not available from the diet or glycogen reserves.
- It maintains the level of intermediates of the TCA cycle even when fatty acids are the main source of acetyl coA in the tissues.
- It clears the lactate produced by muscle and erythrocytes and glycerol produced by adipose tissue.

Cori cycle (the Lactic acid cycle)

- The pathway through which lactate produced by anaerobic glycolysis in skeletal muscle returns liver and is converted to glucose
- This moves back to muscle and is converted to glycogen is called the Cori cycle
- **Site: Liver**
- **Substrate: Lactate**
- **"Occurs due to absence of glucose-6-phosphate in liver"**

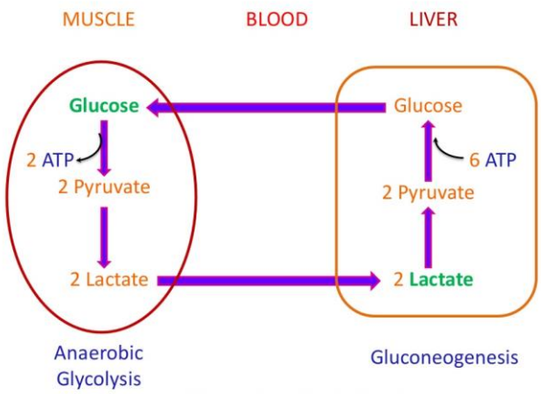


Fig: The Cori Cycle

Significance of Cori cycle

1. Prevents lactic acidosis in the muscle under anaerobic conditions
2. The cycle is also important in producing ATP, an energy source, during muscle activity
3. The Cori cycle is a much more important source of substrate for gluconeogenesis than food.
4. The drug **METFORMIN** can cause lactic acidosis in patients because metformin inhibits the h gluconeogenesis of the Cori cycle

Glucose-Alanine cycle

- When in extrahepatic tissues amino acids are used for energy, pyruvate, derived from GLYCOL is used as amino group acceptor, forming **alanine**
- Alanine diffuses into the bloodstream and reaches the liver
- In the liver, the amino group of alanine is transferred to **α-ketoglutarate** to form pyruvate
- The amino group enters the urea cycle, and in part acts as a nitrogen donor in many biosyn pathways.
- Pyruvate enters the GLUCONEOGENESIS and is used for glucose synthesis
- The newly formed glucose diffuses into the bloodstream and reaches the peripheral tissues wher to GLYCOLYSIS, is converted into pyruvate that can accept amino groups from the free amino thus closing the cycle.

Significance of Glucose-Alanine cycle

1. The Alanine cycle is less productive than the Cori cycle, which uses lactate, since a byproduct of e production from alanine is production of Urea
2. Removal of the urea is energy-dependent, requiring four "high-energy" phosphate bonds
3. This pathway requires the presence of Alanine aminotransferase, restricted to mucle, liver and int
4. Therefore, this pathway is used instead of the Cori cycle only when an aminotransferase is prese a need to transfer ammonia to the liver
5. The alanine cycle also Recycles carbon skeletons between muscle and liver
6. Transports ammonium to the liver and is converted into urea.

Gluconeogenesis and glycolysis

Gluconeogenesis and glycolysis are not identical pathways running in opposite directions, although th share several steps

7 of the 10 enzymatic reactions of gluconeogenesis are the reverse of glycolytic reactions

- three reactions of glycolysis are essentially irreversible in vivo and cannot be used in gluconeoge

1. the conversion of glucose to glucose 6-phosphate by hexokinase,
2. the phosphorylation of fructose 6-phosphate to fructose 1,6-bisphosphate by phosphofructokinas
3. the conversion of phosphoenolpyruvate to pyruvate by pyruvate kinase

Glycolysis VS Gluconeogenesis

The reactions of glycolysis are on the left side, in red; the opposing pathway of gluconeogenesis the right, in blue.

Bypass reactions in Gluconeogenesis

Glycolysis VS Gluconeogenesis

The reactions of glycolysis are on the left side, in red; the opposing pathway of gluconeogenesis is on the in blue.

- Both are energy generating pathways-**NOT identical**
- At one time only one of 2 can take place within a cell
- Glycolysis takes place essentially in the cytoplasm
- Gluconeogenesis occurs mutually in the cytosol and mitochondria
- Requirement of gluconeogenesis-**Glycogen**
- **Gluconeogenesis** from pyruvate share 7 reversible steps of glycolysis
- **Seven out of the 10 enzymes** used in the glycolytic pathway **are** used..
- The 3 irreversible steps of glycolysis that proceed with a large negative free energy chang bypassed during **gluconeogenesis** by using different enzymes.
- These three bypass reactions (during glycolysis) are catalyzed by:
- Pyruvate kinase
- Phosphofructokinase-1 (PFK-1)
- Hexokinase/glucokinase

TABLE 14–2 Free-Energy Changes of Glycolytic Reactions in Erythrocytes

Glycolytic reaction step	$\Delta G'°$ (kJ/mol)	ΔG (kJ/mol)
1 Glucose + ATP ⟶ glucose 6-phosphate + ADP	−16.7	−33.4
2 Glucose 6-phosphate ⇌ fructose 6-phosphate	1.7	0 to 25
3 Fructose 6-phosphate + ATP ⟶ fructose 1,6-bisphosphate + ADP	−14.2	−22.2
4 Fructose 1,6-bisphosphate ⇌ dihydroxyacetone phosphate + glyceraldehyde 3-phosphate	23.8	−6 to 0
5 Dihydroxyacetone phosphate ⇌ glyceraldehyde 3-phosphate	7.5	0 to 4
6 Glyceraldehyde 3-phosphate + P_i + NAD^+ ⇌ 1,3-bisphosphoglycerate + NADH + H^+	6.3	−2 to 2
7 1,3-Bisphosphoglycerate + ADP ⇌ 3-phosphoglycerate + ATP	−18.8	0 to 2
8 3-Phosphoglycerate ⇌ 2-phosphoglycerate	4.4	0 to 0.8
9 2-Phosphoglycerate ⇌ phosphoenolpyruvate + H_2O	7.5	0 to 3.3
10 Phosphoenolpyruvate + ADP ⟶ pyruvate + ATP	−31.4	−16.7

Note: $\Delta G'°$ is the standard free-energy change, as defined in Chapter 13 (pp. 507–508). ΔG is the free-energy change calculated from the actual concentrations of glycolytic intermediates present under physiological conditions in erythrocytes, at pH 7. The glycolytic reactions bypassed in gluconeogenesis are shown in red. Biochemical equations are not necessarily balanced for H or charge (p. 517).

Table 14-2
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

1. Conversion of Pyruvate to Phosphoenolpyruvate

- The first of the bypass reactions in gluconeogenesis is the conversion of pyruva phosphoenolpyruvate (PEP).
- This reaction cannot occur by simple reversal of the pyruvate kinase reaction
- Instead, the phosphorylation of pyruvate is achieved by a sequence of reactions that in eukar requires 2 enzymes in both the **cytosol and mitochondria**.
- Pyruvate is first transported from the cytosol into mitochondria or is generated from alanine v mitochondria by transamination, in which the **alpha -amino group is transferred from al (leaving pyruvate) to an alpha-keto carboxylic acid**
- In mitochondria, pyruvate is converted to oxaloacetate in a biotin (coenzyme)-requiring rea catalyzed by **pyruvate carboxylase**, a mitochondrial enzyme

$$\text{Pyruvate} + HCO_3^- + \text{ATP} \longrightarrow \text{oxaloacetate} + \text{ADP} + P_i$$

- **Pyruvate carboxylase** is the first regulatory enzyme in the gluconeogenic pathway
- **Pyruvate carboxylase requires acetyl-CoA as a positive effector. (Acetyl-CoA is produc fatty acid oxidation)**
- the mitochondrial membrane has no transporter for oxaloacetate, before export to the cytos oxaloacetate formed from pyruvate must be reduced to malate by mitochondrial **m dehydrogenase**, at the expense of NADH

Alternative paths from pyruvate to phosphoenolpyruvate

- Oxalo acetic acid (OAA) produced in mitochondria is impermeable through mitochondrial mem
- Oxaloacetate + NADH + H^+ L-malate + NAD^+
- Therefore, Oxalo acetic acid is converted into malate, exported from the mitochondrion int cytoplasm, **converted back to oxaloacetate** inorder to allow gluconeogenesis
- In both mitochondria and cytoplasm, the transporter of malate is malate dehydrogenase
- In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by **PEP carboxykinase**.
- The CO2 incorporated in the pyruvate carboxylase reaction is lost here as CO2. The decarboxy leads to a rearrangement of electrons that facilitates attack of the carbonyl oxygen of the pyr moiety on the phosphate of GTP

Synthesis of phosphoenolpyruvate from pyruvate.

- (a) In mitochondria, pyruvate is converted to oxaloacetate in **a biotin-requiring reaction** catalyz pyruvate carboxylase.
- (b) In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by PEP carboxykinase.
- The CO2 incorporated in the pyruvate carboxylase reaction is lost here as CO2.
- **The decarboxylation leads to a rearrangement of electrons that facilitates attack of the car oxygen of the pyruvate moiety on the phosphate of GTP**.

Role of biotin in the pyruvate carboxylase reaction.

- The cofactor biotin is covalently attached to the enzyme through an amide linkage to the amino of a Lys residue, forming a biotinylenzyme.
- The reaction occurs in two phases, which occur at two different sites in the enzyme.
- At catalytic site 1, bicarbonate ion is converted to CO2 at the expense of ATP. Then CO2 react biotin, forming carboxybiotinyl- enzyme.
- The long arm composed of biotin and the Lys side chain to which it is attached then carry the C carboxybiotinyl enzyme to catalytic site 2 on the enzyme surface, where CO2 is released and with the pyruvate, forming oxaloacetate and regenerating the biotinyl-enzyme.

Role of biotin in the pyruvate carboxylase reaction.

- The cofactor biotin is covalently attached to the enzyme through an amide linkage to the ´-amino of a Lys residue, forming a biotinylenzyme.
- The reaction occurs in two phases, which occur at two different sites in the enzyme.
- At catalytic site 1, bicarbonate ion is converted to CO2 at the expense of ATP. Then CO2 react biotin, forming carboxybiotinyl- enzyme.
- The long arm composed of biotin and the Lys side chain to which it is attached then carry the C carboxybiotinyl enzyme to catalytic site 2 on the enzyme surface, where CO2 is released and with the pyruvate, forming oxaloacetate and regenerating the biotinyl-enzyme.
- There is a logic to the route of these reactions through the mitochondrion.
- The [NADH]/[NAD+] ratio in the cytosol is 8 X 10-4, about 10 times lower than in mitochondri
- Because cytosolic NADH is consumed in gluconeogenesis (in the conversion of bisphosphoglycerate to glyceraldehyde 3-phosphate
- Glucose biosynthesis cannot proceed unless NADH is available.
- The transport of malate from the mitochondrion to the cytosol and its reconversion there to oxaloa effectively moves NAD to the cytosol, and converts it into NADH.
- This path from pyruvate to PEP therefore provides an important balance between NADH produce consumed in the cytosol during gluconeogenesis.

Alternative paths from pyruvate to phosphoenolpyruvate

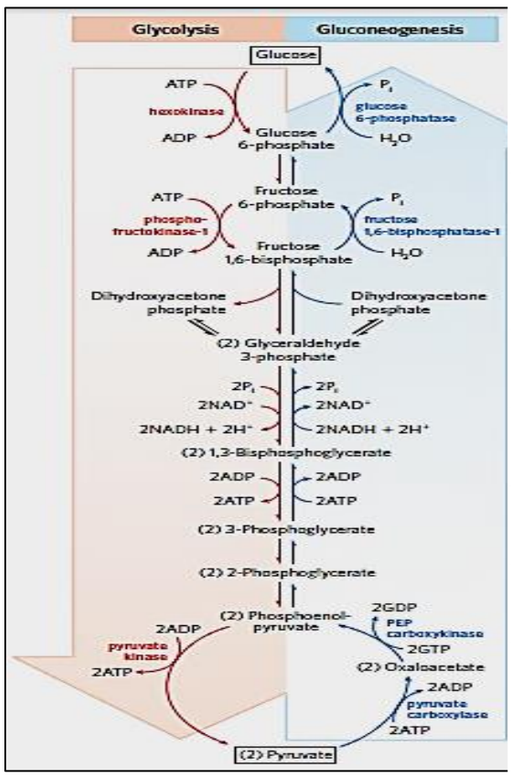
- The relative importance of the two pathways depends on the availability of lactate or pyruvate a cytosolic requirements for NADH for gluconeogenesis.
- The path on the right predominates when lactate is the precursor, because cytosolic NADH is gen in the lactate dehydrogenase reaction and does not have to be shuttled out of the mitochondrion

- The requirements of ATP for pyruvate carboxylase and GTP for PEP carboxykinase are omitte simplicity.

Bypass reactions in Gluconeogenesis-2

Conversion of Fructose 1,6-Bisphosphate to Fructose 6-Phosphate Is the Second Bypass

- The second glycolytic reaction that cannot participate in gluconeogenesis is the phosphorylati fructose 6-phosphate by PFK-1
- Because this reaction is highly exergonic and therefore irreversible, the generation of fructose 6-phosphate from fructose 1,6-bisphosphate is catalyzed by a different enzyme, **fructose 1,6-bisphosphatase (FBPase-1)**
- **fructose 1,6-bisphosphatase (FBPase-1) is** Mg+2-dependent
- **FBPase-1** promotes the essentially irreversible *hydrolysis* of the C-1 phosphate (*not* phosphoryl group transfer to ADP



fructose 1,6-bisphosphate + H20 → fructose 6-phosphate + Pi

Conversion of Glucose 6-Phosphate to Glucose Is the Third Bypass

- The third bypass is the final reaction of gluconeogenesis, the dephosphorylation of glucose 6-phosphate to yield glucose
- Reversal of the hexokinase reaction would require phosphoryl group transfer from glucose 6-phosphate to ADP, forming ATP… An energetically unfavorable reaction
- The reaction catalyzed by **glucose 6-phosphatase** does not require synthesis of ATP; it is a simple hydrolysis of a phosphate ester

Glucose 6-phosphatase + H20 → Glucose + Pi

- This Mg+2-activated enzyme (**glucose 6-phosphatase)** is found on the luminal side c endoplasmic reticulum of hepatocytes, renal cells, and epithelial cells of the small intestine but other tissues, therefore other organs are unable to supply glucose to the blood.
- If other tissues had glucose 6-phosphatase, this enzyme's activity would hydrolyze **the gluc phosphate needed within those tissues for glycolysis**.
- Glucose produced by gluconeogenesis in the liver or kidney or ingested in the diet is delivered to other tissues, including brain and muscle, through the bloodstream.

Gluconeogenesis Is Energetically Expensive, but Essential

- The sum of the biosynthetic reactions leading from pyruvate to free blood glucose is as follows

$$2\ \text{Pyruvate} + 4\text{ATP} + 2\text{GTP} + 2\text{NADH} + 2\text{H}^+ + 4\text{H}_2\text{O} \longrightarrow \text{glucose} + 4\text{ADP} + 2\text{GDP} + 6\text{P}_i + 2\text{NAD}^+ \quad (14\text{–}9)$$

- For each molecule of glucose formed from pyruvate, six high-energy phosphate groups are requ four from ATP and two from GTP.
- In addition, two molecules of NADH are required for the reduction of two molecules of 1,3-bisphosphoglycerate.
- Clearly, Glycolysis is not simply the reverse of the equation for conversion of glucose to pyruva glycolysis, which would require only two molecules of ATP

$$\text{Glucose} + 2\text{ADP} + 2\text{P}_i + \text{NAD}^+ \longrightarrow 2\ \text{pyruvate} + 2\text{ATP} + 2\text{NADH} + 2\text{H}^+ + 2\text{H}_2\text{O}$$

- The synthesis of glucose from pyruvate is a relatively expensive process

- this high energy cost is necessary to ensure the irreversibility of gluconeogenesis
- Under intracellular conditions, the overall free-energy change of glycolysis is at least -63 kJ/mol
- Under the same conditions the overall Delta*G* of gluconeogenesis is -16 kJ/mol.
- Thus both glycolysis and gluconeogenesis are essentially irreversible processes in cells.
- A second advantage to investing energy to convert pyruvate to glucose is **that if pyruvate were instead excreted, its considerable potential for ATP production by complete, aerobic oxida would be lost (more than 10 ATP are produced per pyruvate**

TABLE 14–3 Sequential Reactions in Gluconeogenesis Starting from Pyruvate

Pyruvate + HCO_3^- + ATP ⟶ oxaloacetate + ADP + P_i	×2
Oxaloacetate + GTP ⇌ phosphoenolpyruvate + CO_2 + GDP	×2
Phosphoenolpyruvate + H_2O ⇌ 2-phosphoglycerate	×2
2-Phosphoglycerate ⇌ 3-phosphoglycerate	×2
3-Phosphoglycerate + ATP ⇌ 1,3-bisphosphoglycerate + ADP	×2
1,3-Bisphosphoglycerate + NADH + H^+ ⇌ glyceraldehyde 3-phosphate + NAD^+ + P_i	×2
Glyceraldehyde 3-phosphate ⇌ dihydroxyacetone phosphate	
Glyceraldehyde 3-phosphate + dihydroxyacetone phosphate ⇌ fructose 1,6-bisphosphate	
Fructose 1,6-bisphosphate ⟶ fructose 6-phosphate + P_i	
Fructose 6-phosphate ⇌ glucose 6-phosphate	
Glucose 6-phosphate + H_2O ⟶ glucose + P_i	
Sum: 2 Pyruvate + 4ATP + 2GTP + 2NADH + $2H^+$ + $4H_2O$ ⟶ glucose + 4ADP + 2GDP + $6P_i$ + $2NAD^+$	

The bypass reactions are in red; all other reactions are reversible steps of glycolysis. The figures at the r indicate that the reaction is to be counted twice, because two three-carbon precursors are required to ma molecule of glucose. The reactions required to replace the cytosolic NADH consumed in the glyceralde 3-phosphate dehydrogenase reaction (the conversion of lactate to pyruvate in the cytosol or the transpor reducing equivalents from mitochondria to the cytosol in the form of malate) are not considered in this summary. Biochemical equations are not necessarily balanced for H and charge.

Gluconeogenesis and Glycolysis are Reciprocally regulated

Reciprocal Regulation of Gluconeogenesis and Glycolysis in the Liver

- Gluconeogenesis and glycolysis are coordinated so that within a cell one pathway is relatively ina while the other is highly active.
- The rate of glycolysis is determined by the concentration of glucose
- The rate of gluconeogenesis by the concentrations of lactate and other precursors of glucose
- Pyruvate kinase is inhibited by phosphorylation during starvation.
- The level of fructose 2,6-bisphosphate is high in the fed state and low in starvation.
- The interconversion of fructose 6-phosphate and fructose 1,6-bisphosphate is stringently control
- AMP stimulates phosphofructokinase, whereas ATP and citrate inhibit it.
- Fructose 1,6-bisphosphatase, is inhibited by AMP and activated by citrate.
- A high level of AMP indicates that the energy charge is low and signals the need for ATP gener
- Conversely, high levels of ATP and citrate indicate that the energy charge is high and that biosyn intermediates are abundant.
- Under these conditions, glycolysis is nearly switched off and gluconeogenesis is promoted.
- Phosphofructokinase and fructose 1,6-bisphosphatase are also reciprocally controlled by *fructos bisphosphate in the liver*
- The level of F-2,6-BP is low during starvation and high in the fed state, because of the antago effects of glucagon and insulin on the production and degradation of this signal molecule.
- *Fructose 2,6-bisphosphate strongly stimulates phosphofructokinase and inhibits fructose bisphosphatase*.
- Hence, glycolysis is accelerated and gluconeogenesis is diminished in the fed state.
- During starvation, gluconeogenesis predominates because the level of F-2,6-BP is very low. Gl formed by the liver under these conditions is essential for the viability of brain and muscle.
- The interconversion of phosphoenolpyruvate and pyruvate also is precisely regulated.
- High levels of ATP and alanine, which signal that the energy charge is high and that building b are abundant, inhibit the enzyme in liver.
- Conversely, pyruvate carboxylase, which catalyzes the first step in gluconeogenesis from pyruv activated by Acetyl CoA and inhibited by ADP.
- Likewise, ADP inhibits phosphoenolpyruvate carboxykinase.

- Hence, gluconeogenesis is favored when the cell is rich in biosynthetic precursors and ATP

Regulatory effect of hormones

- The regulators in this case are hormones.
- Hormones affect gene expression primarily by changing the rate of transcription, as well regulating the degradation of mRNA
- Insulin, which rises subsequent to eating, stimulates the expression of phosphofructokinase, pyr kinase, and the bifunctional enzyme that makes and degrades F-2,6-BP.
- Glucagon, which rises during starvation, inhibits the expression of these enzymes and stim instead the production of two key gluconeogenic enzymes, phosphoenolpyruvate carboxykinas fructose 1,6-bisphosphatase.

Pentose Phosphate Pathway (PPP) of Glucose Oxidation

- The *oxidative* pentose phosphate pathway
- **Phosphogluconate pathway**
- **Hexose monophosphate pathway**
- The major catabolic fate of **glucose 6-phosphate** is glycolytic breakdown to **pyruvate,** much of which is then oxidized via the **citric acid cycle**, leading to ATP formation .
- Glucose 6-phosphate does have other catabolic fates, which lead to specialized products needed the cell.
- E.g. the **oxidation of glucose 6-phosphate to pentose phosphates** by the **pentose phosphate pathway.**
- Since it results in the synthesis of pentoses and NADPH, it is considered as an **anaerobic and anabolic pathway.**
- Occurs in cytosol

Outputs of PPP

- **Reducing NADP+ to NADPH and producing pentose phosphates**. .
- brings about **oxidation and decarboxylation** of glucose 6-phosphate (G6P) at C-1, to form pentoses.. (bone marrow, skin, intestinal mucosa) to make RNA, DNA, ATP, NADH, FADH2 a coenzyme A
- NADPH: to counter damaging effects of oxygen radicals (liver, adipose, erythrocytes, cornea)
- Erythrose 4 phosphate (E4P) used in the synthesis of aromatic amino acids

Two phases of PPP

- There are **two phases** of the pentose phosphate pathway:

1. The oxidative phase
2. The non-oxidative phase

1. The Oxidative Phase

- The Oxidative Phase Produces Pentose Phosphates and NADPH
- The first reaction of the pentose phosphate pathway is the oxidation of glucose 6-phosphate by glucose 6-phosphate dehydrogenase (G6PD) to form 6-phosphoglucono-delta-lactone, an intramolecular ester.
- NADP+ is the electron acceptor, and overall equilibrium is in the direction of NADPH formatio
- The lactone is hydrolyzed to the free acid 6-phosphogluconate by a specific enzyme lactonase, tl 6-phosphogluconate undergoes oxidation and decarboxylation by 6-phosphogluconate dehydrog

Pentose Phosphate Pathway NADPH formed in the oxidative phase is used to reduce glutathione to G (Glutathione disulfide) and to support reductive biosynthesis. The other product of the oxidative phase i **ribose 5-phosphate**, which serves as a precursor for nucleotides, coenzymes, and nucleic acids. In cells are not using ribose 5-phosphate for biosynthesis, the **non-oxidative phase recycles six molecules of tl pentose into five molecules of the hexose glucose 6-phosphate,** allowing continued production of NA and converting glucose 6-phosphate to CO2.

- The reaction generates a second molecule of NADPH.
- **Phosphopentose isomerase** converts ribulose 5-phosphate to its aldose isomer, ribose 5-phosph
- In some tissues, the pentose phosphate pathway ends at this point, and its overall equation for reactions is
- **G6-P + 2NADP$^+$ + H_2O ribulose 5-phosphate + CO2 + 2NADPH + 2H$^+$**
- **The net result is the production of NADPH, a reducing agent for biosynthetic reactions, an ribose 5-phosphate, a precursor for nucleotide synthesis.**

Summary of The Oxidative Phase

Reactants	Products	Enzyme	Description
Glucose 6-phosphate + NADP+	→ 6-phosphoglucono-δ-lactone +**NADPH**	glucose 6-phosphate dehydrogenase	Dehydrogenation. The hydroxyl on carbon 1 of glucose 6-phosphate tur into a carbonyl, generati a lactone, and, in the process, NADPH is generated.
6-phosphoglucono-δ-lactone + H_2O	→ 6-phosphogluconate+ H^+	6-phosphogluconolactonase	Hydrolysis
6-phosphogluconate+ $NADP^+$	→ ribulose 5-phosphate +**NADPH** + CO_2	6-phosphogluconate dehydrogenase	Oxidative decarboxylatio $NADP^+$ is the electron acceptor, generating another molecule of NADPH, a CO_2, and ribulose 5-phosphat

2. The Non-oxidative Phase

3. The Non-oxidative Phase Recycles Pentose Phosphates to Glucose 6-Phosphate
4. In tissues that require primarily NADPH, the pentose phosphates produced in the oxidative phase pathway are recycled into glucose 6-phosphate.
5. A series of rearrangements of the carbon skeletons takes place during which 6 five-carbon sugar phosp are converted to 5 six-carbon sugar phosphates, completing the cycle and allowing continued oxidat glucose 6-phosphate with production of NADPH
6. In nonoxidative phase, ribulose 5-phosphate is first epimerized to xylulose 5-phosphate
   - Continued recycling leads ultimately to the conversion of glucose 6-phosphate to six CO_2.
   - **Two enzymes unique to the pentose phosphate pathway act in these interconversions of su** transketolase and transaldolase.

a. **Transketolase**

- Transketolase catalyzes the transfer of a two carbon fragment from **a ketose donor to an a acceptor** In its first appearance in the pentose phosphate pathway, transketolase transfers C-1 an of xylulose 5-phosphate to ribose 5-phosphate, forming the seven-carbon product sedoheptulo phosphate.
- The remaining three carbon fragment from xylulose forms glyceraldehyde 3-phosphate.

The first reaction catalyzed by transketolase. (a) The general reaction catalyzed by transketolase transfer of a two carbon group, carried temporarily on enzyme-bound TPP, (cofactor thymine pyrophosp from a ketose donor to an aldose acceptor.

(b) **Conversion of two pentose phosphates to a triose phosphate and a seven-carbon sugar phosp sedoheptulose 7-phosphate.**

b. Transaldolase

- Transaldolase catalyzes a reaction similar to the aldolase reaction of glycolysis, that is, a **three-ca fragment is removed from sedoheptulose 7-phosphate** and condensed with glyceraldehy phosphate, forming fructose 6-phosphate and the tetrose erythrose 4-phosphate**.**
- Now transketolase acts again, forming fructose 6-phosphate and glyceraldehyde 3-phosphate erythrose 4-phosphate and xylulose 5-phosphate. Two molecules of glyceraldehyde 3-phos formed by two repetition of the process can be converted to a molecule of fructose 1,6-bisphosph in gluconeogenesis
- finally FBPase-1 and phosphohexose isomerase convert fructose 1,6-bisphosphate to gluco phosphate.
- Overall, six pentose phosphates have been converted to five hexose phosphates—the cycle is complete.
- Transketolase requires the cofactor thiamine pyrophosphate (TPP), which stabilizes a two-c carbanion in this reaction

Nonoxidative reactions of the pentose phosphate pathway.

(a) These reactions convert pentose phosphates to hexose phosphates, allowing the oxidative reactio Transketolase and transaldolase are specific to this pathway; the other enzymes also serve in the glycoly gluconeogenic pathways.

(b) A schematic diagram showing the pathway from six pentoses (5C) to five hexoses (6C). Note tha involves two sets of the interconversions shown in (a). Every reaction shown here is reversible; unidirec

arrows are used only to make clear the direction of the reactions during continuous oxidation of gluc phosphate. In the light-independent reactions of photosynthesis, the direction of these reactions is revers

Summary of The Non-Oxidative Phase

Reactants	Products	Enzymes
ribulose 5-phosphate	→ ribose 5-phosphate	Ribulose 5-Phosphate Isomerase
ribulose 5-phosphate	→ xylulose 5-phosphate	Ribulose 5-Phosphate 3- Epimerase
xylulose 5-phosphate + ribose 5-phosphate	→ glyceraldehyde 3-phosphate +sedoheptulose 7-phosphate	transketolase
sedoheptulose 7-phosphate +glyceraldehyde 3-phosphate	→ erythrose 4-phosphate + fructose 6-phosphate	transaldolase
xylulose 5-phosphate + erythrose 4-phosphate	→ glyceraldehyde 3-phosphate + fructose 6-phosphate	transketolase

3. Glucose 6-Phosphate enters (Glycolysis OR the Pentose Phosphate Pathway)

Summary of The Oxidative Phase

Reactants	Products	Enzyme	Description
Glucose 6-phosphate + NADP+	→ 6-phosphoglucono-δ-lactone +**NADPH**	glucose 6-phosphate dehydrogenase	Dehydrogenation. The hydroxyl on carbon 1 of glucose 6-phosphate turns into a carbonyl, generating a lactone, and, in the process, NADPH is generated.
6-phosphoglucono-δ-lactone + H_2O	→ 6-phosphogluconate+ H^+	6-phosphogluconolactonase	Hydrolysis
6-phosphogluconate+ $NADP^+$	→ ribulose 5-phosphate +**NADPH** + CO_2	6-phosphogluconate dehydrogenase	Oxidative decarboxylation. $NADP^+$ is the electron acceptor, generating another molecule of NADPH, a CO_2, and ribulose 5-phosphate.

Summary of The Non-Oxidative Phase

Reactants	Products	Enzymes
ribulose 5-phosphate	→ ribose 5-phosphate	Ribulose 5-Phosphate Isomerase
ribulose 5-phosphate	→ xylulose 5-phosphate	Ribulose 5-Phosphate 3-Epimerase
xylulose 5-phosphate + ribose 5-phosphate	→ glyceraldehyde 3-phosphate +sedoheptulose 7-phosphate	transketolase
sedoheptulose 7-phosphate +glyceraldehyde 3-phosphate	→ erythrose 4-phosphate + fructose 6-phosphate	transaldolase
xylulose 5-phosphate + erythrose 4-phosphate	→ glyceraldehyde 3-phosphate + fructose 6-phosphate	transketolase

Glucose 6-Phosphate Is Partitioned between (Glycolysis and the Pentose Phosphate Pathway)

- Whether glucose 6-phosphate enters glycolysis or the pentose phosphate pathway depends on th current needs of the cell and on the concentration of $NADP^+$ in the cytosol. Without this electro acceptor, the first reaction of the pentose phosphate pathway (catalyzed by G6PD) cannot proce

- When a cell is rapidly converting NADPH to $NADP^+$ in biosynthetic reductions, the level of NA rises, allosterically stimulating G6PD and increasing the flux of glucose 6-phosphate through the pentose phosphate pathway
- When the demand for NADPH slows, the level of $NADP^+$ drops, the pentose phosphate pathwa slows, and glucose 6-phosphate is instead used to fuel glycolysis.
- The ratio of NADPH to $NADP^+$ is 100:1 in liver cells (cytosol is a highly reducing environment

Wernicke-Korsakoff Syndrome

- Wernicke–Korsakoff syndrome results from thiamine (Vitamin B12), deficiency. a component c TPP, by a Defect in Transketolase
- The syndrome is more common among chronic alcoholics
- can be caused by a mutation in the gene for transketolase that results in an enzyme with a lowere affinity for TPP—an affinity one-tenth that of the normal enzyme.
- This defect makes individuals much more sensitive to a thiamine deficiency: even a moderate thiamine deficiency (tolerable in individuals with an unmutated transketolase) can drop the level TPP below that needed to saturate the enzyme.
- The result is a slowing down of the whole pentose phosphate pathway.
- Symptoms: memory loss, mental confusion, ataxia and partial paralysis.

Feeder pathways of GLYCOLYSIS

- Polysaccharides : Glycogen and starch (Complex carbs)
- Disaccharides: Maltose, lactose, trehalose, sucrose
- Monosaccharides: Fructose, mannose, galactose

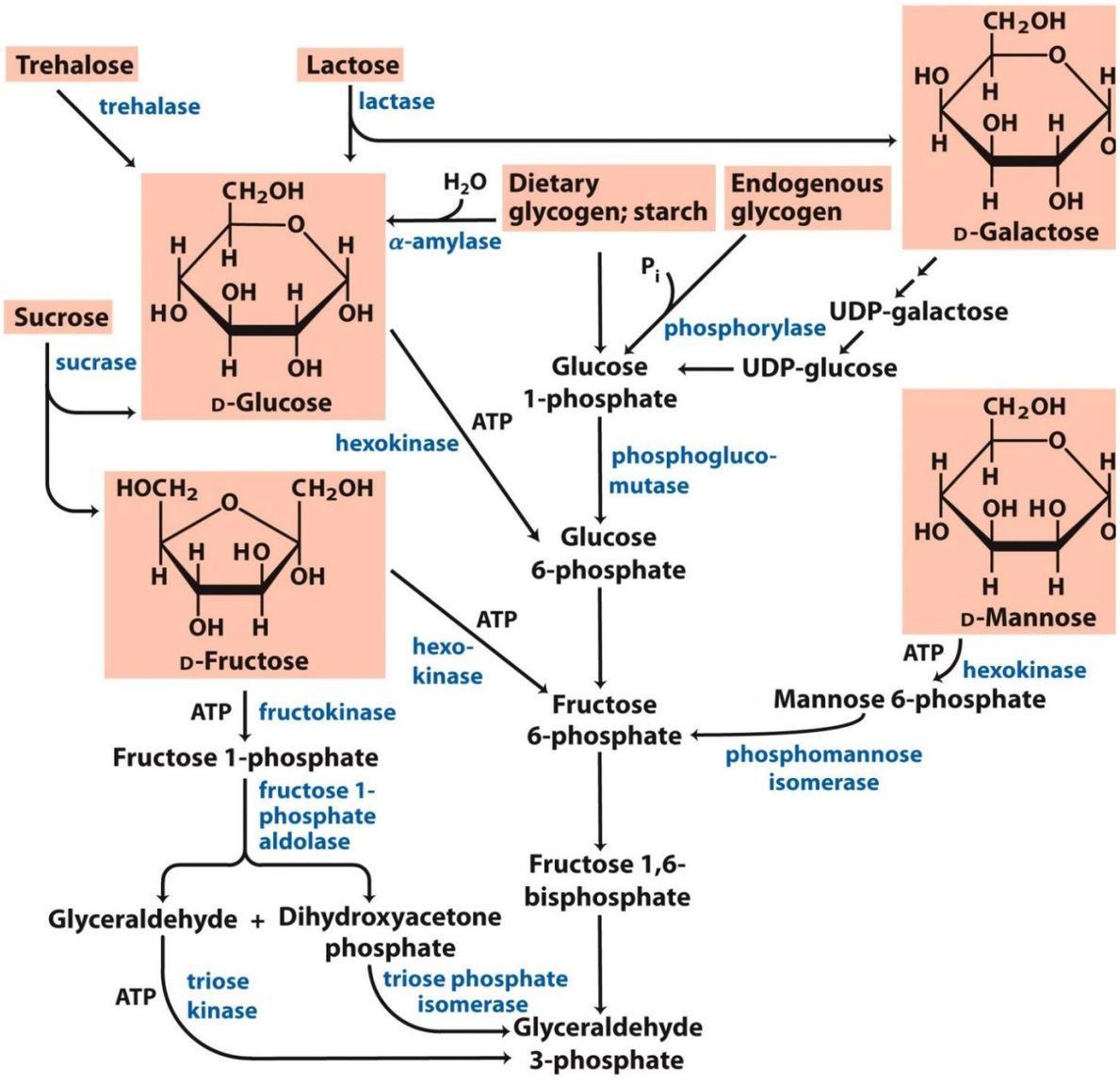


Figure 14-11
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

1. **Catabolism of Polysachharides**

- Alpha amylase in saliva and intestine breaks starch to oligosaccharides, maltose, dextrins, and f glucose
- In ruminants, in the rumen, symbiotic microorganisms that produce cellulase break down cellulos glucose molecules.
- These microorganisms use the resulting glucose an anaerobic fermentation that produces propionate, used in gluconeogenesis to produce lactc milk
- Glycogen phosphorylase (starch phosphorylase in plants) catalyzes an attack by Pi on the (a glycosidic linkage that joins the last two glucose residues at a nonreducing end, generating gluc phosphate and a polymer shortened by one re Glucose 1-phosphate produced by glycogen phosphorylase is converted to glucose 6-phospha **phosphoglucomutase,** which catalyzes the reversible reaction

$$\text{Glucose 1-phosphate} \rightleftharpoons \text{glucose 6-phosphate}$$

1. **Glycolysis**
2. **Pentose phosphate pathway**

2. Metabolism of dietary polysaccharides and disaccharides

$$\text{Dextrin} + n\text{H}_2\text{O} \xrightarrow[\text{dextrinase}]{} n \text{ D-glucose}$$

$$\text{Maltose} + \text{H}_2\text{O} \xrightarrow[\text{maltase}]{} 2 \text{ D-glucose}$$

$$\text{Lactose} + \text{H}_2\text{O} \xrightarrow[\text{lactase}]{} \text{D-galactose} + \text{D-glucose}$$

$$\text{Sucrose} + \text{H}_2\text{O} \xrightarrow[\text{sucrase}]{} \text{D-fructose} + \text{D-glucose}$$

$$\text{Trehalose} + \text{H}_2\text{O} \xrightarrow[\text{trehalase}]{} 2 \text{ D-glucose}$$

Feeder pathways of GLYCOLYSIS-2

- In most organisms, hexoses other than glucose can undergo glycolysis after conversion phosphorylated derivative.

FRUCTOSE:

- D-Fructose, present in free form in many fruits and formed by hydrolysis of sucrose in the intestine of vertebrates, is phosphorylated by hexokinase
- Major pathway of fructose entry into glyc in the muscles and kidney.
- The liver enzyme fructokinase catalyzes the phosphorylation of fructose at C-1 rather than C-6
- The fructose 1-phosphate is then cleaved to glyceraldehyde and dihydroxyacetone phospha fructose **1-phosphate aldolase**

MANNOSE:

- D-Mannose, released in the digestion of various polysaccharides and glycoproteins of foods, c phosphorylated at C-6 by hexokinase
- Mannose 6-phosphate is isomerized by phosphomannose isomerase to yield fructose 6-phospha **intermediate of glycolysis.**

$$\text{Mannose} + \text{ATP} \xrightarrow{Mg^{2+}} \text{mannose 6-phosphate} + \text{AD}$$

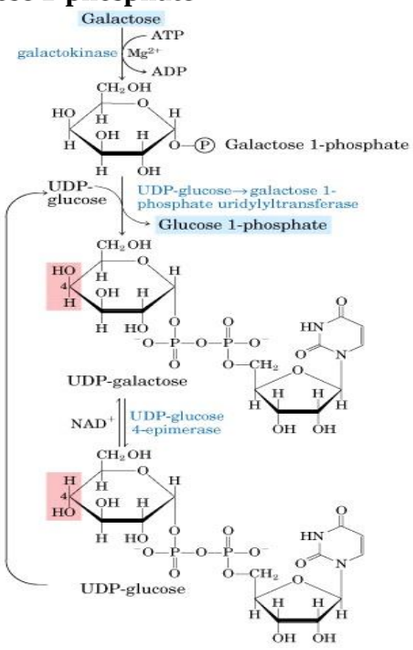
GALACTOSE:

- D-Galactose, a product of the hydrolysis of lactose (milk sugar), passes in the blood from the intestine to the liver, where it is first phosphorylated at C-1, at the expense of ATP, by the enzyme galactokinase
-

$$\text{Galactose} + \text{ATP} \xrightarrow{Mg^{2+}} \text{galactose 1-phosphate} + \text{ADP}$$

-

- **Conversion of galactose to glucose 1-phosphate**



Feeder pathways of GLYCOLYSIS-2

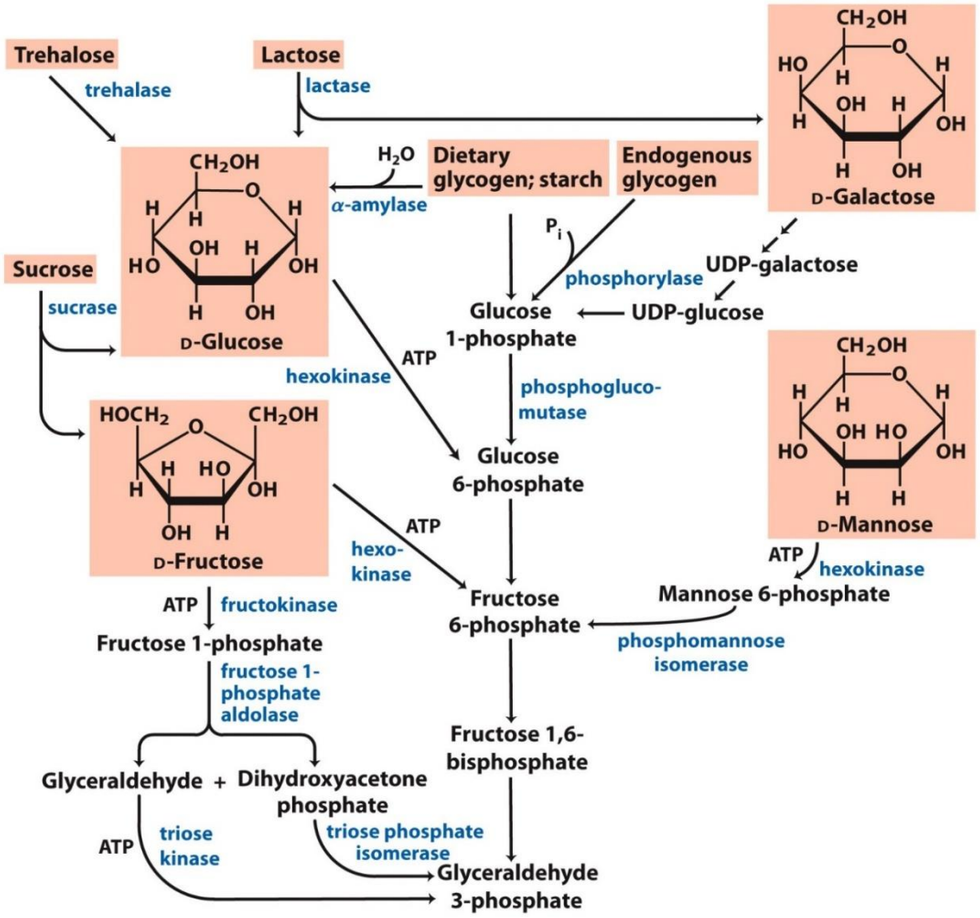


Figure 14-11
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Galactosemia

- A defect in any of the three enzymes in this pathway causes galactosemia

1. Galactokinase deficiency galactosemia
   - high conc. are found in blood and urine.
   - Infant develop cataracts, caused by deposition metabolite galactitol in the lens

2. Transferase deficiency galactosemia
   - more serious
   - characterized by poor growth in children, mental deficiency and liver damage that may be fat

3. Epimerase deficiency galactosemia
   - similar symptoms, less severe when dietary galactose is carefully controlled

Lactose intolerance

- due to the lack of lactase synthesis
- common among adults
- Without intestinal lactase, lactose cannot be completely digested and absorbed in the small inte and it passes into the large intestine, where bacteria convert it to toxic products that cause abdo cramps and diarrhea.
- The problem is further complicated because undigested lactose and its metabolites increas osmolarity of the intestinal contents, favoring retention of water in the intestine.
- In most parts of the world where lactose intolerance is prevalent, milk is not used as a food by a although milk products predigested with lactase are commercially available in some countries.
- The digestive disturbances can sometimes be minimized by a controlled diet

Mutase:

A mutase is an enzyme of the isomerase class that catalyzes the transfer of a functional group from one po to another within the same molecule

Nucleotides and Nucleic Acids

DNA Denaturation

- Separation of the two strands of the double helix when hydrogen bonds between the paired bas disrupted.
- Disruption can occur in the laboratory if
  - the pH or the salt concentration of the DNA solution is altered
  - if the solution is heated above 80°C
- Disruption of the hydrogen bonds between paired bases and of base stacking causes unwinding double helix to form two single strands complete separation of DNA stands along the entire leng
- No covalent bonds in the DNA are broken
- RNA duplexes are more stable than DNA duplexes.
- At neutral pH denaturation of a double helical RNA often requires temperatures 20° C or highe those required for denaturation of a DNA molecule with a comparable sequence.
- T_m: When DNA is heated, the temperature at which one half of the helical structure is lost is de as the melting temperature.
- A specific sequence of DNA has a characteristic denaturation temperature, or melting point (T_m)
- Careful determination of the melting temperature of a DNA specimen, under fixed conditions and ionic strength, can yield an estimate of a DNA base composition.

- **DNA Denaturation (Contd.)**
- Concomitant with this denaturation of the DNA molecule is an increase in the optical absorbar the purine and pyrimidine bases—a phenomenon referred to as hyperchromicity of denaturation.

The absorbance of double-stranded DNA (dsDNA) at 260 nm is less than that of either single-stranded (ssDNA) or the free bases. This is called **"hyperchromism."**

- So single-stranded DNA has a higher relative absorbance at this wavelength than does double-str DNA.
- Denaturation can be monitored by measuring its absorbance at 260 nm

Factors affecting Tm

- The Tm is influenced by
- the base composition of the DNA
- the salt concentration of the solution
- The higher the content of GC base pairs, the higher the melting point of the DNA.
- This is because GC base pairs, with three hydrogen bonds, require more heat energy to dissociate AT base pairs.
- An *increase* in salt concentration *increases* and a decrease in salt concentration decreases the Tm

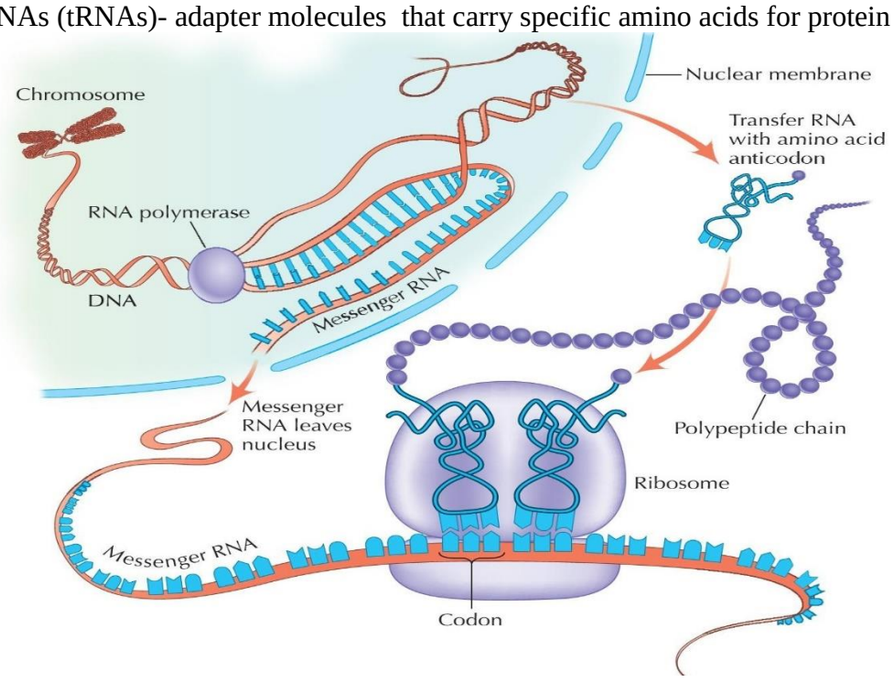
DNA Renaturation

- Under appropriate conditions *(temp. & salt concentration)*, separated strands of DNA will renat reassociate and form the double helix by the process called **renaturation (or reannealing).**
- This reannealing process is also referred to as **hybridization.**

- When the temperature or pH is returned to the range in which most organisms live, the unw segments of the two strands spontaneously rewind, or anneal, to yield the intact duplex
- Renaturation of a DNA molecule is a rapid one-step process.
- The **rate of reassociation** depends upon the concentration of the complementary strands.
- At a given temperature and salt concentration, a particular nucleic acid strand will associate t only with a complementary strand
- This property is utilized in analyzing nucleotide sequencing of a given nucleic acid.
- Under appropriate conditions DNA will form a hybrid with a complementary DNA or w complementary RNA
- Hybridization is combined with gel electrophoresis techniques that separate nucleic acids by size
- Coupled with radioactive or fluorescent probe labeling to provide a detection of a nucleotide sequ

RNA

- The genetic master plan is contained in the nucleotide sequence of DNA.
- It is through the ribonucleic acid (RNA)—the "working copies" of the (DNA) — that the maste is expressed
- RNA is a polymer of ribonucleotides of Adenine, Uracil, Guanine and Cytosine, joined together l 5´phosphodiester bonds.
- RNA does not contain thymine except in rare cases.
- The pentose sugar of RNA is D-ribose.
- Location: RNA is found in the nucleolus, ribosomes, mitochondria, and cytoplasm.
- The genetic material for some animal and plant viruses is RNA rather than DNA.
- There is a wide variety of RNAs,
- messenger RNAs (mRNAs)- transfer genetic information from DNA to the protein-synthes machinery.
- ribosomal RNAs (rRNAs)- contribute to the formation and function of ribosomes
- transfer RNAs (tRNAs)- adapter molecules that carry specific amino acids for protein synthesis



Differences between DNA and RNA

- Although sharing many features with DNA, RNA possesses several specific differences.
- Size: They are considerably smaller than DNA
- Sugar**:** In RNA, the sugar moiety to which the phosphates and purine and pyrimidine bases are att is ribose rather than the 2'-deoxyribose of DNA
- Pyrimidine: The pyrimidine components of RNA differ from those of DNA.
- Instead of thymine, RNA contains the ribonucleotide of uracil.
- Thymine is present in the rare case of tRNA
- Single Strand**:**
- RNA typically exists as a single strand whereas DNA exists as a double-stranded helical molecu
- However, given the proper complementary base sequence with opposite polarity, the single stra RNA is capable of self- folding like a hairpin and thus acquiring double-stranded characteristi pairing with C, and A with U.
- Chargaff's Rules Do Not Apply:
- Since the RNA molecule is a single strand complementary to only one of the two strands of a ge

- G≠C
- A≠U
- Purines ≠ Pyrimidines

- Hydrolysis**:** RNA can be hydrolyzed by alkali to 2',3' cyclic diesters of the mononucleo compounds that cannot be formed from alkali-treated DNA because of the absence of a 2'-hyc group.
- Location: In addition to nucleus, RNA is found in cytoplasm.
- Reverse Transcription**:** DNA forms RNA by transcription whereas the process by which RNA DNA is called reverse transcription.

RNA Structural Hierarchy

- RNA has no simple, structural hierarchy that serves as a reference point, as does the double hel DNA.
- The three-dimensional structures of many RNAs, like those of proteins, are complex and unique

Primary Structure of RNA

- It is defined as the number and sequence of ribonucleotides in the RNA chain.
- The sequence is complementary to the template strand of the gene from which it was transcribed
- The ribonucleotides are held together by 3´- 5´ phosphodiester bonds.
- 3´-OH group of one nucleotide is bound to 5´-PO_4 of the other nucleotide and form a linear strar
- The ribosyl moieties are attached to the nucleobases by *N*-glycosidic bonds
- Similar to DNA RNA polymer also has polarity

Secondary Structure of RNA

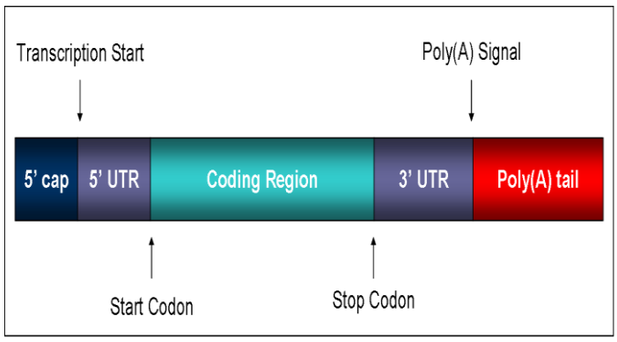
- Secondary structure involves coil formation of the polyribonucleotide chain.
- The coiled structures are stabilized by
- Hydrophobic interactions between purine and pyrimidine bases.
- Intra-chain hydrogen bonds between G-C and A-U.
- Weak interactions, especially base-stacking interactions.
- Where complementary sequences are present, the predominant double-stranded structure is form
- Common RNA secondary structures
- Internal loops: a short series of unpaired bases in a longer paired helix and
- bulges: regions in which one strand of a helix has "extra" inserted bases with no counterparts opposite strand

Tertiary Structure of RNA

- It is the folding of the molecule into three dimensional structure.
- cross-linking at various sites stabilized by hydrophobic and hydrogen bonds produces a comp coiled globular structure.
- The stacking of helices, together with specific helix–helix contacts or helix–loop interactions, l compact tertiary structure of the RNA assemblies, generally in the presence of divalent io polyamines

Messenger RNA (mRNA)

- This class is the most heterogeneous in
- Abundance
- Size (500-6000 nucleotides)
- base sequence
- Stability
- mRNA comprise about 2–5% of total cellular RNA
- mRNA molecules are formed with the help of DNA template strand (3´-5´) during the process transcription.
- In addition to the protein coding regions in the mature eukaryotic mRNA that can be translated,
- there are untranslated regions at its 5´ and 3´ ends
- Moreover, there is a 5´ cap and
- a poly A tail at 3´ end



Function of mRNA

- The members of this class function as messengers to convey the information in a gene to the p synthesizing machinery.
- The mRNA carries genetic information from the nuclear DNA to the cytosol, where it is use template for protein synthesis.

Transfer RNA (tRNA)

- t RNA is the smallest of the three major species of RNA (4S).
- They are single stranded globular molecules.
- They remain largely in cytoplasm.
- They are generated by nuclear processing of a precursor molecule.
- tRNAs compose roughly 20% of total cellular RNA
- There are at least 20 species of tRNA molecules in every cell.
- Although each specific tRNA differs from the others in its sequence of nucleotides, the molecules as a class have many features in common

Primary structure

- t RNA molecules consist of 74-95 nucleotides in a particular sequence.
- The t RNA molecules contain not only the usual bases like adenine, guanine, cytosine, uracil bu contain **unusual bases**
- These unusual bases(also called modified bases) include
- Dihydrouracil
- Pseudouridine
- Thymine.

Secondary Structure Pseudouridine

- Each single stranded

t RNA is folded extensively.

- Extensive intra chain base pairing which leads to a characteristic CLOVER-LEAF structure.
- These folds are stabilized by hydrogen bonds between complementary bases of the same strand.

Transfer RNA (tRNA) (Countinued)

Arms or loops of tRNA

- All tRNA molecules contain 4 main arms or loops.
- 1-Acceptor arm: This is made up of unpaired sequences of cytosine-cytosine-adenine (CCA) 3´end.
- The 3´OH group of adenine binds with the carboxylic group of a specific amino acid and carrie ribosomes for protein synthesis.
- **2-Anticodon arm:** It is in the form of a loop and carries specific sequences of three bases constitute the anticodon.
- The bases of anticodon are bonded with three complementary bases of codon on mRNA.
- **3-D arm:** It contains the base dihydrouridine.
- **4-TΨC arm:** It contains thymine, pseudouridine and cytosine.
- The extra arm and the TΨC arms help to define a specific tRNA.

Function of tRNA

- The t RNA molecules serve as ADAPTERS for the translation of information in the sequer nucleotides of the mRNA into specific amino acids.
- There is at least one (and often several) specific type of tRNA molecule for each of the amino commonly found in proteins.
- Each t RNA carries its specific amino acid to the site of protein synthesis.

- There it recognizes the genetic code word on mRNA (codon) and this specifies the addition of its a acids to the growing peptide chain.

Transfer RNA (tRNA) (Countinued)

Arms or loops of tRNA

- All tRNA molecules contain 4 main arms or loops.
- 1-Acceptor arm: This is made up of unpaired sequences of cytosine-cytosine-adenine (CCA) 3´end.
- The 3´OH group of adenine binds with the carboxylic group of a specific amino acid and carrie ribosomes for protein synthesis.
- **2-Anticodon arm:** It is in the form of a loop and carries specific sequences of three bases v constitute the anticodon.
- The bases of anticodon are bonded with three complementary bases of codon on mRNA.
- **3-D arm:** It contains the base dihydrouridine.
- **4-TΨC arm:** It contains thymine, pseudouridine and cytosine.
- The extra arm and the TΨC arms help to define a specific tRNA.

Function of tRNA

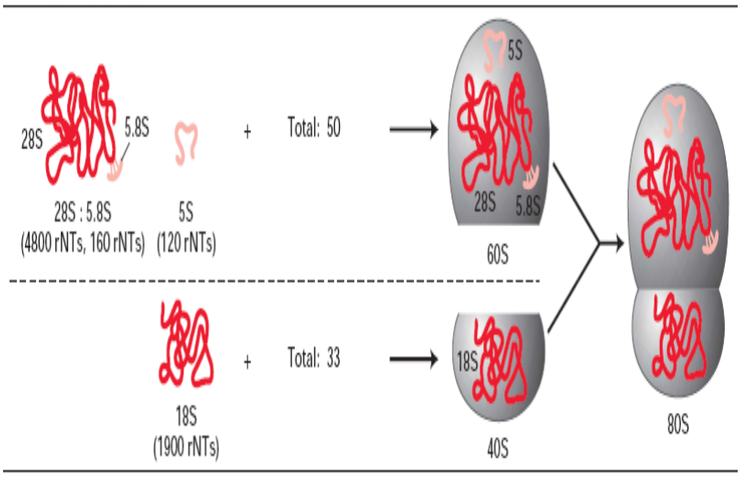
- The t RNA molecules serve as ADAPTERS for the translation of information in the sequer nucleotides of the mRNA into specific amino acids.
- There is at least one (and often several) specific type of tRNA molecule for each of the amino commonly found in proteins.
- Each t RNA carries its specific amino acid to the site of protein synthesis.
- There it recognizes the genetic code word on mRNA (codon) and this specifies the addition of its a acids to the growing peptide chain.

Ribosomal RNA (rRNA)

- found in association with several proteins as a component of ribosomes--- a cytoplasmic nucleop structure that acts as the machinery for the synthesis of proteins from the mRNA template.
- RNAs make up 80% of the total RNA in the cell.
- The ribosomal subunits are defined according to their sedimentation velocity in **Svedberg units**
- Svedberg unit is related to the molecular weight and shape of the compound.
- The bases in r RNA are mainly adenine, guanine, cytosine and uracil and a few pseudouridine

Eukaryotic Ribosome

- The mammalian ribosome contains two major nucleoprotein subunits:
- a larger one with 60S
- a smaller one with 40S.
- The **60S** subunit contains
- a 5S rRNA
- a 5.8S rRNA
- a 28S rRNA
- more than 50 specific polypeptides.
- The **40S** subunit is smaller and contains
- a single 18S rRNA
- Approx. 30 distinct polypeptide chains.



Other types of RNA

Small nuclear RNA (snRNA)

- Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic
- Most of them are complexed with proteins to form ribonucleoproteins.

- They are distributed in the nucleus, in the cytoplasm or in both.
- They are significantly involved in rRNA and mRNA processing and gene regulation.

Large & Small Noncoding Regulatory RNAs

- One of the most exciting discoveries in the last decade of eukaryotic regulatory biology has bee identification and characterization of regulatory nonprotein coding RNAs (ncRNAs).
- NcRNAs exist in two general size classes,
- small consisting of microRNA (miRNAs) and silencing (siRNAs) and
- Large consisting of long noncoding RNAs (lncRNAs)
- The small ncRNAs termed microRNA (miRNAs) and silencing (siRNAs) typically inhibit expression at the level of specific protein production by
- targeting mRNAs through one of several distinct mechanisms.
- Both siRNAs and miRNAs typically hybridize, via the formation of RNA–RNA hybridization tc targeted mRNAs
- **long noncoding RNAs (lncRNAs).**
- LncRNAs, which as their name implies, do not code for protein (ie, the mRNA encoding genes).
- ncRNAs make up a significant portion of eukaryotic transcription
- ncRNAs play many roles ranging from contributing to structural aspects of chromatin to regulat mRNA gene transcription by RNA polymerase II.

Enzymes

IUB Classification of Enzymes

- International Union of Biochemists (IUB) developed an unambiguous system of enzyme nomenc in which each enzyme has a
- unique name and
- code number
- As an example, the formal systematic name of the enzyme (hexokinase) catalyzing the reaction
- ATP + D-glucose→ ADP + D-glucose-6 phosphate is
- ATP:glucose phosphotransferase,
- Its Enzyme Commission (E.C.) number is 2.7.1.1.
- (2) denotes the class name (transferase)
- (7) the subclass phosphotransferase
- (1) denotes a hydroxyl group as acceptor
- (1) D-glucose as the phosphoryl group acceptor.
- In the systematic naming system, enzymes are divided into six major classes each with num subgroups

CLASSIFICATION OF ENZYMES

Group of Enzyme	Reaction Catalysed	Examples
1. Oxidoreductases	Transfer of hydrogen and oxygen atoms or electrons from one substrate to another.	Dehydrogenases Oxidases
2. Transferases	Transfer of a specific group (a phosphate or methyl etc.) from one substrate to another.	Transaminase Kinases
3. Hydrolases	Hydrolysis of a substrate.	Estrases Digestive enzymes
4. Isomerases	Change of the molecular form of the substrate.	Phospho hexo Isomerase, Fumarase
5. Lyases	Nonhydrolytic removal of a group or addition of a group to a substrate.	Decarboxylases Aldolases
6. Ligases (Synthetases)	Joining of two molecules by the formation of new bonds.	Citric acid synthetase

CLASSIFICATION OF ENZYMES		
Group of Enzyme	**Reaction Catalysed**	**Examples**
1. Oxidoreductases	Transfer of hydrogen and oxygen atoms or electrons from one substrate to another.	Dehydrogenases Oxidases
2. Transferases	Transfer of specific group (a phosphate or methyl etc.) from one substrate to another.	Transaminase Kinases
3. Hydrolases	Hydrolysis of a substrate.	Estrases Digestive enzymes

CLASSIFICATION OF ENZYMES		
Group of Enzyme	**Reaction Catalysed**	**Examples**
4. Isomerases	Change of the molecular form of the substrate.	Phospho hexo Isomerase, Fumarase
5. Lyases	Nonhydrolytic removal of a group or addition of a group to a substrate.	Decarboxylases Aldolases
6. Ligases (Synthetases)	Joining of two molecules by the formation of new bonds.	Citric acid Synthetase

1) Oxidoreductases

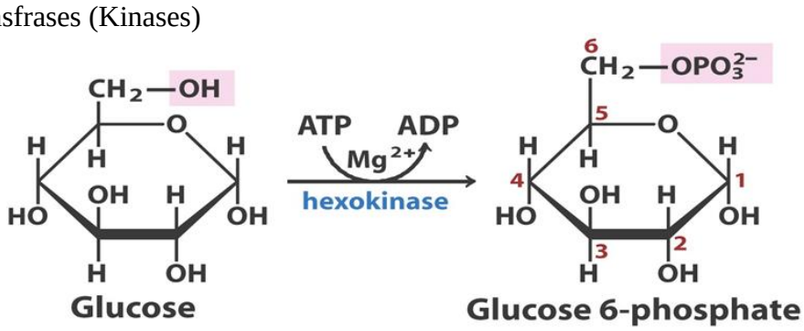
- catalyze oxidation reduction reactions
- further divided into four subgroups;
- Oxidase,
- Dehydrogenases,
- Hydroperoxidases
- Oxygenases.

Two reactions both catalyzed by Xanthine oxidase are given

- Hypoxanthine → xanthine
- Xanthine → Uric acid

2) Transferases

- These bring about a transfer of functional groups such as
- phosphate and
- amino group
- from one molecule to another molecule called donor and acceptor molecules respectively.
- The common examples of this group are
- Transminases
- Phosphotransfrases (Kinases)

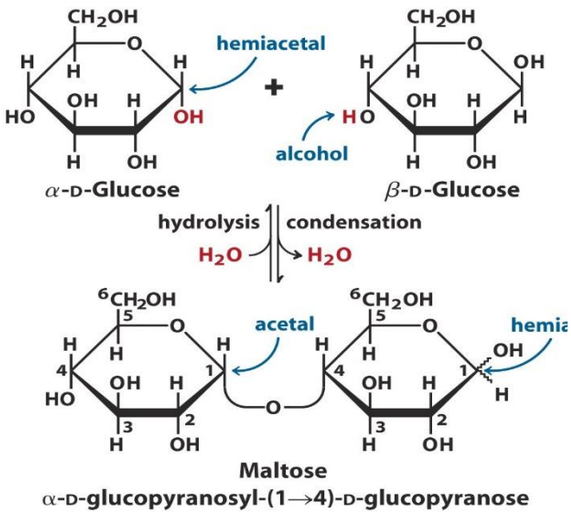


- Hexokinase is a phosphotransfease which catalyze the transfer of phosphate groups.
- Glucose + ATP → Glucose 6-phosphate + ADP.

IUB Classification of Enzymes (Contd.)

3) Hydrolases

- These enzymes catalyze hydrolysis, i.e.
- add water molecule to the substrate which is simultaneously decomposed; the functional gro substrate is transferred to water.
- Common example of hydrolases are:
- Protein hydrolyzing Enzymes (peptidases).
- Carbohydrases
- Lipid hydrolyzing enzymes e.g. Lipases and
- Phospholipases.

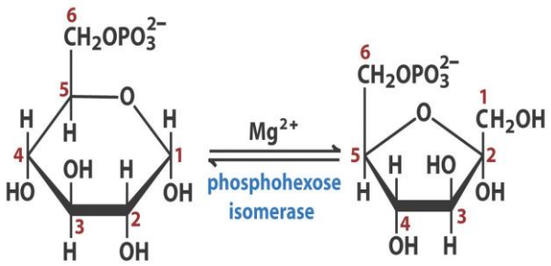


4) Lyases

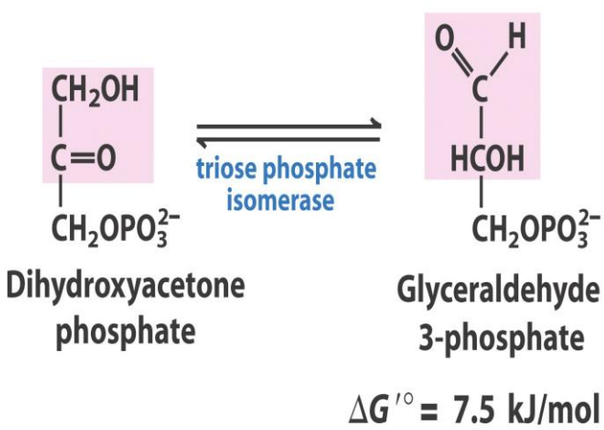
- Theses enzymes catalyze the addition of
- NH_3,
- H_2O or
- CO_2 to double bonds or
- the removal of these groups leaving behind double bonds.
- Lyases are included in a separate class because they catalyze these reactions by means other than hydrolysis or oxidation.

5) Isomerases

- These enzymes catalyze the structural change within a single molecule by the transfer of groups v it, resulting in the formation of an isomeric form of substrate.



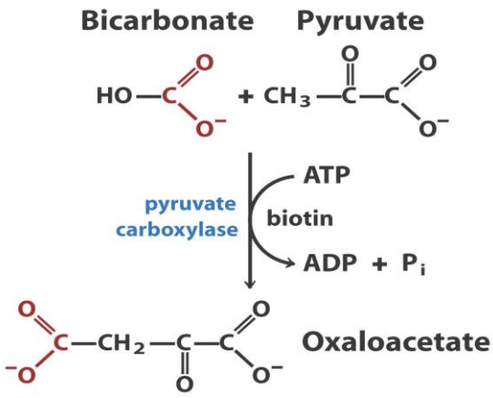
$\Delta G'^{\circ}$ = 1.7 kJ/mol



$\Delta G'^{\circ}$ = 7.5 kJ/mol

6) Ligases

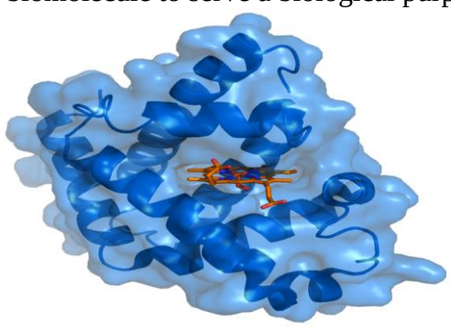
- These enzymes catalyze condensation reactions joining two molecules by forming
- C-O,
- C-S,
- C-N and
- C-C bonds.



- The energy for condensation is provided by cleavage of high energy phosphates, e.g. ATP, GTP

Ligand

- In biochemistry and pharmacology, a **ligand** (from the Latin *ligandum*, *binding*) is a substance (us a small molecule),
- that forms a complex with a biomolecule to serve a biological purpose.



- In a narrower sense, it is a signal triggering molecule, binding to a site on a target protein
- A molecule bound reversibly by a protein is called a **ligand.**
- Ligands include substrates, inhibitors, activators, and neurotransmitters
- A ligand may be any kind of molecule, including another protein.
- A ligand binds at a site on the protein called the **binding site,**

binding site is complementary to the ligand in

- size,
- shape,
- charge, and
- hydrophobic or
- hydrophilic character.
- Furthermore, the interaction is **specific**: the protein can discriminate among the thousands of dif molecules in its environment and selectively bind only one or a few.

The binding of a protein and ligand is often coupled to a **conformational change** in the protein that r the binding site more complementary to the ligand, permitting tighter binding called **induced fit.**

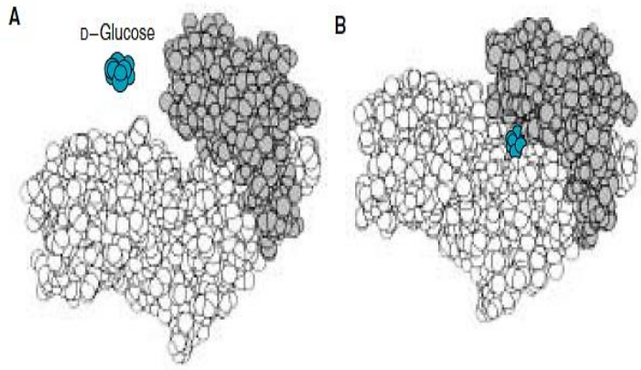
- Furthermore, the interaction is **specific**: the protein can discriminate among the thousands of dif molecules in its environment and selectively bind only one or a few.
- The binding of a protein and ligand is often coupled to a **conformational change** in the protei makes the binding site more complementary to the ligand, permitting tighter binding called **ind fit.**
- A given protein may have **separate binding sites** for several different ligands.
- In a multi-subunit protein, a conformational change in one subunit often affects the conformati other subunits.
- Interactions between **ligands** and proteins may be regulated, through interactions with addi ligands.
- These other ligands may cause conformational changes in the protein that affect the binding of the **ligand**.

Mechanism of Enzyme Action

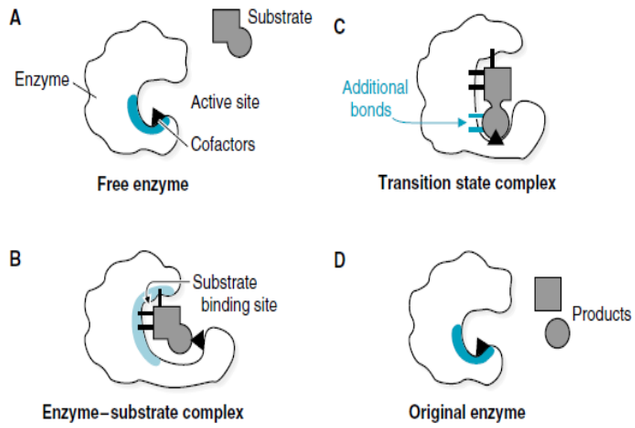
Enzymes bind and chemically transform other molecules— they catalyze reactions.

- The molecules acted upon by enzymes are called reaction **substrates** rather than ligands.
- Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 10^5 to
- The distinguishing feature of an enzyme-catalyzed reaction is that it takes place within the confi a pocket on the enzyme called the **active site.**

- The surface of the active site is lined with **amino acid** residues with side chains that complemer bind the substrate and catalyze its chemical transformation.
- Often, the active site encloses a substrate, sequestering it completely from solution.
- The enzyme substrate complex, whose existence is central to the action of enzymes.



- Enzyme-catalyzed reactions have three basic steps:
- binding of substrate: **E+S↔ES**
- conversion of bound substrate to bound product: **ES ↔ EP**
- release of product : **EP ↔ E+P**

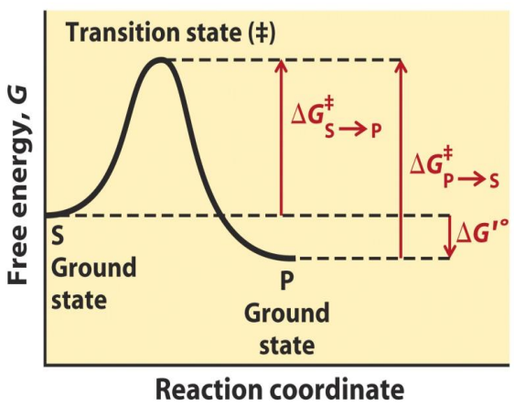


- To understand catalysis, we must first appreciate the important distinction between
- reaction equilibria and
- reaction rates.

Mechanism of Enzyme Action (Contd.)

Reaction Equilibria

- The function of a catalyst is to increase the rate of a reaction.
- Catalysts *do not affect reaction equilibria*
- Any reaction, such as S ↔ P, can be described by a reaction coordinate diagram
- The free energy of the system is plotted against the progress of the reaction **S → P**.
- A diagram of this kind is a description of the energy changes during the reaction.
- The horizontal axis (reaction coordinate) reflects the progressive chemical changes (e.g., breakage or formation) as S is converted to P.



Activation energies, *G‡,*

- The activation energies, *G‡, for* the
- **S → P and P → S** reactions are indicated.
- *G′ 0 is the overall standard* free-energy change in the direction **S → P**

Free-energy change ΔG

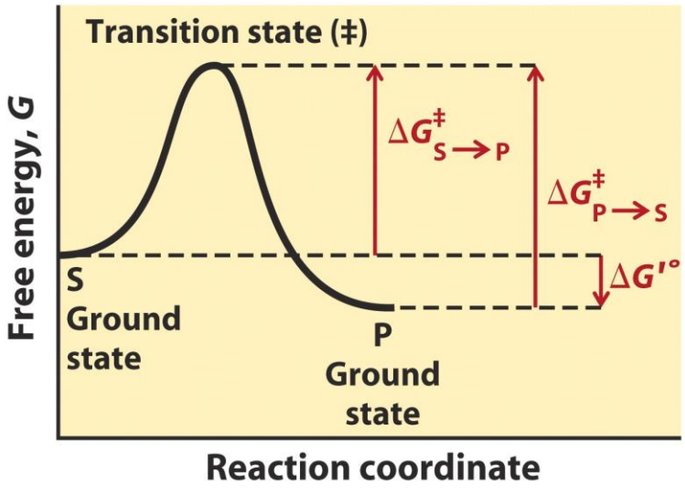
- When a reacting system is not at equilibrium, the tendency to move toward equilibrium repres driving force
- the magnitude of which can be expressed as the free-energy change for the reaction, ΔG.

Standard free-energy change, ΔG^o

- Under standard conditions (298 K = 25 C)
- when reactants and products are initially present at 1 M concentrations or
- for gases, at partial pressures of 101.3 (kPa), or 1 atm
- the force driving the system toward equilibrium is defined as the standard free-energy change, Δ
- However, because the conditions in the body systems are different from standard conditions
- energy in biological systems is described in terms of free energy, G'^{0}.

Ground State & Transition State

- The starting point for either the forward or the reverse reaction is called the **ground state**
- The equilibrium between S and P reflects the difference in the free energies of their ground state
- The free energy of the ground state of P is **lower** than that of S
- So G'^{0} *for the* reaction is negative and the equilibrium favors P.
- The position and *direction* of equilibrium are *not affected by* any catalyst.

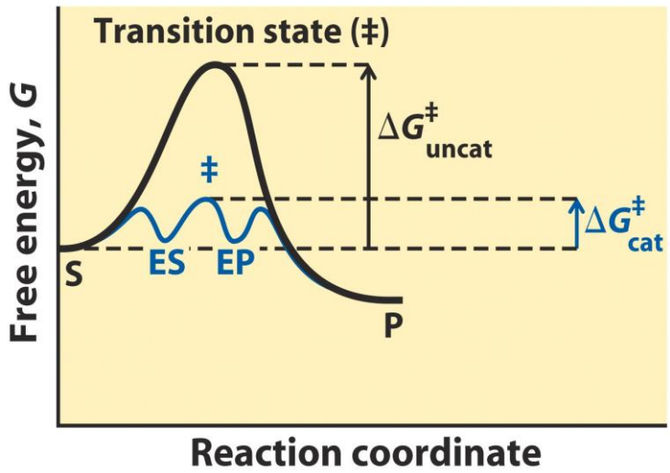


- But there is an **energy barrier** between S and P**:** The energy required for
- alignment of reacting groups
- formation of transient unstable charges
- bond rearrangements
- and other transformations
- This is illustrated by the energy "hill"
- To undergo reaction, the molecules must overcome this barrier and therefore must be raised to a h energy level
- This is called the **transition state.**
- It is simply a fleeting molecular moment in which events such as
- bond breakage,
- bond formation, and
- charge development
- have proceeded to the precise point at which decay to either substrate or product is equally likel
- The difference between the energy levels of the ground state and the transition state is the **activ energy, $G‡$.**
- At the top of the energy hill is a point at which decay to the S or P state is equally probable downhill either way)
- A substance that modifies the transition state to lower the activation energy is termed a catal biological catalyst is termed an enzyme.

Activation energy $\Delta G^{‡}$.

- The difference between the energy levels of the ground state and the transition state is the **activ energy $\Delta G^{‡}$.**
- The rate of a reaction reflects this activation energy: a higher activation energy corresponds to a s reaction.
- Reaction rates can be increased by raising the temperature, thereby increasing the number of mole with sufficient energy to overcome the energy barrier.
- Alternatively, the activation energy can be lowered by adding a catalyst.

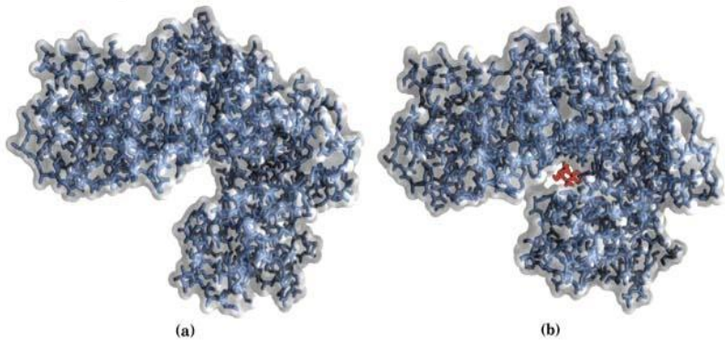
- Catalysts enhance reaction rates by lowering activation energies.



- The role of enzymes is to accelerate the inter-conversion of S and P.
- i.e ***enzymes lower the energy of activation,*** $\Delta G^{\ddagger}$, ***of a reaction.***
- The enzyme is not used up in the process, and the equilibrium point is unaffected.
- However, the reaction reaches equilibrium much faster when the appropriate enzyme is present,
- because the rate of the reaction is increased.

The Induced Fit Hypothesis

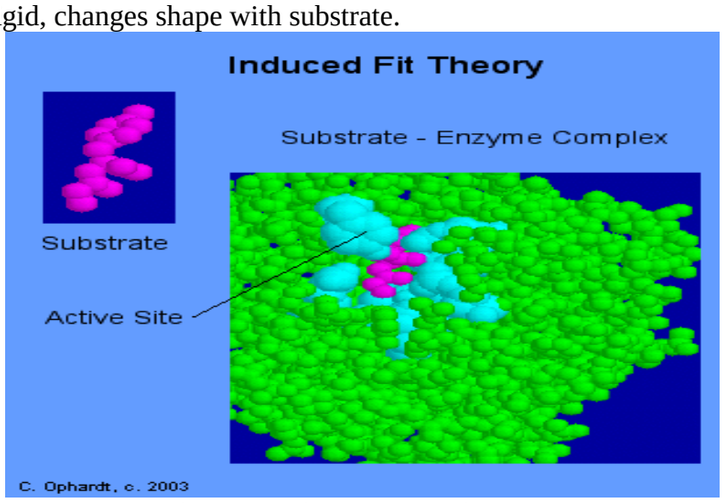
- Some proteins can change their shape (conformation)
- When a substrate combines with an enzyme, it induces a change in the enzyme's conformation
- This change in conformation when the substrate binds is induced by multiple weak interactions the substrate.
- There may also be rearrangements of covalent bonds during an enzyme-catalyzed reaction.
- This conformational change is referred to as **induced fit.**



- Chemical reactions of many types take place between substrates and enzyme's functional g (specific amino acid side chains, metal ions, and coenzymes
- Induced fit serves to bring specific functional groups on the enzyme into the proper position to cat the reaction.

Induced Fit Theory

- Enzyme is not rigid, changes shape with substrate.



- The active site is also moulded into a precise conformation
- Making the chemical environment suitable for the reaction

Cofactors, Coenzymes and Prosthetic groups

- Some enzymes require no chemical groups for activity other than their amino acid residues.
- Whereas some enzymes require molecules other than proteins for enzymic activity.
- If the non-protein moiety is a metal ion such as Zn^{2+} or Fe^{2+}, it is called a cofactor.
- If it is a complex organic molecule or metallo-organic compound it is termed a coenzyme.

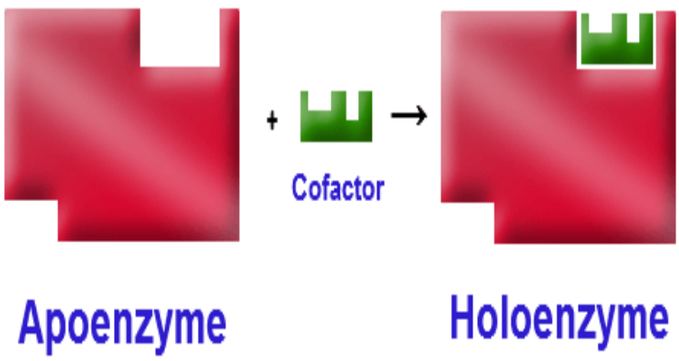
Some Coenzymes That Serve as Transient Carriers

Coenzyme	*Examples of chemical groups transferred*
Biocytin	CO_2
Coenzyme A	Acyl groups
5′-Deoxyadenosylcobalamin (coenzyme B_{12})	H atoms and alkyl groups
Flavin adenine dinucleotide	Electrons
Lipoate	Electrons and acyl groups
Nicotinamide adenine dinucleotide	Hydride ion ($:H^-$)

Some Inorganic Elements That Serve as Cofactors for Enzymes

Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^+	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase

- A coenzyme or metal ion that is very tightly or even covalently bound to the enzyme protein is a prosthetic group.
- The term holoenzyme refers to the active enzyme with its non-protein component,
- whereas the enzyme without its non-protein moiety is termed an apoenzyme (apoprotein) a inactive.



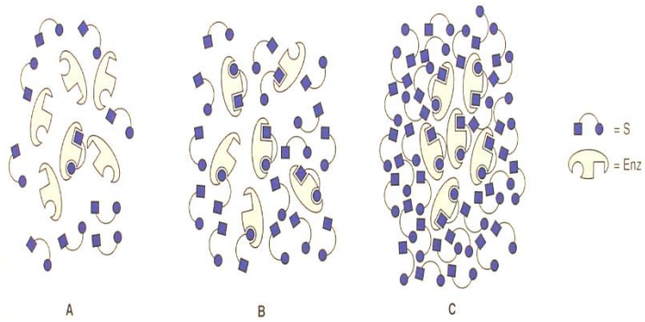
- Coenzymes serve as recyclable shuttles that transport many substrates from one point within th to another.
- The function of these shuttles is twofold.
- First, they stabilize species
- such as hydrogen atoms (FADH) or hydride ions (NADH)

- that are too reactive to persist for any significant time in the presence of the water or organic mole that permeate cells.
- Second, they serve as an adaptor or handle that facilitates the
- recognition and binding of small chemical groups, such as acetate (coenzyme A) or glucose (UD their target enzymes

Reaction Rates and Order of Reactions

Reaction Velocity (v)

- The rate or velocity of a reaction (v) is the number of substrate molecules converted to product pe time;
- Velocity is usually expressed as **µmol of product** formed **per minute**.
- The rate of an enzyme-catalyzed reaction increases with substrate concentration until a ma velocity (Vmax) is reached- reflecting the **saturation** with substrate of **all available binding site occupied** on the enzyme molecules present.



- An equilibrium such as S ↔ P is described by an **equilibrium constant, Keq**, or simply **K.**
- Under the standard conditions used to compare biochemical processes, an equilibrium const denoted ***K´eq*** (or ***K´*)**.
- The rate of any reaction is determined by the concentration of the reactant (or reactants) and by a **constant,** usually denoted by **k.**
- For the uni molecular reaction S → P, the rate (or velocity) of the reaction, V—representing the ar of S that reacts per unit time—is expressed by a **rate equation:** V = k[S].
- In this reaction, the rate depends only on the concentration of S.
- This is called a first-order reaction.
- The factor k is a proportionality constant that reflects the probability of reaction under a given conditions (pH, temperature, and so forth).
- For the uni molecular reaction S → P, the rate (or velocity) of the reaction, V—representing the ar of S that reacts per unit time—is expressed by a **rate equation:** V = k[S].
- In this reaction, the rate depends only on the concentration of S.
- This is called a first-order reaction.
- The factor k is a proportionality constant that reflects the probability of reaction under a given conditions (pH, temperature, and so forth).
- If a reaction rate depends on the concentration of two different compounds.
- or if the reaction is between two molecules of the same compound, the reaction is second order.
- The rate equation then becomes
- **$V = k[S_1][S_2]$**

Factors Affecting Enzymatic Activity

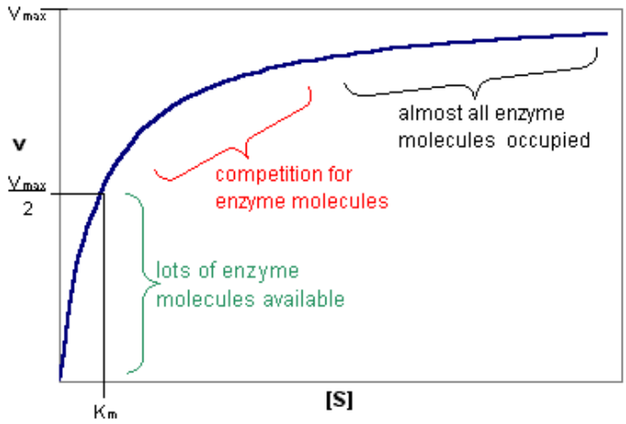
- Enzymes can be isolated from cells, and their properties studied in a test tube (that is, in vitro).

Factors Affecting Enzymatic Activity

- Different enzymes show different responses to changes in;
- substrate concentration
- temperature, and
- pH.
- Enzymic responses to these factors give us valuable clues as to how enzymes function in living (that is, in vivo)

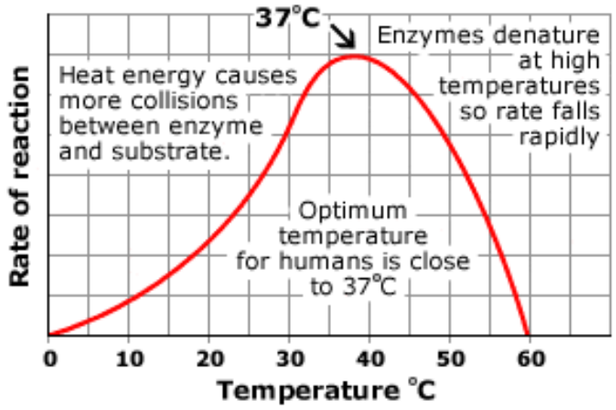
Substrate concentration

- The rate of an enzyme-catalyzed reaction increases with substrate concentration until maximal ve (Vmax) is reached
- The leveling off of the reaction rate at high substrate concentrations
- reflects the saturation with substrate
- of all available binding sites on the enzyme molecules present.



Temperature

- The reaction velocity increases with temperature until a peak velocity is reached
- This increase is the result of the increased number of molecules having sufficient energy to pass the energy barrier and form the products.



- Further elevation
- of the temperature results in a decrease in reaction velocity as a
- result of temperature-induced denaturation of the enzyme

Thank You
and
Regard in Your Prayers